Inspection of Pressure Vessels

API RECOMMENDED PRACTICE 572
FIRST EDITION, FEBRUARY 1992

American Petroleum Institute
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Inspection of Pressure Vessels

Refining Department

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FOREWORD

This recommended practice covers the inspection of pressure vessels. It is based on the accumulated knowledge and experience of engineers and other personnel in the petroleum industry. The information contained in this publication was previously presented as Chapters VI and VII of the *Guide for Inspection of Refinery Equipment,* which is currently being reorganized as individual recommended practices. The information in this recommended practice does not constitute and should not be construed as a code of rules, regulations, or minimum safe practices. The practices described in this publication are not intended to supplant other practices that have proven satisfactory, nor is this publication intended to discourage innovation and originality in the inspection of refineries. Users of this recommended practice are reminded that no book or manual is a substitute for the judgment of a responsible, qualified person.

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Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

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CONTENTS

	Page
SECTION 1—GENERAL	1
1.1 Scope	1
1.2 Referenced Publications	1
SECTION 2—TYPES OF PRESSURE VESSELS	1
2.1 Description	1
2.2 Methods of Construction	2
2.3 Materials of Construction	2
2.4 Internal Equipment	2
2.5 Uses of Pressure Vessels	3
SECTION 3—CONSTRUCTION STANDARDS	3
SECTION 4—MAINTENANCE INSPECTION	4
SECTION 5—REASONS FOR INSPECTION	5
5.1 General	5
5.2 Safety	6
5.3 Continuity	6
5.4 Reliability	6
SECTION 6—CAUSES OF DETERIORATION	7
6.1 General	7
6.2 Corrosion	8
6.3 Erosion	8
6.4 Metallurgical and Physical Changes	8
6.5 Mechanical Forces	8
6.6 Faulty Material	9
6.7 Faulty Fabrication	10
6.7.1 General	10
6.7.2 Poor Welding	10
6.7.3 Improper Heat Treatment	10
6.7.4 Dimensional Intolerance	10
6.7.5 Improper Installation	10
6.7.6 Improper Fit	10
SECTION 7—FREQUENCY AND TIME OF INSPECTION	10
7.1 Factors Governing Frequency of Inspection	10
7.2 Opportunities for Inspection	11
7.3 Inspection Schedule	14
SECTION 8—METHODS OF INSPECTION AND LIMITS	14
8.1 General	14
8.2 Safety Precautions and Preparatory Work	17
8.2.1 Safety	17
8.2.2 Preparatory Work	18
8.3 External Inspection	19
8.3.1 General	19
8.3.2 Ladders, Stairways, Platforms, and Walkways	19
8.3.3 Foundations and Supports	19
8.3.4 Anchor Bolts	20
8.3.5 Concrete Supports	20
8.3.6 Steel Supports	20
8.3.7 Guy Wires	21

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Page

8.3.8 Nozzles 21
8.3.9 Grounding Connections 22
8.3.10 Auxiliary Equipment 22
8.3.11 Protective Coatings and Insulation 22
8.3.12 External Metal Surfaces 22
8.3.13 External Evidence of Corrosion 23
8.4 Internal Inspection 24
8.4.1 General 24
8.4.2 Surface Preparation 24
8.4.3 Preliminary Visual Inspection 25
8.4.4 Detailed Inspection 25
8.4.5 Inspection of Metallic Linings 29
8.4.6 Inspection of Nonmetallic Linings 30
8.4.7 Thickness-Measuring Methods 31
8.4.8 Special Methods of Detecting Mechanical Defects 31
8.4.9 Metallurgical Changes and In-Situ Analysis of Metals 31
8.5 Testing 32
8.5.1 Hammer Testing 32
8.5.2 Pressure and Vacuum Testing 32
8.5.3 Testing Exchangers 33
8.6 Limits of Thickness 34

SECTION 9—METHODS OF REPAIR 34

SECTION 10—RECORDS AND REPORTS 35
10.1 Records 35
10.2 Reports 36

APPENDIX A—EXCHANGERS 37
APPENDIX B—SAMPLE RECORD FORMS 47

Figures
1—Type-316 Stainless-Clad Vessel 3
2—Weld Metal Surfacing 3
3—Strip-Lined Vessel 3
4—Principal Strip-Lining Methods 4
5—Reinforced Refractory Lining for Regenerator Lines and Slide Valves 4
6—Heat Exchanger Parts 5
7—Heat Exchanger Types 9
8—Catalyst Storage Hoppers 10
9—Horizontal Drum 11
10—Spheres 12
11—Horton Spheroid (Noded) 12
12—Process Towers and Drums 13
13—Fluid Catalytic Regenerator 14
14—Caustic-Stress Corrosion 14
15—Severe Graphitic Corrosion of Floating-Head Cover 15
16—Plug-Type Dezincification 15
17—Layer-Type Dezincification 15
18—Tube Sheet Fouled With Marine Growth 15
19—Tube Sheet Corroded Beneath Marine Growth 16
20—Internal Vessel Corrosion 16
21—Erosion 16

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Page

22—Condensate Grooving of Tube in Area Adjoining Tube Sheet 16
23—Intergranular Corrosion 17
24—Thermal and Pressure Damage 17
25—Fatigue Cracking at Nozzle 18
26—Exchanger Installation and Foundation 20
27—Severe Deterioration of Anchor Bolts 21
28—Method of Obtaining Vessel Profile Measurements 24
29—Pitting in Channel 25
30—Crack in Shell Weld 26
31—Catalytic-Reactor Internals—Cyclones 27
32—Hydrogen Blistering 27
33—Cross Section Through Hydrogen Blisters Showing Various Types of Crack Propagation 28
34—Corrosion Tab Method of Determining Metal Loss on Vessel Linings 29
35—Strip-Liner Deterioration 30
36—Deteriorated Refractory-Tile Lining 30
37—Steps in Using Special Equipment to Test Individual Tubes 33
A-1—Properly Rolled Tube 37
A-2—Tube-Bundle Type of Tank Heater Entirely Inside Tank 39
A-3—Air-Cooled Exchangers 40
A-4—Clean-Service Double-Pipe Coils 40
A-5—Tank Suction Heater With Everything But Forward End Enclosed; Shell Suction Nozzle Enclosed in Far End 41
A-6—Fin-Type Tubes in Double-Pipe Coil 42
A-7—Plate-Type Exchanger 43
A-8—Tubes Thinned at Baffles 44
A-9—Erosion-Corrosion Attack at Tube Ends 45

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Inspection of Pressure Vessels

SECTION 1—GENERAL

1.1 Scope

This recommended practice covers the inspection of pressure vessels. It includes a description of the various types of pressure vessels and of standards for their construction and maintenance. The reasons for inspection, the causes of deterioration, the frequency and methods of inspection, the methods of repair, and the preparation of records and reports are covered. Safe operation is emphasized.

1.2 Referenced Publications

The latest editions of the following standards, codes, and specifications shall, to the extent specified herein, form a part of this recommended practice:

ASME[1]

Boiler and Pressure Vessel Code, Section VIII, "Pressure Vessels"

API

510 *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*

RP 571 *Conditions Causing Deterioration and Failure* (in press)

RP 574 *Inspection of Piping, Valves, and Fittings*

RP 575 *Inspection of Atmospheric and Low-Pressure Storage Tanks* (in press)

RP 576 *Inspection of Pressure-Relieving Devices* (in press)

Std 660 *Shell-and-Tube Heat Exchangers for General Refinery Services*

Std 661 *Air-Cooled Heat Exchangers for General Refinery Service*

Publ 920 *Prevention of Brittle Fracture of Pressure Vessels*

Publ 945 *Avoiding Environmental Cracking in Amine Units*

Publ 2007 *Safe Maintenance Practices in Refineries*

Publ 2214 *Spark Ignition Properties of Hand Tools*

Publ 2217 *Guidelines for Work in Confined Spaces in the Petroleum Industry*

Publ 2217A *Guidelines for Work in Inert Confined Spaces in the Petroleum Industry*

NBBPVI[2]

NB-23 *National Board Inspection Code*

TEMA[3]

Standards of Tubular Exchanger Manufacturers Association

SECTION 2—TYPES OF PRESSURE VESSELS

2.1 Description

A pressure vessel is a container that falls within the scope of Section VIII of the ASME Code and that is subject to an external or internal design pressure greater than 15 pounds per square inch gauge (103 kilopascals). External pressure on a vessel can be caused by an internal vacuum or by fluid pressure between an outer jacket and the vessel wall. Vessels subject to external pressure are usually inspected in the same manner as those subject to internal pressure. Columns, towers, drums, reactors, heat exchangers, condensers, air coolers, bullets, spheres, and accumulators are common types of industry pressure vessels. [See Appendix A for an introduction to exchangers. Storage vessels subject to internal pressures up to 15 pounds per square inch gauge (103 kilopascals) are covered by API Recommended Practice 575.]

Pressure vessels are designed with various shapes. They may be cylindrical (with flat, conical, toriconical, torispherical, semiellipsoidal, or hemispherical heads), spherical, spheroidal, boxed (with flat rectangular or flat square plate heads as for the headers of air-cooled exchangers), or lobed. They may be of modular construction.

Cylindrical vessels, including exchangers and condensers, may be either vertical or horizontal and may be supported by steel columns, cylindrical plate skirts, or plate lugs attached to the shell. Spherical vessels are usually supported by steel columns attached to the shell or by skirts. Spheroidal vessels are partially or completely supported by resting on the ground. Jacketed vessels are those built with a casing or outer shell that forms an annular space between itself and the main shell.

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.

[2]National Board of Boiler and Pressure Vessel Inspectors, 1055 Crupper Avenue, Columbus, Ohio 43229.

[3]Tubular Exchanger Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591.

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2.2 Methods of Construction

Prior to the development of welding, riveting was the most common method of construction. Seams were either lapped and riveted or butted with butt straps and then riveted. To prevent leakage, the edges of the seams and rivet heads were caulked. At high temperatures, it was difficult to keep this caulking tight. After the technique of welding was developed, a light bead of weld was frequently applied to the caulking edges. Although some vessels of this type can still be found in older refineries, this method of construction is seldom used today.

Today, several different methods are used to construct pressure vessels. Most pressure vessels are constructed with welded joints.

Shell rings are usually made by rolling plate at either elevated or ambient temperature. The cylinder is formed by welding the ends of the rolled plate together. This yields a cylinder with a longitudinal weld.

Hot forging is another method of making cylindrical vessels. Some vessel manufacturers use the formation of cylindrical shell rings by hot forging for high-pressure, heavy-wall vessels such as those used for hydrotreater or hydrocracker reactors. This method does not produce a longitudinal seam in the cylinder.

In the multilayer method, the cylindrical section is made up of a number of thin concentric cylinders fabricated together, one over the other, until the desired thickness is obtained. Multilayer construction is sometimes used for heavy-wall reactors and vessels subject to high pressure.

2.3 Materials of Construction

Carbon steel is the most common material used to construct pressure vessels. For special purposes, a suitable austenitic or ferritic alloy, Monel alloy, nickel, titanium, high-nickel alloys, or aluminum may be used. Copper and copper alloys (except Monel) are seldom used in refinery vessels but may be found in petrochemical plant vessels.

Materials used to construct the various parts of exchangers are selected to safely handle the service and heat load required. Materials that will most economically resist the type of corrosion expected are selected.

Exchanger shells are usually made of carbon steel but may be made of a corrosion-resistant alloy or clad with a corrosion-resistant material.

Tubes for exchanger bundles may be made of any of a variety of materials. Where water is used as a cooling or condensing medium, they are generally made of copper-based alloys or steel. Where the exchange is between two different hydrocarbons, the tubes may be made of steel or a suitable corrosion-resistant alloy. Duplex tubing, consisting of an inner layer of one material and an outer layer of a different material, may in some cases be required to resist two different corrosive mediums.

Tube sheets for exchanger bundles are made of a variety of materials. Where water is the cooling or condensing medium, they are usually made of naval brass or steel. Where the exchange of heat is between two hydrocarbons, the tube sheets may be composed of steel or a suitable corrosion-resistant alloy. In some cases, it may be necessary to face one side of the tube sheet with a material different from that facing the other to resist two different corrosive mediums.

Exchanger channels and baffles are made of carbon steel or a suitable resistant alloy material, usually similar to the material of the tubes.

If carbon steel would not resist the corrosion or erosion expected or would cause contamination of the product, vessels may be lined with other metals or nonmetals. A lined vessel is usually more economical than one built of a solid corrosion-resistant material. However, when the pressure vessel will operate at a high temperature, a high pressure, or both, solid alloy steels may be both necessary and economical.

Metallic liners are installed in various ways. They may be an integral part of the plate material rolled or explosion bonded before fabrication of the vessel. They may instead be separate sheets of metal fastened to the vessel by welding. Corrosion-resistant metal can also be applied to the vessel surfaces by various welding processes. Metallic liners may be made of a ferritic alloy, Monel alloy, nickel, lead, or any other metal resistant to the corrosive agent.

Nonmetallic liners may be used to resist corrosion and erosion or to insulate and reduce the temperature on the walls of a pressure vessel. The most common nonmetallic lining materials are reinforced concrete, acid brick, refractory material, insulating material, carbon brick or block, rubber, glass, and plastic. Figures 1–4 show various methods of applying metallic linings. Figure 5 shows reinforced refractory lining for regenerator lines and slide valves.

2.4 Internal Equipment

Many pressure vessels have no internals. Others have internals such as baffles, trays, mesh- or strip-type packing grids, catalyst bed supports, cyclones, pipe coils, spray nozzles, demister pads, and quench lines. Large spheroids may have internal bracing and ties, and most vacuum vessels have either external or internal stiffening rings. Some pressure vessels have heat exchangers or reboilers in the lower shell area.

Exchangers have internal tube bundles with baffles or support plates, which vary with the service and heat load the exchanger is designed to handle. Pass partitions are usually installed in the channels and sometimes installed in the floating tube-sheet covers to provide multiple pass flow through the tubes. The flow through the shell may be single pass, or longitudinal baffles may be installed to provide

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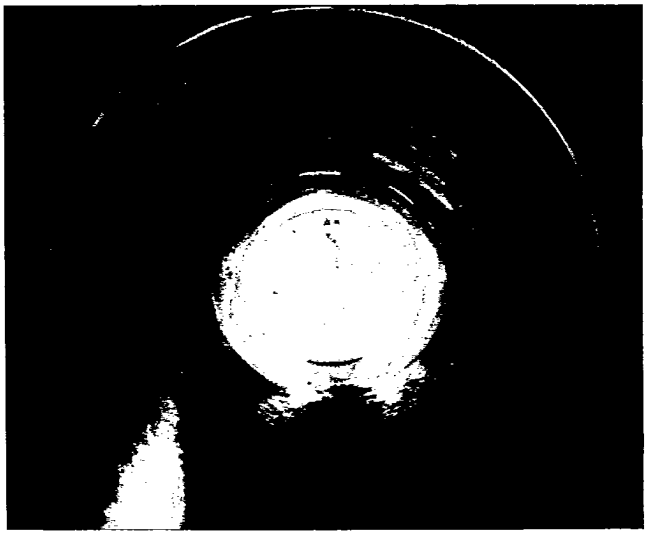


Figure 1—Type-316 Stainless-Clad Vessel

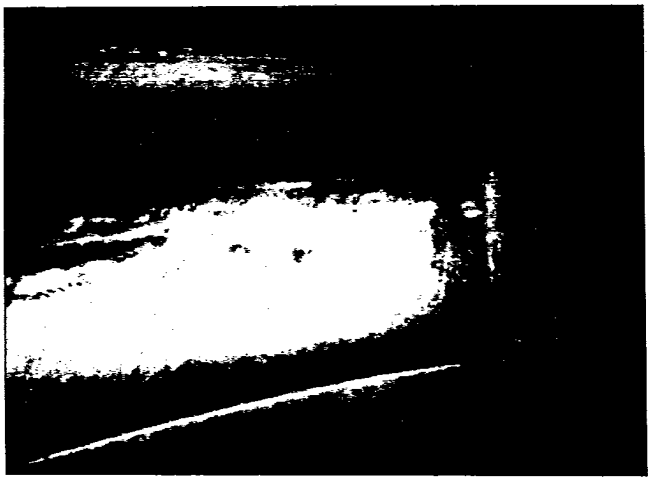


Figure 2—Weld Metal Surfacing

multiple passes. The baffling used in the shell determines the location and number of shell nozzles required. Figures 6 and 7 show various channel and shell baffle arrangements. Frequently, an impingement baffle or plate is placed below the shell inlet nozzle to prevent impingement of the incoming fluid on the bundle tubes.

2.5 Uses of Pressure Vessels

Pressure vessels are used in most processes in a refinery or petrochemical plant. They are used to store volatile or dangerous liquids. A pressure vessel can be used as a thermal reactor or a catalytic reactor to contain the chemical change required by a process; as a fractionator to separate various constituents produced in a reaction; as a separator to separate gases, chemicals, or catalyst from a product; as a surge drum for liquids; as a chemical treating unit; as a settling drum to permit separation of a chemical from a treated product; as a regenerator to restore a catalyst or chemical to its original properties; or as an exchanger, condenser, cooler, or other type of vessel for any of various other purposes. Figures 8–13 illustrate various types of pressure vessels.

SECTION 3—CONSTRUCTION STANDARDS

Prior to the early 1930s, most unfired pressure vessels for refineries were built to the design and specifications of the user or manufacturer. Later, most pressure vessels in the U.S. were built to conform either to the API/ASME *Code for Unfired Pressure Vessels for Petroleum Liquids and Gases* or to Section VIII of the ASME Code. Publication of the API/ASME *Code for Unfired Pressure Vessels for Petroleum Liquids and Gases* was discontinued December 31, 1956, and it is no longer used for new vessels.

Section VIII of the ASME Code is divided into two parts. Section VIII, Division 2, of the ASME Code provides alternative and more stringent rules for the design, fabrication, and inspection of vessels than those found in Section VIII, Division 1, of the ASME Code. Most pressure vessels for U.S. refineries are now built to conform to the latest edition of Section VIII, Division 1, of the ASME Code.

Some high-pressure vessels are designed and built in accordance with the specifications of Section VIII, Division 2, of the ASME Code.

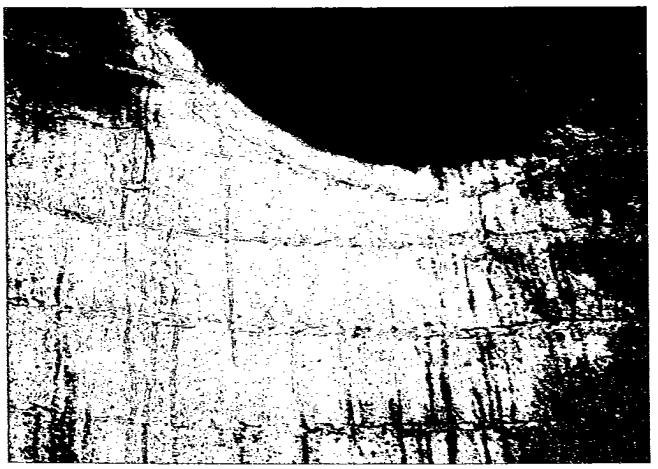


Figure 3—Strip-Lined Vessel

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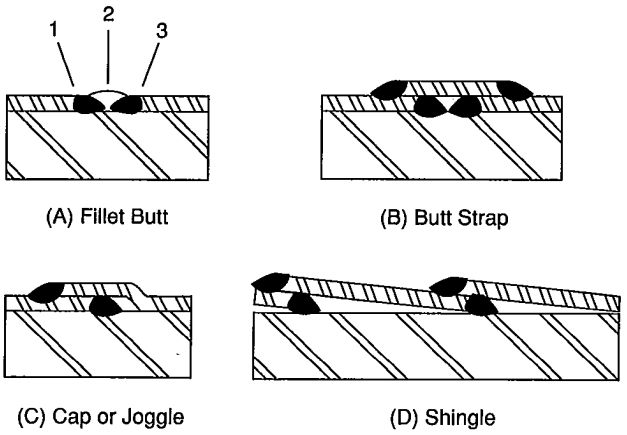


Figure 4—Principal Strip-Lining Methods

In the U.S., heat exchangers and condensers are designed and built in accordance with the ASME Code, the TEMA Standards, API Standard 660, and API Standard 661. (Other countries may have equipment design requirements other than those of ASME, TEMA, and API.)

Both Division 1 and Division 2 of Section VIII of the ASME Code require the manufacturer of a vessel to have a quality control system. Before the manufacturer can obtain a certificate of authorization from ASME, a written manual must be provided, and the system must be implemented. The quality control system requires detailed documentation of examinations, tests, and data regarding the vessel and makes available a history of the construction of the vessel. This documentation and any other supplementary documentation the manufacturer may have can be helpful in the evaluation of vessels in service.

The ASME Code lists materials that may be used for construction, gives formulas for calculating thicknesses, provides rules on methods of manufacture, and specifies the procedures for testing completed vessels. Inspection is required during the construction and testing of vessels. The code also prescribes the qualifications of the persons who perform the construction inspections.

After a qualified construction inspector certifies that a vessel has been built and tested as required by the ASME Code, the manufacturer is empowered to stamp the vessel with the appropriate symbol of the ASME Code. The symbol stamped on a pressure vessel is an assurance that the vessel has been designed, constructed, tested, and inspected as required by the ASME Code.

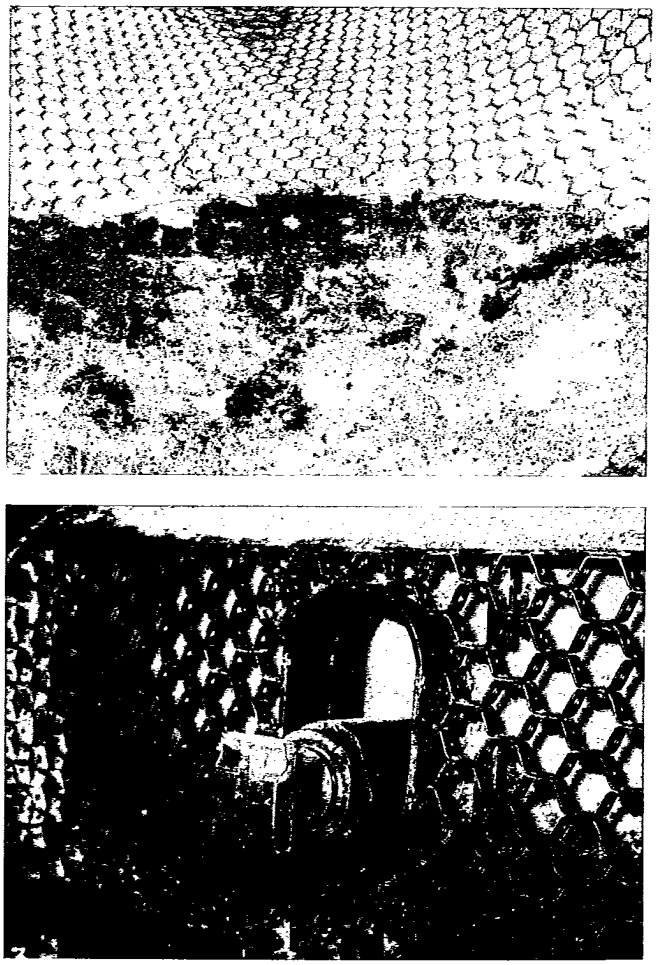


Figure 5—Reinforced Refractory Lining for Regenerator Lines and Slide Valves

Some states and cities and many countries have laws, other than the regulations of the ASME Code (and other codes), that govern the design, construction, testing, installation, inspection, and repair of pressure vessels used in their localities. These codes may supersede the ASME Code's (and other code's) minimum requirements.

SECTION 4—MAINTENANCE INSPECTION

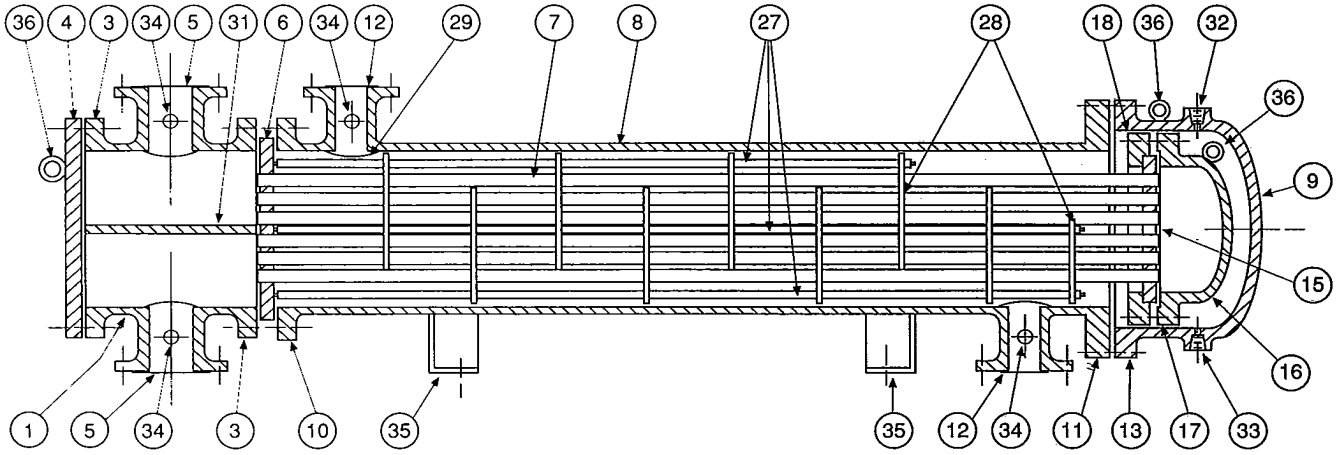
Construction codes are periodically revised as the designs of pressure vessels improve and as new construction materials become available. A pressure vessel should be maintained according to the requirements under which it was designed and constructed; if re-rated, it should be maintained according to the requirements under which it was re-rated. A refinery inspector should be familiar not only with the latest editions of codes but also with previous editions of the codes

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Legend

1. Stationary Head—Channel
2. Stationary Head—Bonnet
3. Stationary-Head Flange—Channel or Bonnet
4. Channel Cover
5. Stationary-Head Nozzle
6. Stationary Tube Sheet
7. Tubes
8. Shell
9. Shell Cover
10. Shell Flange—Stationary-Head End
11. Shell Flange—Rear-Head End
12. Shell Nozzle
13. Shell-Cover Flange
14. Expansion Joint
15. Floating Tube Sheet
16. Floating-Head Cover
17. Floating-Head Flange
18. Floating-Head Backing Device
19. Split Shear Ring
20. Slip-On Backing Flange
21. Floating-Head Cover—External
22. Floating Tube-Sheet Skirt
23. Packing Box
24. Packing
25. Packing Gland
26. Lantern Ring
27. Tie Rods and Spacers
28. Transverse Baffles or Support Plates
29. Impingement Plate
30. Longitudinal Baffle
31. Pass Partition
32. Vent Connection
33. Drain Connection
34. Instrument Connection
35. Support Saddle
36. Lifting Lug
37. Support Bracket
38. Weir
39. Liquid Level Connection



AES

Note: *AES* refers to heat exchanger type (refer to Figure 7).

Figure 6—Heat Exchanger Parts

and with other specifications under which any vessels he inspects were built. The inspector should also familiarize himself with any regulations [including city, county, parish, provincial, state, or national (such as OSHA[4]) regulations] governing inspection and maintenance of pressure vessels in the refinery. The inspector should be familiar with the contents of API 510 and NBBPVI NB-23, where applicable.

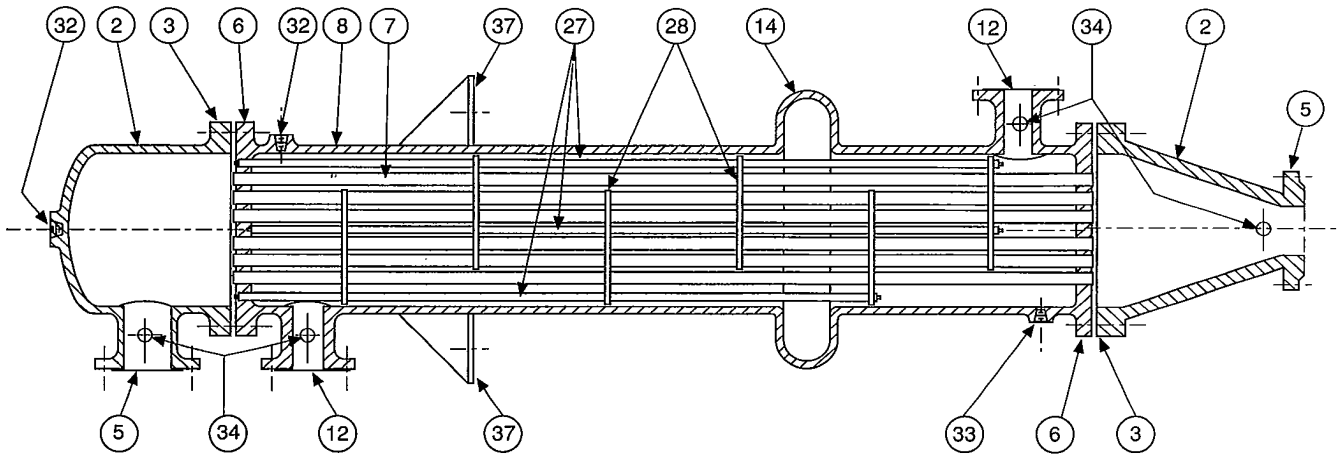
SECTION 5—REASONS FOR INSPECTION

5.1 General

The basic reasons for inspection are to determine the physical condition of the vessel and to determine the type, rate, and causes of deterioration. This information should be carefully documented after each inspection. With this data, safety can be maintained, the period of operation without a shutdown can be extended, the rate of deterioration can sometimes be reduced, and future repair and replacement requirements can be estimated.

[4]Occupational Safety and Health Administration, Francis Perkins Department of Labor Building, 200 Constitution Avenue, N.W., Washington, D.C. 20210.

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BEM

Note: *BEM* refers to heat exchanger type (refer to Figure 7).

Figure 6 (Continued)

5.2 Safety

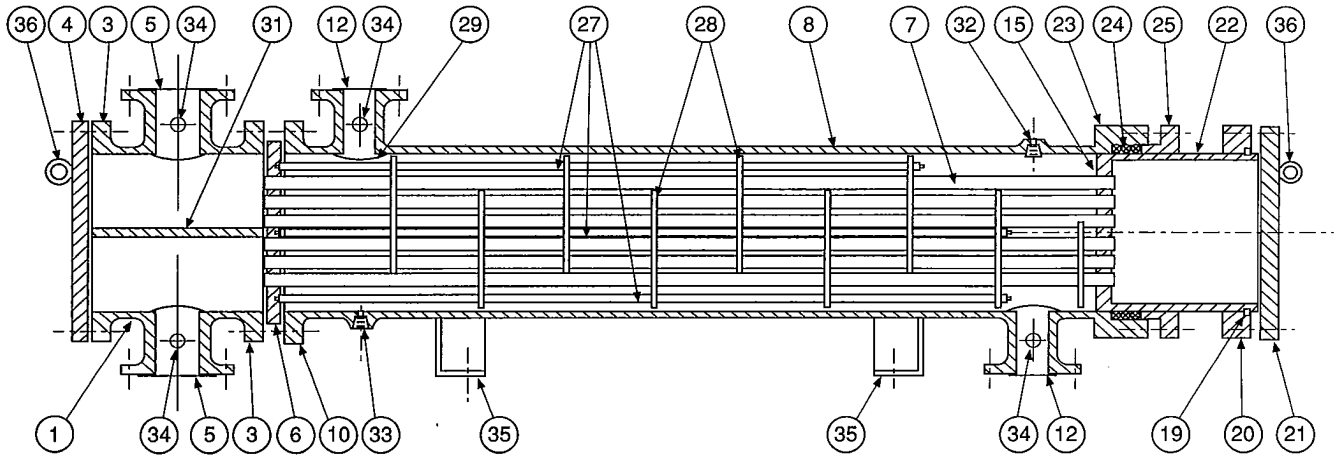
Periodic scheduled inspections can reveal conditions that might result in emergency or unscheduled shutdowns, leaks, or even vessel failures if not corrected.

5.3 Continuity

Periodic inspection can lead to a well-planned maintenance program. The corrosion rates and remaining corrosion allowances determined by inspection form the normal bases for predicting replacement or repair needs. These predictions support planned maintenance and continuity of operation and help to ensure a safe, reliable plant.

5.4 Reliability

External inspections made during equipment operation with acoustic, ultrasonic, or radiographic instruments or the tools of other nondestructive techniques may reveal important information without requiring the equipment to be opened. Defects such as leaks, cracks, improper installation of parts, plugged lines, undue vibration, unusual noises, and other evidence of malfunctioning may be found. If these symptoms are properly analyzed and corrective steps are taken, the overall reliability of operations will improve.

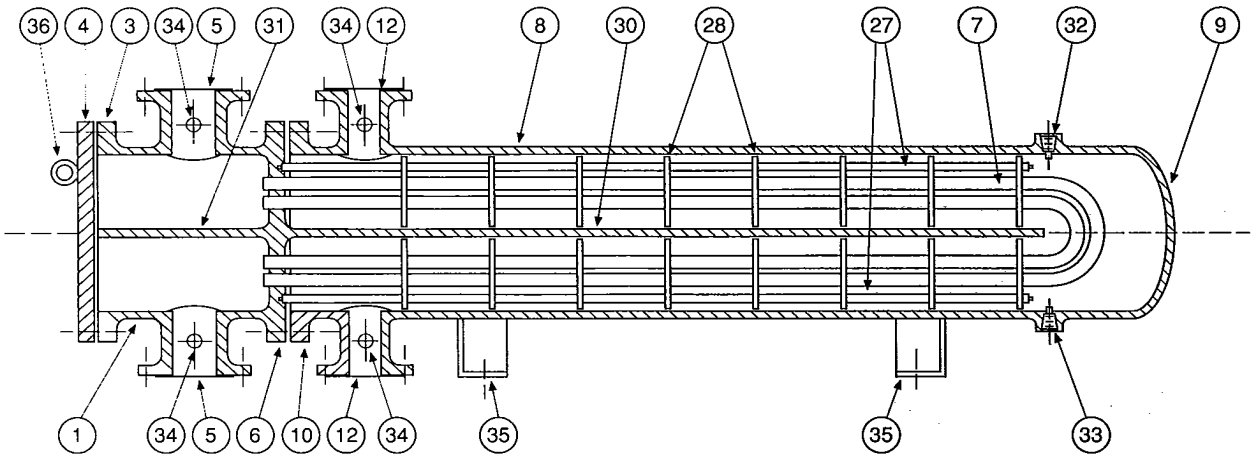


AEP

Note: *AEP* refers to heat exchanger type (refer to Figure 7).

Figure 6 (Continued)

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CFU

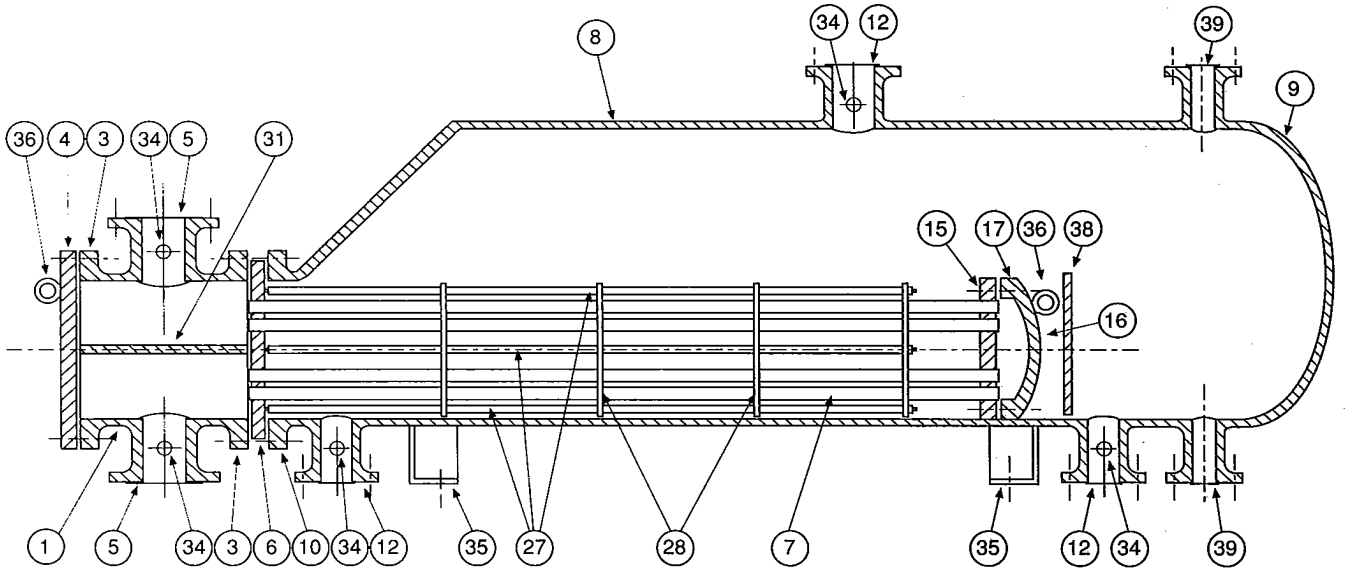
Note: *CFU* refers to heat exchanger type (refer to Figure 7).

Figure 6 (Continued)

SECTION 6—CAUSES OF DETERIORATION

6.1 General

Deterioration is possible on all vessel surfaces in contact with any of a wide range of organic and inorganic compounds, with contaminated or fresh water, with steam, or with the atmosphere. The form of the deterioration may be electrochemical, chemical, mechanical, or a combination of the three. The deterioration may be accelerated by temperature, stress, fatigue (including that due to vibration), impingement, or high velocity or irregularity of flow. API Recommended Practice 571 covers this subject in detail.

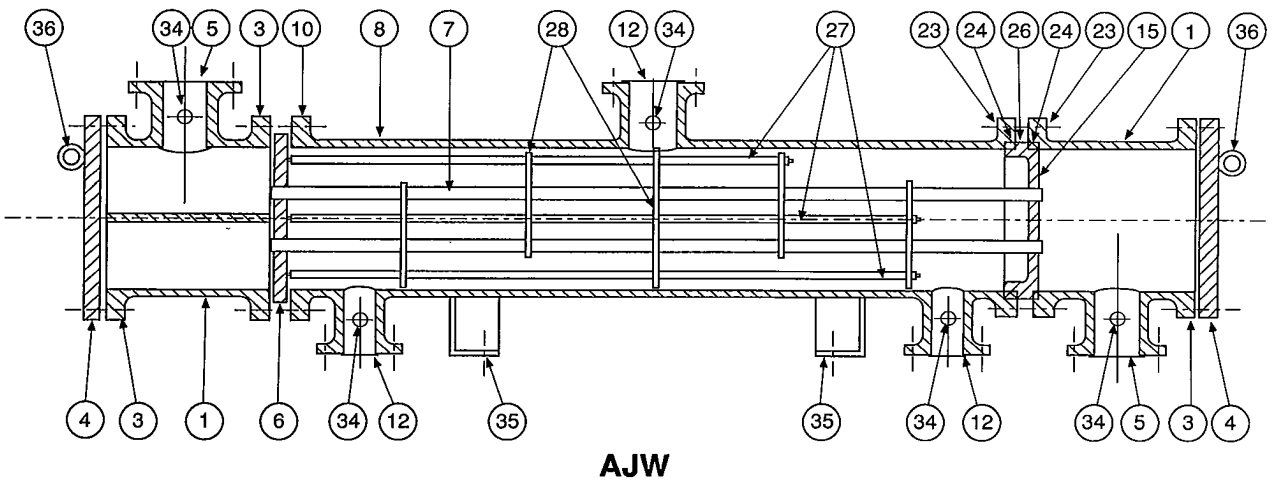


AKT

Note: *AKT* refers to heat exchanger type (refer to Figure 7).

Figure 6 (Continued)

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Note: *AJW* refers to heat exchanger type (refer to Figure 7).

Figure 6 (Continued)

6.2 Corrosion

Corrosion is the prime cause of deterioration in a pressure vessel and may occur on any part of the vessel. The severity of the deterioration is influenced by the concentration, temperature, and nature of the corrosive agents in the fluid and by the corrosion resistance of the construction materials. API Recommended Practice 571 covers corrosion in detail.

The most common internal corrodents in refineries are sulfur and chloride compounds. Caustics, inorganic acids, organic acids, water (especially water with a low pH), deposit or cellular attack chemicals, and other chemicals used in a particular process may also corrode vessels.

Figure 14 shows stress-corrosion cracking caused by a caustic. Figure 15 shows severe graphitic corrosion of a floating-head cover. Figures 16 and 17 show plug-type and layer-type dezincification of exchanger tubes. Figures 18 and 19 show fouling with and corrosion beneath marine growth.

External corrosion, especially under insulation, varies with atmospheric conditions and increases with humidity. It is often accelerated by contaminants in the atmosphere. Atmospheric conditions in industrial areas and seashore locations are especially corrosive.

6.3 Erosion

Erosion is the attrition of a surface from the impingement of solid particles or liquid drops on the surface. It is usually found where restrictions in flow, changes in direction, or disturbances result in local high velocities. Erosion is typically found at inlet and outlet nozzles, on internal piping, on grid or tray sections, on vessel walls opposite inlet nozzles, on internal support beams, and on impingement baffles. API Recommended Practice 571 covers erosion in detail.

Occasionally, corrosion and erosion combine to increase rates of deterioration. The combination of corrosion and erosion may occur opposite nozzles in the bottom of fractionators of fluid catalytic cracking units. Figures 20 and 21 illustrate corrosion and erosion. Figure 22 illustrates condensate grooving of an exchanger tube in the area adjoining a tube sheet; it also shows a combination of erosion and corrosion. API Recommended Practice 571 covers erosion-corrosion in detail.

6.4 Metallurgical and Physical Changes

Pressure vessel metals are exposed to service conditions that may cause microstructural or metallurgical changes in the metal. These changes often affect the mechanical properties of the metal and may result in cracking or other deterioration. Microstructural changes may result either from improper heating and cooling of the metal or from metallurgical chemical changes in the metal. Examples of these changes are graphitization, high-temperature hydrogen attack, carbide precipitation, intergranular corrosion (see Figure 23), and embrittlement. More detailed descriptions of metallurgical and physical changes may be found in API Recommended Practice 571.

6.5 Mechanical Forces

Mechanical forces can cause a vessel to fail or to operate inefficiently unless adequate provision for the forces has been made. Thermal shock (Figure 24 shows thermal and pressure damage), cyclic temperature changes, vibration (Figure 25 shows fatigue cracking caused by vibration), excessive pressure surges from any cause, and external loads

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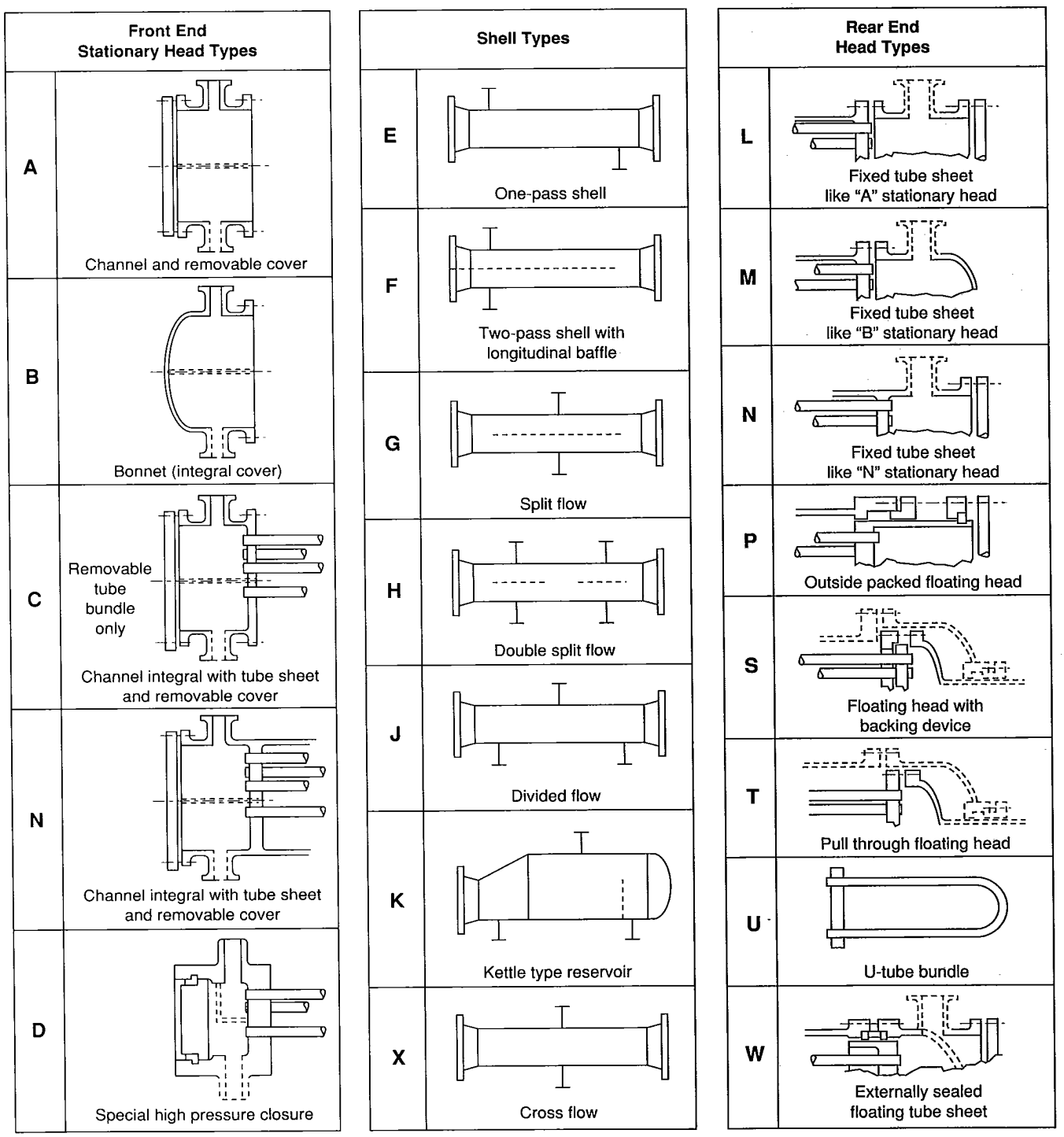


Figure 7—Heat Exchanger Types

are examples of sources of mechanical forces. Cracks, bulges, distortion, and upset internal equipment are visual signs of the application of mechanical forces. API Recommended Practice 571 covers mechanical forces in detail.

6.6 Faulty Material

Many of the problems that may develop in pressure vessels can be traced to faulty material or fabrication. Some

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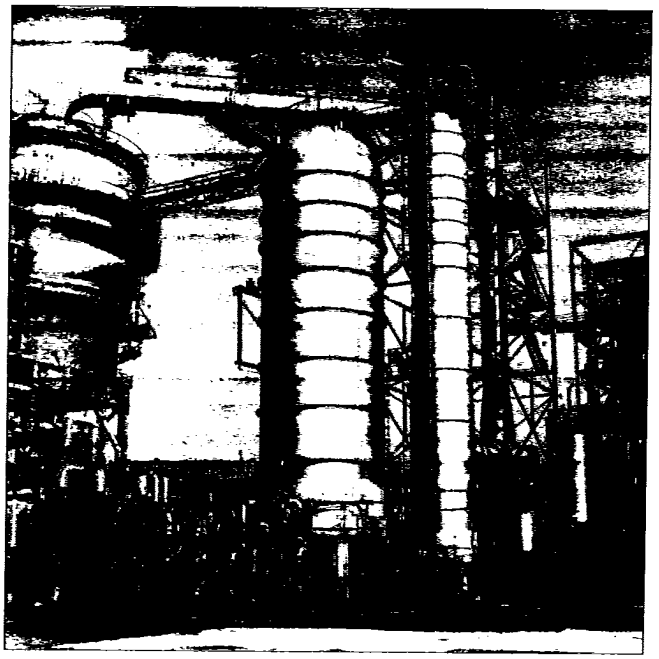


Figure 8—Catalyst Storage Hoppers

of the problems due to faulty material or fabrication are cracking, leakage, blockage, and excessive corrosion.

The material used to construct a vessel may contain laminations away from the edges of a plate, and these laminations may not be discovered before or during fabrication. After the vessel has been in operation, a lamination may open, especially if it reaches the surface of the plate, and show itself as a crack.

Castings may contain defects not visible on the surface. After corrosion has occurred, such defects may result in leaks. If the leaks are of sufficient magnitude or if many occur in close proximity, they may result in failure.

Lack of experience with a new process may result in the selection of an improper material that leads to excessive corrosion. Materials with physical properties unsuitable for repair work may also lead to excessive corrosion or to failure.

6.7 Faulty Fabrication

6.7.1 GENERAL

Faulty fabrication includes poor welding, improper heat treatment, fabrication with dimensions outside the tolerances allowed by the ASME Code, improper installation of internal equipment, and the assembly of flanged or threaded joints that results in an improper fit.

6.7.2 POOR WELDING

The use of improper welding techniques or careless welding may result in incomplete penetration, lack of fusion, cracking, undercutting, slag inclusion in the welds, and the development of porous welds. Any of these conditions may result in cracking or failure.

6.7.3 IMPROPER HEAT TREATMENT

Improper heat treatment may leave high residual stresses near welds and may affect the physical properties and corrosion resistance of the metal. It may also result in a hard material that may crack under shock.

High residual stresses may produce latent cracks, particularly around nozzle and reinforcement attachments, that may not be detected until the vessel has been operated for an extensive period of time. Under corrosive conditions, high residual stress may also lead to stress-corrosion cracking. A change in the properties of corrosion resistance may also result.

Materials heated above the proper heat-treating temperature or held too long at the proper temperature can be damaged so that their physical properties may not meet specification.

6.7.4 DIMENSIONAL INTOLERANCE

Poor fabrication techniques can cause tolerances outside of the range permitted by the ASME Code, which can lead to stress concentrations and subsequent failure.

6.7.5 IMPROPER INSTALLATION

Improper installation of internal equipment may result in inefficient operation, the blockage of passages, and, in the case of a pressure surge, the displacement of the internal equipment.

6.7.6 IMPROPER FIT

Improper fitting or tightening of flanged or threaded joints may result in leaks and, in the case of threaded joints, complete failure.

SECTION 7—FREQUENCY AND TIME OF INSPECTION

7.1 Factors Governing Frequency of Inspection

The frequency with which a pressure vessel should be inspected depends on several factors. The most important factor is the rate of deterioration and the remaining corrosion allowance (see API 510).

Corrosion rates vary markedly with the types of crude oil or stocks processed, the temperature of exposure, and the mate-

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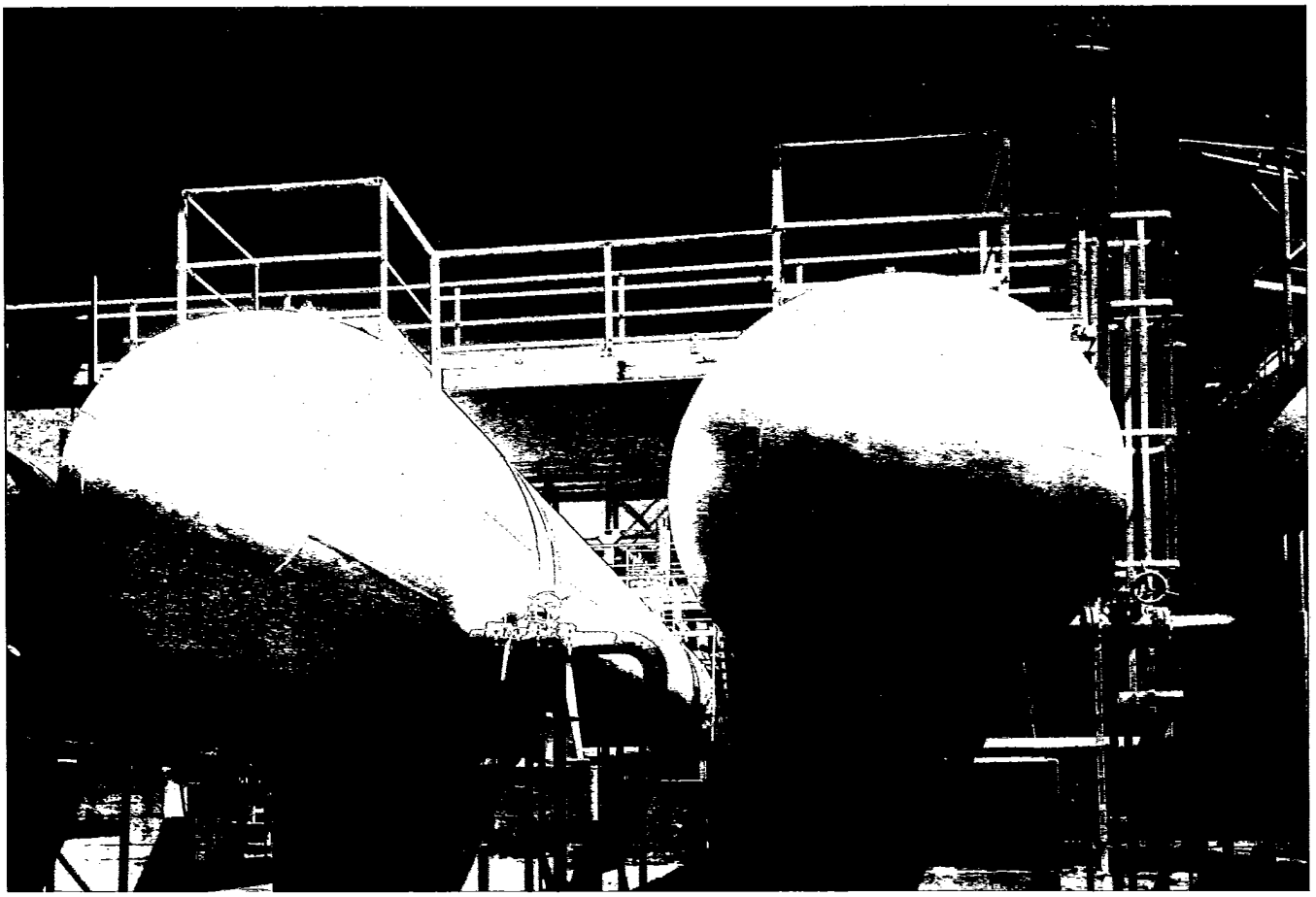


Figure 9—Horizontal Drums

rials of construction. Each must be individually appraised to establish the initial inspection period for new equipment.

A service history record should be established after the first inspection by on-stream methods or internal examination. On the basis of this history, an inspection interval can be set in accordance with API 510 or jurisdictional requirements. The period between inspections is normally planned so that at least half the corrosion allowance should remain at the next scheduled inspection. The predetermined frequency of inspection should allow for unanticipated changes in corrosion rates where appropriate.

Insurance and legal requirements may also affect the inspection of pressure vessels. Those responsible for inspection and maintenance of equipment should familiarize themselves with the applicable requirements. Recommendations in API 510, in NBBPVI NB-23, and in jurisdictional requirements should be followed where appropriate.

When changes in process operations are expected, they should be reviewed to determine whether they may affect the deterioration rate. When a change in the deterioration rate occurs or is anticipated, the recommended inspection interval should be changed accordingly.

Visual checks of the external parts of a vessel should be made periodically. Such inspections can be made without removing the vessel from service. These inspections may be made at comparatively short intervals, the interval depending on the service and previous condition of the particular equipment involved. Thorough external inspection of unfired pressure vessels should be conducted at the quarter life, in accordance with API 510.

On most units, operating needs such as the minimum permissible internal cleanliness, especially for towers and exchangers, and the maintenance of required heat-transfer rates, for exchangers or coolers, may determine the length of a unit run.

7.2 Opportunities for Inspection

The actual time for inspection will usually be determined through the collaboration of process, mechanical, and inspection groups or by the mandate of a jurisdiction.

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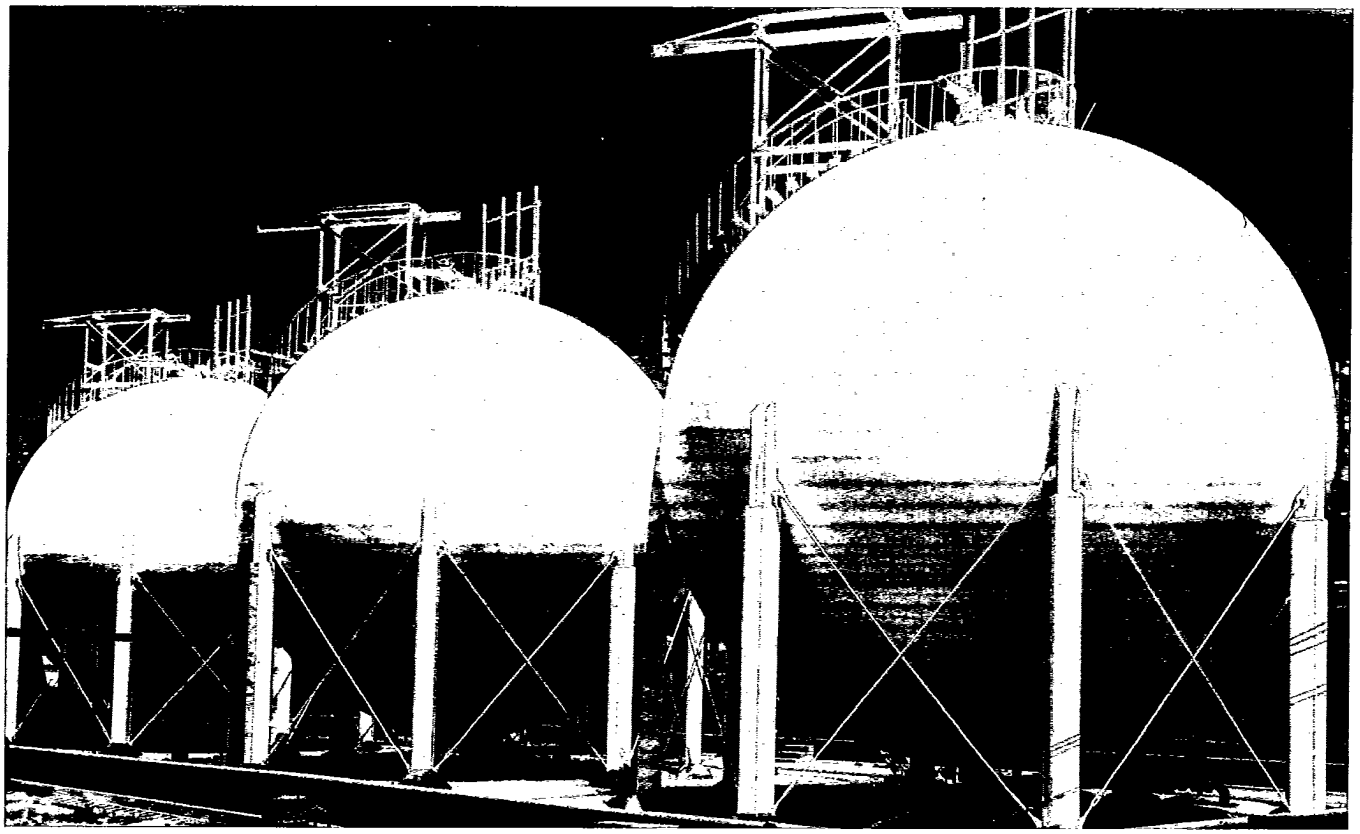


Figure 10—Spheres

Unscheduled shutdowns due to mechanical or process difficulties often present opportunities for checks of vessel areas where rapid corrosion, erosion, or other deterioration is known or suspected to occur. Partial shutdowns of units for process reasons also provide opportunities for making some internal inspections to determine conditions and verify on-stream inspection findings and for making needed repairs.

Inspections during the following opportunities are possible:

a. *When exchangers are normally taken out of service for cleaning.* When it is anticipated that an exchanger will have to be cleaned at more frequent intervals than permitted by the normal run of the process unit, it is customary to install spare exchangers so valved that any exchanger can be bypassed and opened for cleaning. Advantage should be taken of the opportunity to inspect exchangers so bypassed and removed from service during operation so that the work load when the process unit is shut down for a turnaround is reduced.

b. *When vessels or process towers are removed from service to clean trays and the like.*

c. *When a vessel or an exchanger is removed from service at other than the scheduled time.* When this occurs, the most critical parts of the vessel or exchanger could be inspected.

d. External inspections may be made *while a vessel or an exchanger is in service.* Inspection work performed while the equipment is in service will reduce the work load when it is out of service. These inspections should cover the condition of the foundation, supports, insulation, paint, ladders, platforms, and other structural elements. The existence and location of abnormally high metal temperatures or hot spots on internally insulated units can also be detected. On-stream inspection methods may be used to detect defects and to measure wall thicknesses. For example, thicknesses can be determined by using ultrasonic equipment or profile radiography where applicable.



Figure 11—Horton Spheroid (Noded)

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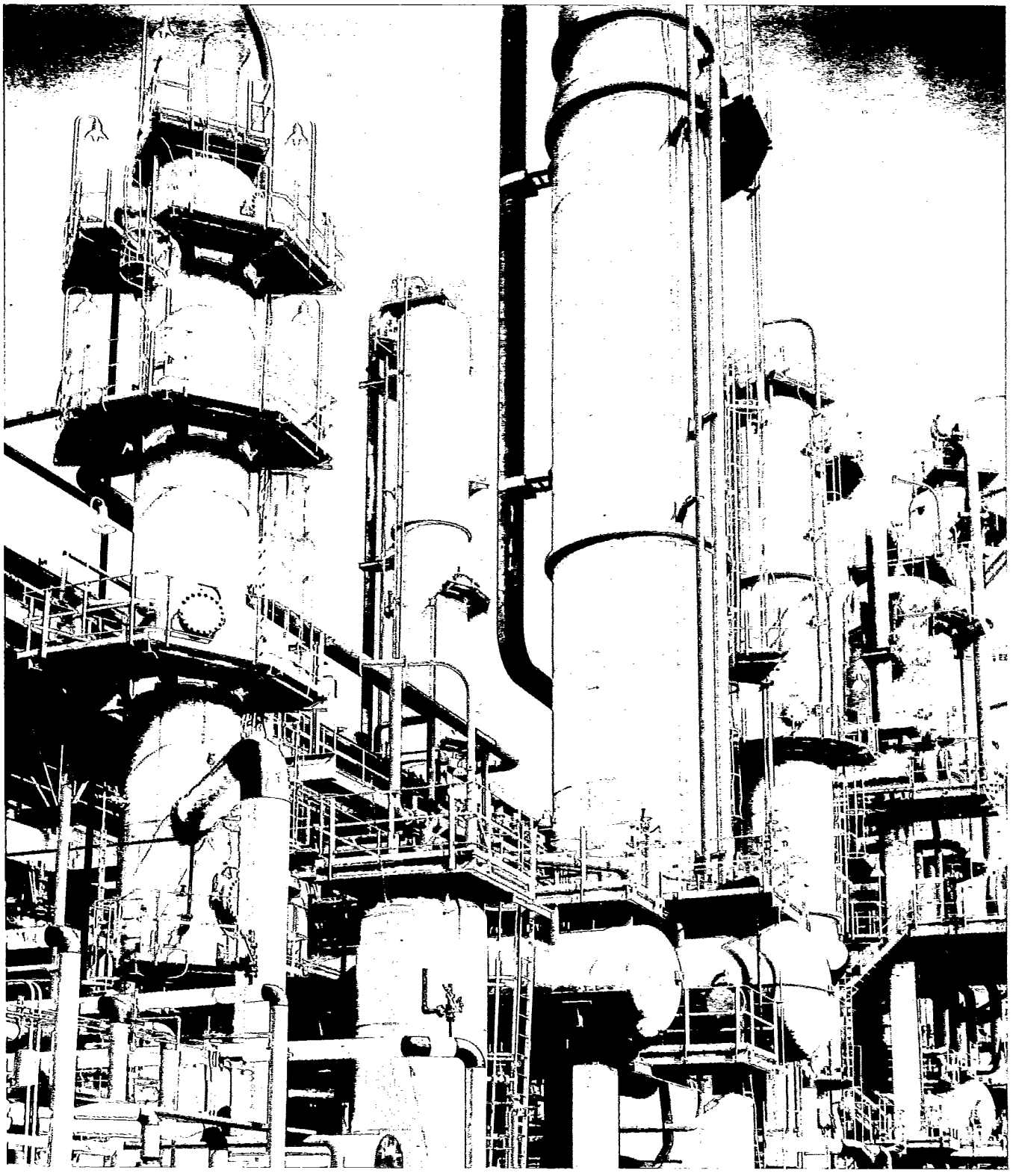


Figure 12—Process Towers and Drums

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Figure 13—Fluid Catalytic Regenerator

e. An occasional check of the operating record *while equipment is in service* is sometimes helpful in determining and locating the cause of functional deterioration. An increased drop in pressure may indicate blockage from excessive corrosion deposits. A reduced exchange of heat from exchangers or coolers may indicate heavy corrosion deposits on or in the bundle tubes. The inability to draw the product of fractionation or distillation from certain trays may indicate fouling or loss of tray parts in a process tower. Product deterioration may indicate loss of trays, tray parts, or other internal equipment in process vessels. The inspector should always keep in close touch with operations.

7.3 Inspection Schedule

Scheduling of shutdowns for maintenance or inspection is usually arranged through the collaboration of process, maintenance, and inspection groups or as mandated by a jurisdiction. Efforts should be made to schedule unit shutdowns evenly throughout the year to distribute the workload on the inspection and maintenance groups.

The safety and the reliability of operation are the most important considerations in scheduling the shutdown of units for inspection. Occasionally, seasonal demands for certain products may make some units available for inspection and maintenance work without serious interruption of supply. New vessels should be inspected at a reasonable time interval after being placed in service. This interval will depend on the service. The past records on vessels in similar units may be used as a guide.

SECTION 8—METHODS OF INSPECTION AND LIMITS

8.1 General

Before starting the inspection of a pressure vessel, especially one in severe service, the inspector should determine the pressure and temperature conditions under which the vessel has been operated since the last inspection. He should also ascertain the character of the vessel contents and the function the vessel serves in the process. He should confer with operations to determine pertinent data,

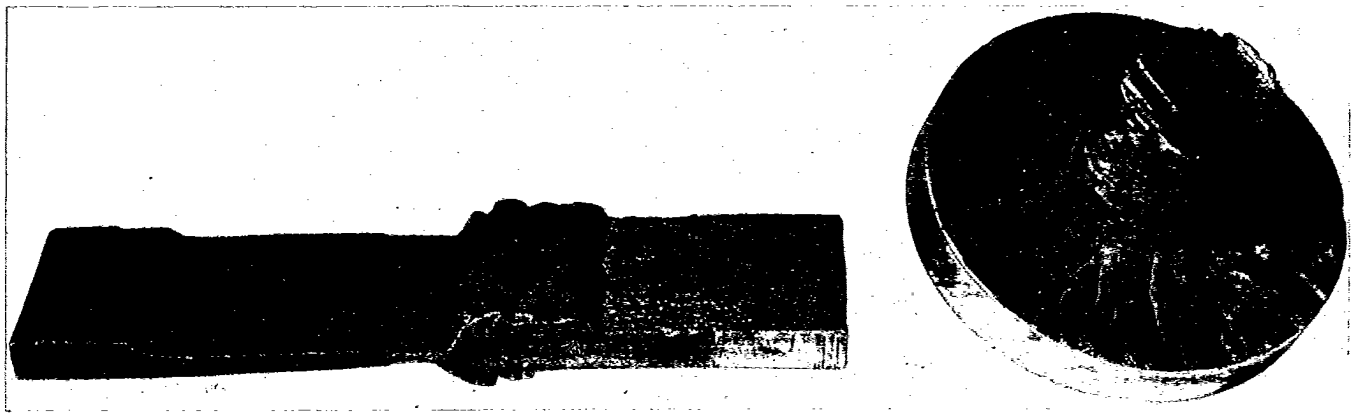


Figure 14—Caustic-Stress Corrosion

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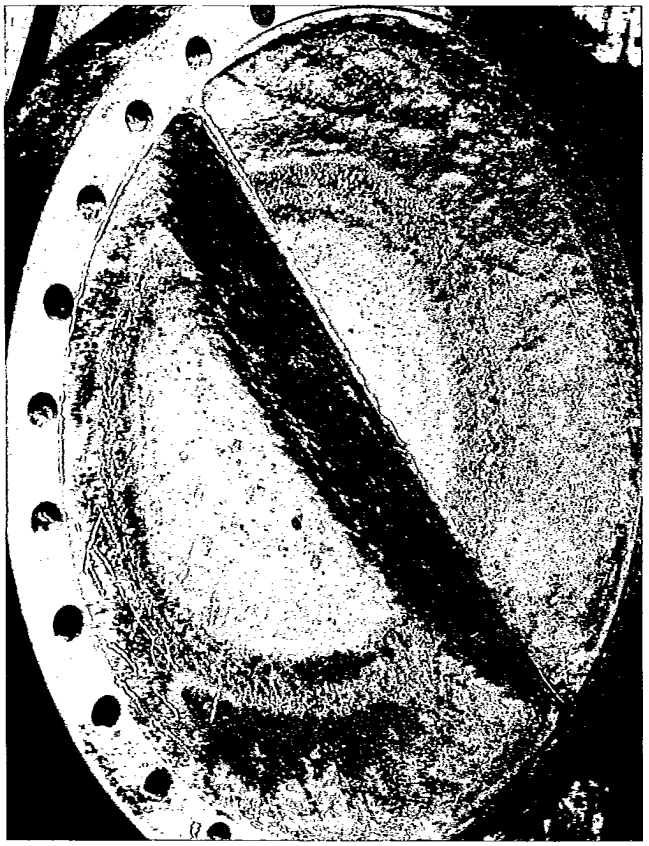


Figure 15—Severe Graphitic Corrosion of Floating-Head Cover

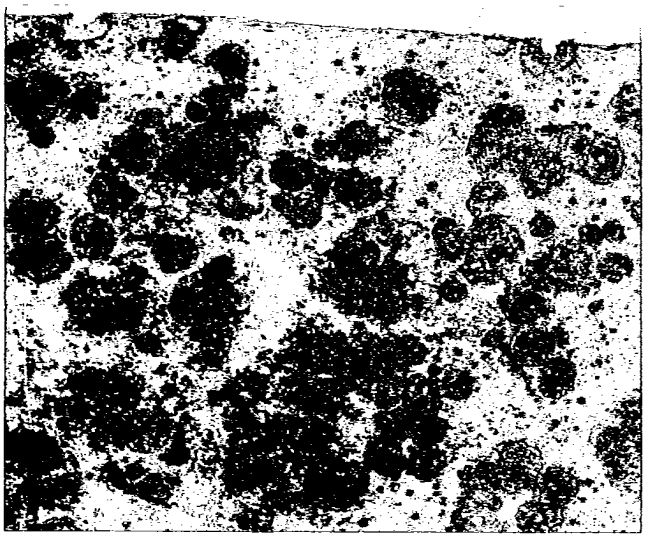


Figure 16—Plug-Type Dezincification

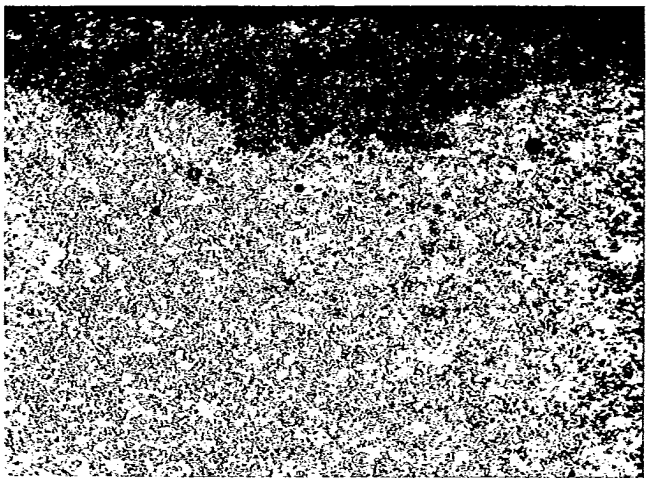


Figure 17—Layer-Type Dezincification

including whether there have been any abnormal conditions or disturbances such as excessive pressures or temperatures. This data may offer valuable clues to the types and locations of corrosion and other forms of deterioration—such as scaling, bulging, and warping—that may have occurred. The inspector should develop and exercise sound judgment on the extent and kinds of inspection required for each vessel.

Careful visual inspection of every vessel is of paramount importance to determine other forms of in-

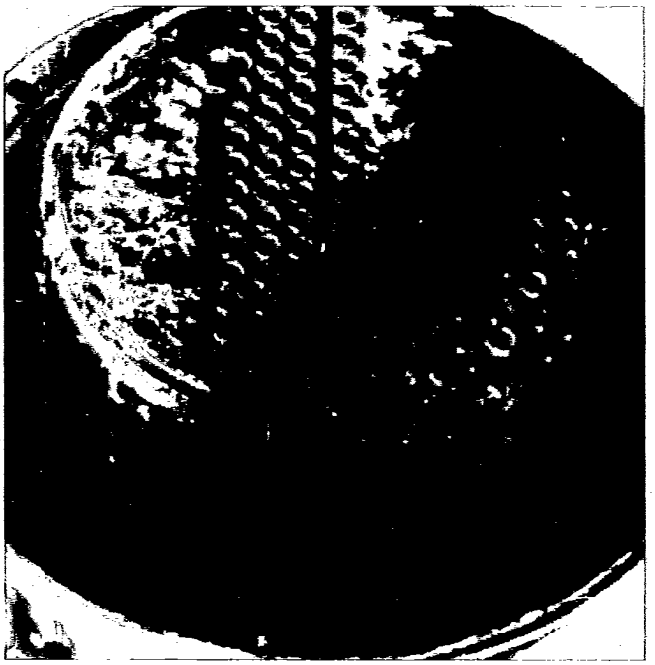


Figure 18—Tube Sheet Fouled With Marine Growth

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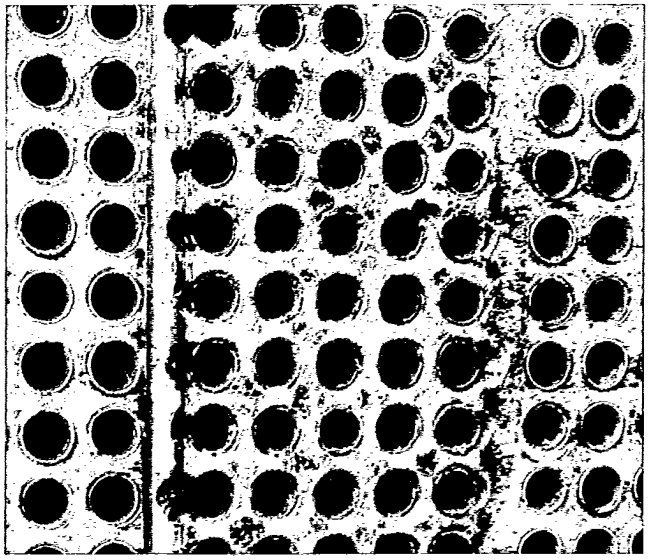


Figure 19—Tube Sheet Corroded Beneath Marine Growth

spection that may need to be made. Adequate surface preparation is essential to proper visual examination and to the satisfactory application of some auxiliary proce-



Figure 20—Internal Vessel Corrosion

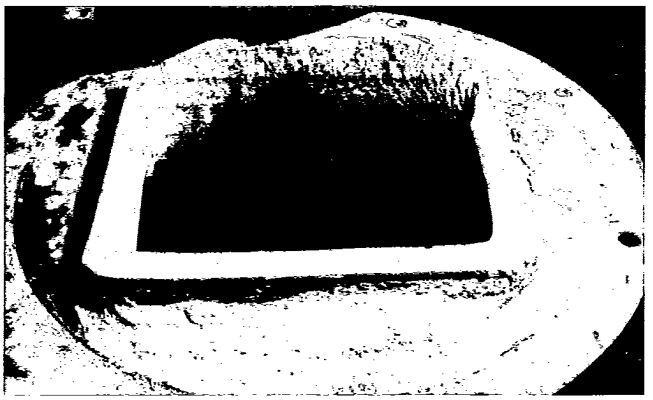


Figure 21—Erosion

dures, such as magnetic-particle (wet or dry) and dye-penetrant inspection methods. The extent to which special surface preparation may be required depends on the particular circumstances involved. Wire brushing, sandblasting, high-pressure water blasting, chipping, grinding, or a combination of these operations may be required in addition to routine cleaning.

If external or internal coverings—such as insulation, refractory linings, or corrosion-resistant linings—are in good condition and without evidence of an unsafe condition behind them, it is not necessary to remove them for inspection of the vessel. However, in the case of nonmetallic coverings, it may sometimes be advisable to remove small portions to investigate their condition and the condition of the metal behind them. When any metallic or nonmetallic covering is found to be defective, a sufficient amount of the covering in the vicinity of the defect should be removed to find out whether the base metal is deteriorating and to determine the extent of the deterioration.

Where operating deposits such as coke are normally permitted to remain on a vessel surface, it is important to determine the condition of the vessel surface behind the deposits. This may require thorough removal of the deposit in selected critical areas for spot check examination.

Where vessels are equipped with removable internals, the internals need not be completely removed provided reasonable assurance exists that deterioration is not oc-

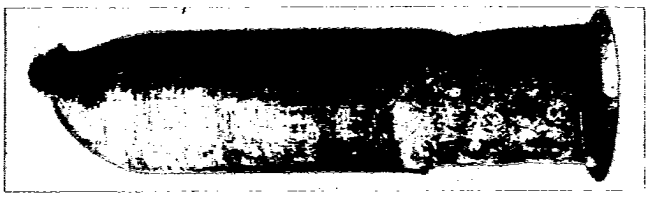


Figure 22—Condensate Grooving of Tube in Area Adjoining Tube Sheet

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Figure 23—Intergranular Corrosion

curring beyond that found in more readily accessible parts of the vessel.

The number of thickness determinations needed will depend largely on the corrosion pattern found in the vessel. In an extensively corroded area, an average of the least thicknesses found in the area may sometimes be considered the thickness of the corroded area. Scattered or isolated corrosion pits should be evaluated on the basis of their depth, number, and rate of progression. See API 510 for methods of determining the minimum acceptable thickness where localized corrosion has been detected.

8.2 Safety Precautions and Preparatory Work

8.2.1 SAFETY

Safety precautions must be taken before entering a vessel. Because of limited access and confined spaces, safety precautions are probably more important in vessel inspection work than in the inspection of any other type of equipment.

The vessel should be isolated from all sources of liquids, gases, or vapors, using blinds or blind flanges of suitable

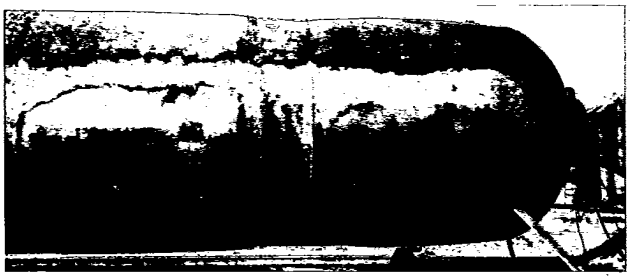


Figure 24—Thermal and Pressure Damage

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Figure 25—Fatigue Cracking at Nozzle

pressure and temperature rating. The vessel should be drained, purged, cleaned, and gas tested before it is entered. This preparation will minimize danger from toxic gases, oxygen deficiency, explosive mixtures, and irritating chemicals. Clothing that will protect the body and eyes from the hazards in the vessel to be entered should be worn. Details of the precautions to be followed are covered by API Publication 2217, Publication 2217A, and Publication 2007.

On occasion, it may be desirable to enter a vessel before it has been properly cleaned to search for internal causes of poor operation. In this case, the inspector should exercise the special precautions for such entry given in API Publication 2217A.

The use of nondestructive devices for inspection is subject to safety requirements customarily met in gaseous atmospheres, which are listed in API Publication 2007 and Publication 2214.

Before the inspection starts, all persons working around a vessel should be informed that people will be working inside the vessel. The posting of tags on the manways of tall towers is a worthwhile precaution. Usually a safety guard is stationed at the manway nearest the area under inspection. Workers inside a vessel should be informed when any work will be done on the exterior of the vessel to prevent their becoming alarmed by unexpected or unusual noise.

8.2.2 PREPARATORY WORK

The tools needed for vessel inspection, including those tools and equipment needed for personnel safety, should be checked for availability and proper working condition prior to the inspection. Any necessary safety signs should be installed prior to work in vessels.

Some of the tools that should be available for pressure vessel inspections follow:

a. Portable lights, including a flashlight.
b. A flashlight with a bulb on flexible cable.
c. A thin-bladed knife.
d. A broad chisel or scraper.
e. A pointed scraper.
f. Mirrors.
g. Inside calipers.
h. Outside calipers.
i. A pocket knife.
j. Steel tape [50 feet (15 meters)].
k. A flange square.
l. An inspector's hammer or ball peen hammer (4 ounce or 8 ounce).
m. Ultrasonic thickness-measuring equipment.
n. Tube gauges (inside diameter).
o. A steel rule.
p. A pit depth gauge.
q. Paint or crayons.
r. A notebook and pencils.
s. A straightedge.
t. A wire brush.
u. A plumb bob and line.
v. A magnet.
w. A magnifying glass.
x. A hook gauge.
y. Plastic bags for corrosion product samples.

The following tools should be available if required:

a. A surveyor's level.
b. A carpenter's or plumber's level.
c. Magnetic-particle inspection equipment.
d. A micrometer.
e. Radiographic equipment.
f. A Megger ground tester.
g. Sandblasting equipment.
h. High-pressure water-blasting equipment.
i. Portable hardness-testing equipment.
j. Eddy-current-testing equipment.
k. Sonic and radiation-measuring equipment.
l. Fiber optic flexible scopes.
m. A surveyor's transit.
n. Temperature-indicating crayons.
o. Thermocouples.
p. Metal sample-cutting equipment.
q. A material identification kit or machine.
r. A camera.
s. Ultrasonic flaw-detection equipment.
t. Dye-penetrant inspection equipment.
u. Test-hole-drilling equipment (drill, tap, and plugs).
v. Sand- or water-blasting equipment.
w. A borescope.
x. Plumb lines and levels.
y. A spotting scope or binoculars.

Other related equipment that might be provided for inspection includes planking, scaffolding, bosun's chairs, chain or rope ladders, safety devices for climbing flares or ladders without cages, stages for lifting by cranes, radios, and

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portable ladders. If external scaffolding is necessary, it may be possible to erect it before the inspection starts.

8.3 External Inspection

8.3.1 GENERAL

As indicated in Section 7, much of the external inspection can be made while the vessel is in operation. Any inspection made during vessel operation will reduce the period during which the vessel will be out of service.

8.3.2 LADDERS, STAIRWAYS, PLATFORMS, AND WALKWAYS

The external inspection of pressure vessels should start with ladders, stairways, platforms, or walkways connected to or bearing on the vessel.

A careful visual inspection should be made for corroded or broken parts, cracks, the tightness of bolts, the condition of paint or galvanized material, the wear of ladder rungs and stair treads, the security of handrails, and the condition of flooring on platforms and walkways. Visual inspection should be supplemented by hammering and scraping to remove oxide scales and other corrosion products; floor plates can be removed to check their supporting members. The tightness of bolts can be determined by tapping them with an inspector's hammer or a small ball peen hammer or by trying the nuts with a wrench. Wear on metal stair treads and flooring may not only weaken them but also make them slippery if worn smooth. Depressions in platforms should be closely checked, because water lying in depressions can accelerate corrosion. Crevices should be checked by picking at them with a pointed scraper. Loose or broken parts are easily found by tapping with a small ball peen hammer or an inspector's hammer. If desired, thickness measurements of the platforms and structural members can be made with transfer calipers.

Corrosion is most likely to occur where moisture can collect. On ladders and stairs, corrosion is likely to concentrate where rungs or treads fit into the runners or stringers. Crevice corrosion may exist around the heads of bolts and nuts, at bracket connections between stair treads and angle supports, and at connections between intermediate supports and the vessel wall. Welded bracket connections are especially susceptible to corrosion, as the welds are usually rough, and it is difficult to apply a good, void-free paint coating to them. Corrosion may exist beneath a paint film and will be indicated by rust stains showing through the paint or by a blistering or a general lifting of the paint film.

The condition of most parts can be determined by hammering. Where corrosion appears to be severe, the actual thickness should be determined by calipering or by other means.

8.3.3 FOUNDATIONS AND SUPPORTS

Foundations for vessels are almost always constructed of steel-reinforced concrete or structural steel fireproofed with concrete. They should be inspected for deterioration such as spalling, cracking, and settling.

The foundations for exchangers usually consist of steel cradles on concrete piers. Occasionally the supports are made entirely of steel. Figure 26 shows a typical exchanger foundation.

The crevice formed between an exchanger shell or a horizontal vessel and a cradle support should be carefully checked. Moisture lying in the crevice can cause rapid attack on carbon steel and on low-chrome-molybdenum steels. If the cradle is sealed with a mastic compound, this seal should be checked by judiciously picking at the mastic with a scraper to make sure that it is intact. Cradles are often seal-welded to vessel shells to prevent moisture from accumulating in the crevice and causing corrosion.

Spalling can be caused by excessive heat, mechanical shock, the corrosion of reinforcing steel, or the freezing of entrapped moisture. Inspection for this type of damage should consist of visual observation and scraping. Measurements of the depth of such damage can usually be made with a straightedge or a steel rule.

Cracks in concrete or fireproofing may be caused by excessive heat, poor design or material, mechanical shock, or unequal settlement. Inspection for cracks should be mostly visual. Some picking with a pointed scraper may be helpful.

Very small openings or cracks in concrete or fireproofing caused by a high temperature or by temperature changes can usually be identified by their hair-like appearance. Such cracks are not usually serious unless they expose the steel to corrosion.

When major cracks appear and propagate, and measurements indicate that no settlement has taken place, the cracks are probably the result of poor design or poor material. A complete check or engineering study may be required. If such investigations show that the design is correct, the cracks are most likely caused by the use of poor concrete material. Careful visual examination and minor chipping with a hammer will usually confirm or contradict the diagnosis, but removal of a core for testing may be required.

Some settling is expected in any foundation. When settlement is even and of a nominal amount, no problem should be experienced. However, if it is excessive or uneven, the settlement should be corrected before serious damage occurs. When foundation or support settlement has occurred, the condition of connection pipe lines should be checked.

Records of settlement should be maintained on vessels known to be settling. A rough check for uneven settling can be made with a plumb line and steel rule. When accurate measurements are desired, a surveyor's level may be used. When settlement is appreciable, it can be observed by noting the misalignment of the foundation with the surrounding paving or ground. The frequency with which settlement

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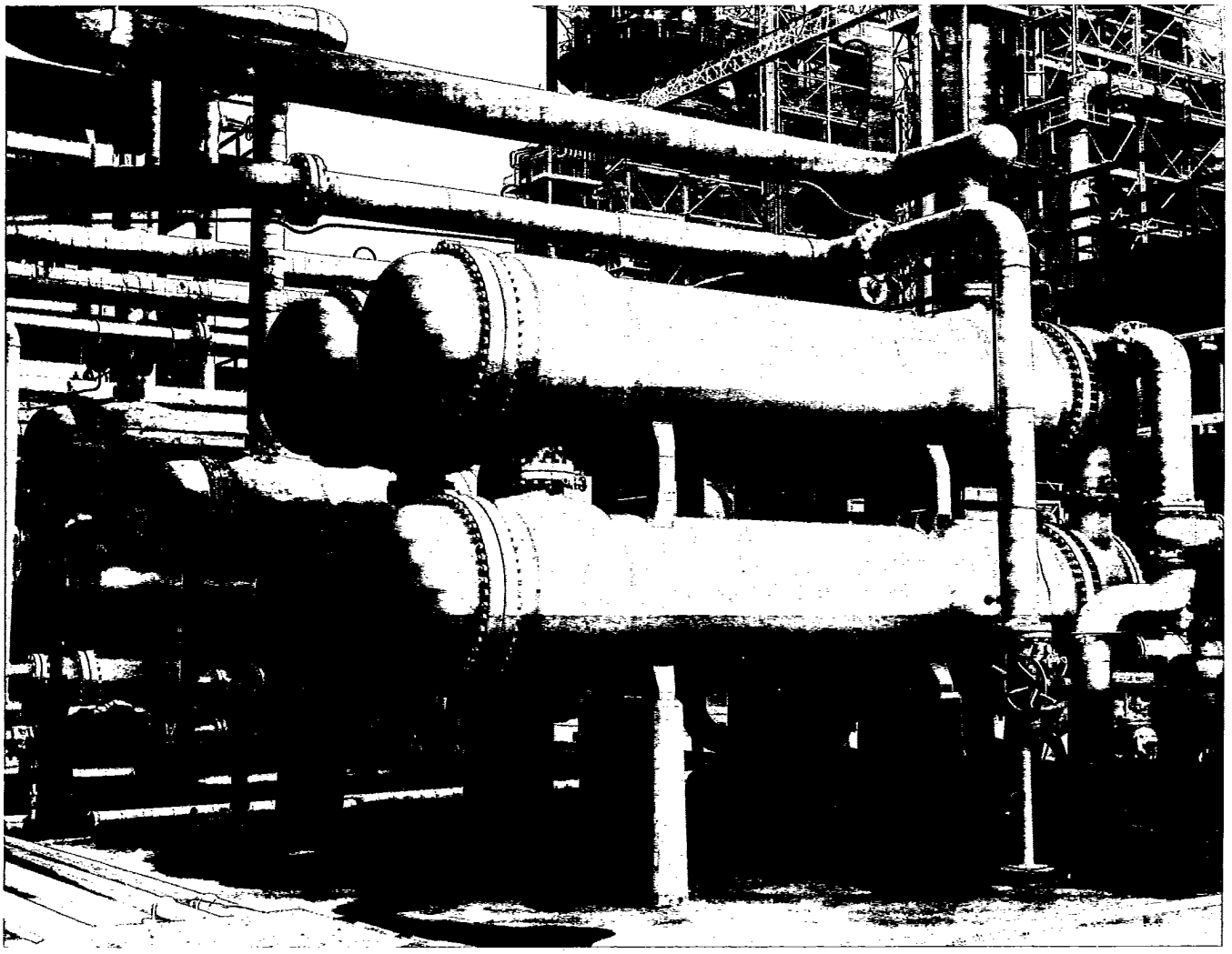


Figure 26—Exchanger Installation and Foundation

measurements should be taken depends on the rate and the seriousness of the settlement. Measurements should be taken until the settlement stops. Vessels supported on long concrete slabs or on two or more separate foundations are more likely to undergo uneven settlement.

8.3.4 ANCHOR BOLTS

Although the condition of anchor bolts cannot always be completely determined by visual inspection, the area of contact between the bolts and any concrete or steel should be scraped and closely examined for corrosion. Although this will not reveal the condition below the top surface of the baseplate or lugs, a sideways blow with a hammer may reveal complete or nearly complete deterioration of the anchor bolt below the baseplate (see Figure 27). Distortion of anchor bolts may indicate serious foundation settlement. The nuts on anchor bolts should be inspected to determine whether they are properly tightened. Ultrasonics may also be used to test bolts.

8.3.5 CONCRETE SUPPORTS

Inspection of concrete supports is similar to inspection of concrete foundations. The opening between concrete supports and a vessel shell or head should be sealed to prevent water from seeping between the supports and the vessel. A visual inspection with some picking and scraping should disclose the condition of the seal. A concentration cell could develop there and cause rapid corrosion.

8.3.6 STEEL SUPPORTS

Steel supports should be inspected for corrosion, distortion, and cracking.

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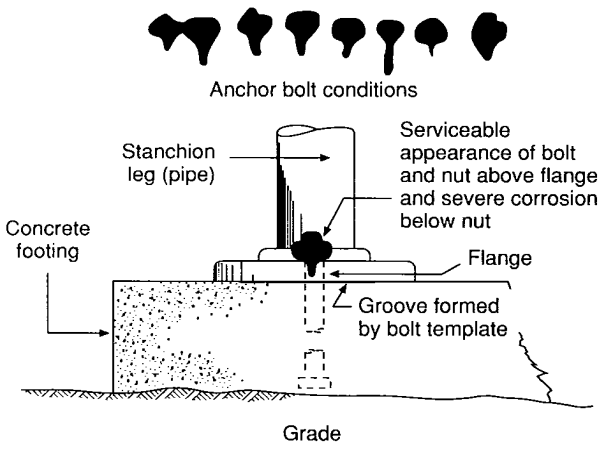


Figure 27—Severe Deterioration of Anchor Bolts

The remaining thickness of corroded supporting elements (skirts, columns, and bracing) is of primary importance. It can usually be determined by readings with transfer or indicating calipers in the most severely corroded areas. The readings should be compared with the original thickness (if known) or with the thickness of uncorroded sections to establish a corrosion rate. Visual examination of the support surfaces should be supplemented by wire brushing, picking, and tapping with a hammer. On large skirt supports, ultrasonic thickness-measuring devices can be used. Often corrosion of structural elements can be virtually eliminated simply by keeping the structural elements properly painted. Galvanizing is one of the best methods of protecting steel structures from corrosion.

Columns and load-carrying beams should be inspected for buckling or excessive deflection. They can be inspected visually with the aid of a straightedge or plumb line. Distortion of cylindrical skirts can be detected by diameter measurements at several points approximately 60 degrees (1.0 radian) apart.

The inside surface of a skirt sheet is often subject to attack by condensed moisture, especially when the temperature in the enclosed area is less than approximately 100°F (40°C) or when steam is put in the skirt to warm the bottom of the vessel. Visual inspection will usually disclose the condition of the metal surface. If a scale or rust layer has built up, it should be wire brushed or scraped off before the inspection is made.

Vessel support lugs should be inspected to see that they are sound. Scraping will usually reveal corrosion. Tapping with a hammer will disclose extreme thinning. Connecting fasteners should be checked for corrosion and general tightness. Any crevices found should be examined for crevice corrosion by picking. Cracks can occur in all types of supporting structures and lugs. However, they are most likely to appear in welded structures. The welds and the areas adjacent to the welds are the most common locations of cracks. If the vessel is in service, the inspection will probably be limited to visual methods of detecting cracks. Magnetic-particle (wet or dry), dye-penetrant, or ultrasonic shear-wave methods may be used to supplement visual examination. These methods will often require further surface preparation.

If supporting skirts are insulated, the insulation should be inspected. Visual inspection will usually disclose any deterioration of the insulation. If there is reason to suspect that water or moisture is seeping through to the steel, enough insulation should be removed to determine the extent of any corrosion.

The fireproofing on support beams and skirts should be inspected. It is usually made of bricks or concrete. Visual examination aided by scraping will disclose most defects. Very light taps with a hammer will disclose lack of bond between concrete fireproofing and the protected steel. If moisture can get behind the fireproofing, the steel may corrode and cause the fireproofing to bulge. The bulge in the fireproofing would indicate the corrosion. Rust stains on the surface of the fireproofing would indicate possible corrosion of the metal underneath.

8.3.7 GUY WIRES

Most vessels are self-supporting structures. Some towers or columns are guyed for support by steel cables. These cables radiate to the ground and end in the concrete deadman anchors beneath the ground surface.

The connections to the tower and to each ground anchor point should be inspected for tightness and correct tension. Visual examination should be sufficient. If there is a question regarding the correct tension in the cables, a structural engineer should be consulted.

The cable should be inspected for corrosion and broken strands. The threaded parts of any turnbuckles are subject to crevice corrosion. Picking with a pointed scraper will disclose this corrosion.

The wire rope clips on the guy wire cable—at the tower and at the ground anchor point—should be checked for correct installation. The clips should be attached to the cable with the base against the live or long end and the U bolt against the dead or short end of the wire rope. The clips should be spaced at least six rope diameters apart to insure maximum holding power. The number of clips necessary for each wire rope end depends on the diameter of the wire rope. This number can be found in wire rope catalogs and in engineering handbooks.

8.3.8 NOZZLES

If any settling of the vessel has occurred, nozzles and adjacent shell areas should be inspected for distortion and cracking. Excessive pipeline expansions, internal explosions, earthquakes, and fires may also damage piping connections.

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Flange faces may be checked with a flange square for distortion. If there is any evidence of distortion or cracks in the area around the nozzles, all seams and the shell in this area should be examined for cracks. The area should be abrasive-grit blasted or wire brushed. Magnetic-particle (wet or dry), dye-penetrant, ultrasonic shear-wave, or replication techniques may be used to supplement visual examination. (Catalytic reformer equipment operating at temperatures of more than 900°F (482°C) may experience creep embrittlement damage during operation. Replication is a useful technique in detecting this damage).

Nozzles should be internally inspected when possible for corrosion, cracking, and distortion. The inspection can be visual with a scraper and a flashlight.

Exposed gasket surfaces should be checked for scoring and corrosion. The surfaces should be cleaned thoroughly and carefully for a good visual inspection.

The grooves of ring joint flanges should be checked for cracks due to excessive bolt tightening. Also, stainless steel ring joint grooves should be checked for stress-corrosion cracking. Nondestructive testing (NDT) methods such as magnetic-particle (wet or dry), dye-penetrant, or ultrasonic shear-wave techniques may be used to supplement visual examination.

Lap joint flanges or slip flanges such as Van Stone flanges should be checked for corrosion between the flange and the pipe. The check can be made from inside the pipe by special probes and ultrasonic thickness-measuring devices. The flanges can also be moved for inspection after bolt removal.

Wall thicknesses of nozzles should be measured. Calipers, ultrasonic instruments, or radiography may be used. Internal diameter measurements may be taken with inside calipers to monitor corrosion; the pipe does not have to be removed for this measurement, but the vessel must be open and approved for internal inspection. These measurements should be recorded and compared with previous or original thickness readings. Any losses should be analyzed, and appropriate action—such as renewal if thickness is near or at the minimum, consideration of lining installation if feasible, monitoring at shorter intervals, and use of corrosion inhibitors—should be taken.

Leaks are likely to occur at piping attachments to the vessel wall. Leaks can best be located visually while the vessel is in service or under test conditions.

8.3.9 GROUNDING CONNECTIONS

Grounding connections should be visually examined to see that good electrical contact is maintained. These connections provide a path for the harmless discharge of lightning or static electricity into the ground. The system usually consists of a stranded copper conductor with one end bolted to the vessel and the other end brazed or bolted to an iron or copper rod placed deep in the ground. The cable connections should be checked for tightness and positive bonding to the vessels. The continuity of all ground wires should be checked. No break should exist in the grounding circuit. Test the system to see that the resistance to ground does not exceed the accepted values in the area. Recommended resistance is 5 ohms or less, and resistance is not to exceed 25 ohms; jurisdictional requirements in some areas differ from these values, will govern, and need to be checked.

8.3.10 AUXILIARY EQUIPMENT

Auxiliary equipment, such as gauge connections, sight glasses, and safety valves, may be visually inspected while the unit is in service. Undue vibration of these parts should be noted. The vibrations should be arrested by adding supports, or calculations should be performed by a qualified engineer to assure that the vibrations will not cause a fatigue failure.

8.3.11 PROTECTIVE COATINGS AND INSULATION

The condition of the protective coating or insulation on a vessel shell should be determined. Rust spots, blisters, and film lifting are the types of paint failures usually found. Rust spots and blisters are easily found by visual examination. Film lifting is not easily seen unless the film has bulged appreciably or has broken. It can be found by picking at the film with a scraper or knife in the suspected areas. Scraping paint away from blisters and rust spots often reveals pits in the vessel walls. The depth of such pitting can be measured with a pit gauge or a depth gauge. The most likely spots to search for paint failure are in crevices, in constantly moist areas, and at welded or riveted vessel seams. The bottom heads of vessels supported on skirts in humid locations are other likely points of paint failure.

Visual examination of insulation is normally sufficient to determine its condition. A few samples may be removed to better determine the condition of the insulation and the metal wall under it. The supporting clips, angles, bands, and wires should all be examined visually for corrosion and breakage. Occasionally, special blocks of insulation may be installed so that they are easily removable. These blocks are installed where it is desirable to make periodic inspections, usually at welded seams.

8.3.12 EXTERNAL METAL SURFACES

The external metal surfaces of a vessel may be inspected visually by picking, scraping, and limited hammering to locate corroded areas.

CAUTION: Extreme care should be used on operating equipment containing hot, harmful, or high-pressure material.

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If conditions warrant, scaffolding may be erected around a vessel to permit access to all surfaces.

The degree of surface preparation required for external inspection depends on the type and extent of deterioration expected. Under normal conditions, thorough cleaning to bare metal will be needed only at those points where ultrasonic thickness measurements are taken. When cracking or extensive pitting is suspected, thorough cleaning of a large area (possibly the entire vessel shell) may be required.

Hand tools such as a pointed scraper, an inspector's hammer, a wire brush, a scraper, and a file can be used to clean small spots. For larger areas, power wire brushing or abrasive blasting will usually be cheaper and more effective than the use of hand tools.

Any evidence of corrosion should be investigated and the depth and extent of the corrosion should be determined.

Thickness measurements of the vessel walls, heads, and nozzles are usually required at each complete vessel inspection. Whether these measurements are taken from the outside of a vessel or the inside will depend on the location and accessibility of the corroded areas.

Under normal conditions, at least one measurement in each shell ring and one measurement on each head should be taken. However, if much corrosion is evident, several readings should be taken in the most corroded areas. Also, if no history of a particular vessel exists, getting readings in each quadrant of each shell ring and head should be considered. Ultrasonic instruments may be used for these measurements.

8.3.13 EXTERNAL EVIDENCE OF CORROSION

Certain types of corrosion may be found on external surfaces of a vessel. Among these are atmospheric corrosion, caustic embrittlement, hydrogen blistering, and soil corrosion. These types of corrosion are covered in detail by Recommended Practice 571.

The extent of atmospheric corrosion on the outside of a vessel will vary with local climatic and service conditions. In humid areas and in areas where corrosive chemical vapors are present in the air, corrosion of external shell surfaces may be a problem. Vessels operating in a temperature range that will permit moisture to condense are most susceptible. Corrosion of this type is usually found by visual inspection.

If a caustic material is stored or used in a vessel, the vessel should be checked for caustic embrittlement. This type of attack is most likely to occur at connections for internal heating units and in areas of residual or other high stress. The more susceptible areas are around nozzles and in or next to welded seams. Frequently, visual inspection will disclose this type of attack. The caustic material seeping through the cracks will often deposit white salts that are readily visible. Magnetic-particle (wet or dry), dye-penetrant, and ultrasonic shear-wave examinations may also be used to check for caustic embrittlement.

Those areas below the liquid level in vessels containing acidic corrodents are more likely than other areas to be subject to hydrogen blistering. Hydrogen blistering is more often found inside a vessel than outside but may be found either inside or outside depending on the location of the void that causes the blistering. Blisters are found most easily by visual examination. A flashlight beam directed parallel to the metal surface will sometimes reveal blisters. When many small blisters occur, they can often be found by running the fingers over the metal surface.

Attention should be given to metal surfaces in contact with concrete saddles. In humid atmospheres, severe attack at the points of support may require weld repairs and subsequent application of protective coatings.

Vessels that are partially or completely underground are subject to soil corrosion wherever they are in contact with the ground. This corrosion will be especially intense in areas where cinder fills were used or where acid splash-over has occurred. Inspection of the vessel surface will require thorough cleaning. Abrasive blasting will usually provide the best surface preparation. Visual examination, supplemented by picking and tapping, will disclose most faults. The location of any deep pitting should be recorded. Good judgement should be used in determining how much of the surface should be uncovered to permit this inspection. The most severe corrosion will usually be found at ground level or several inches below.

The external surfaces of the vessels should be examined not only for corrosion but also for leaks, cracks, buckles, bulges, defects in the metal plates, and deformation and corrosion of any external stiffeners. If the vessel is insulated, small sections of insulation should be removed, particularly where moisture might accumulate, to gain a general idea of whether external corrosion is occurring.

Unless readily visible, leaks are best found by pressure or vacuum testing the vessel. If there are visual or other indications that a leak comes through a crack, more thorough methods of examination should be employed.

In welded vessels, cracks are most commonly found at nozzle connections, in welded seams, and at bracket and support welds. In riveted vessels, the most common location is at metal ligaments between the rivets. Usually, close visual inspection with some picking or scrapping will disclose most cracks. When cracking is suspected in an area, the entire area should be cleaned by an appropriate method—such as wire brushing, high-pressure water blasting, or abrasive-grit blasting—to facilitate inspection. If visual inspection is not sufficient (often the case in the detection of amine and deaerator cracking), wet or dry magnetic-particle, shear-wave ultrasonic, dye-penetrant, or acoustic emission analysis may be used to locate and provide additional information on the structural significance of cracks or other discontinuities. The wet fluorescent magnetic-particle analysis is more sensitive than the dry fluorescent magnetic-particle analysis.

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Buckles and bulges will normally be quite evident. Small distortions can be found and measured by placing a straightedge against the shell of the vessel. Determining the cause of distortion is very important. Causes of distortion such as internal vapor explosions or excessive internal corrosion will be disclosed by the internal inspection. Settlement, earthquakes, extensive distortion in connected piping, and other sources can often be determined by external inspection. The extent of bulging or buckling can be determined by measuring the changes in circumferences or by making profiles of the vessel wall. Profiles are made by taking measurements from a line parallel to the vessel wall (see Figure 28). A surveyor's transit or an 180-degree optical plummet may also be used.

Hot spots that have developed on the shell or heads of vessels that are internally insulated should be inspected at frequent intervals while the vessel is in service. Evidence of bulging should be noted and recorded. A check of the skin temperature of the metal in the hot spot area can be made by using a portable thermocouple, infrared equipment, or temperature-indicating crayons or special paints. A complete dimensional check in the hot spot area should be made when the vessel is shut down. Using replication techniques or taking a material sample (a boat or other sample) should be considered if the temperatures were for a long period of time above 800°F (427°C) [or 500°F (260°C) if hydrogen attack is possible]; an experienced metallurgist or engineer should be consulted.

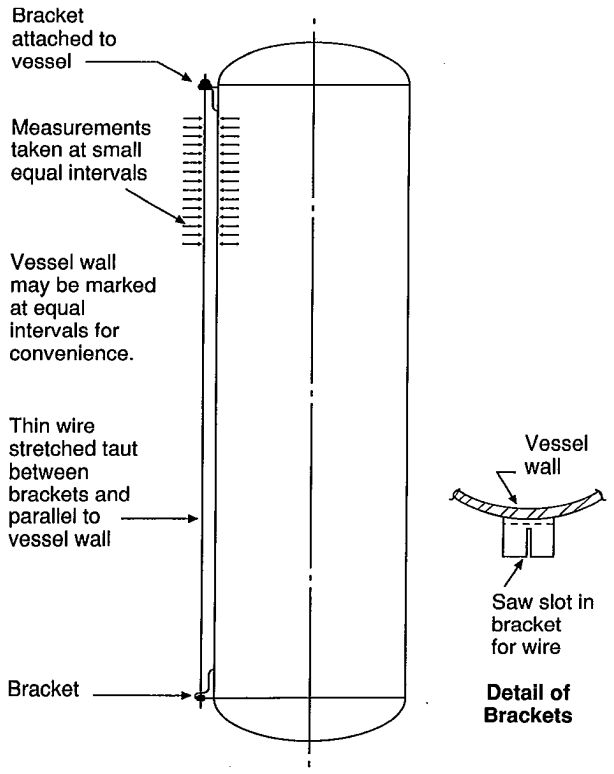


Figure 28—Method of Obtaining Vessel Profile Measurements

The external surfaces should be checked for laminations and mechanical damage. Laminations that come to the surface can be detected by visual inspection. Mechanical damage such as gouges and dents should be inspected. The depth and extent of any surface gouge should be measured when the gouge seems large enough to affect the strength of the vessel. All gouges should be reported.

Usually a certain amount of external auxiliary equipment is attached to a pressure vessel. This equipment includes items such as the following:

a. Drain lines and other connected piping (see API Recommended Practice 574).
b. Gauges for liquid level, pressure, and temperature and other instruments.
c. Safety and relief valves (see API Recommended Practice 576).
d. External water sprays and other fire-fighting equipment.

8.4 Internal Inspection

8.4.1 GENERAL

All necessary inspection equipment, including tools, ladders, and lights, should be assembled in advance to minimize downtime for the vessel.

8.4.2 SURFACE PREPARATION

The degree of surface preparation needed for internal inspection will vary with several factors. Foremost among these factors are the following:

a. The type of deterioration expected.
b. The location of any deterioration.

Usually the cleanliness required by the vessel operators will be sufficient for inspection purposes. This would entail the usual cleaning methods of washing with hot water, steaming, using solvents, and ordinary scraping. Where better cleaning is needed, the inspector's hand tools will sometimes be adequate.

The cleaning methods mentioned should be supplemented by power wire brushing, abrasive-grit blasting, grinding, high-pressure water blasting [8000–12,000 psi (55,158–82,737 kilopascals)], or power chipping when warranted by circumstances. These extra cleaning methods are necessary when stress-corrosion cracking, wet sulfide cracking, hydrogen attack, or other metallurgical forms of degradation are suspected. Extensive cracking, deep pitting, and extensive weld deterioration require thorough cleaning over wide areas. If the entire inside of the vessel is not accessible from one opening, the procedures discussed in 8.4.3 will have to be followed at each access opening.

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8.4.3 PRELIMINARY VISUAL INSPECTION

If an inspection is not the initial one, the first step in an internal inspection is to review the previous records of the vessel to be inspected.

When possible, a preliminary general visual inspection is the next step. The type of corrosion (pitted or uniform), its location, and any other obvious data should be established. In refinery process vessels, certain areas corrode much more rapidly than others. This uneveness of corrosion is covered in detail by API Recommended Practice 571. Data collected for vessels in similar service will aid in locating and analyzing corrosion in the vessel being inspected.

The bottom head and shell of fractionators processing high-sulfur crude oils are susceptible to sulfide corrosion. This corrosion will usually be most intense around the inlet lines.

The upper shells and the top heads of the fractionation and distillation towers are sometimes subject to chloride attack. The liquid level lines at trays in towers and in the bottom of overhead accumulators are points of concentrated attack. Corrosion in the form of grooving will often be found at these locations.

Fractionation and distillation towers, knockout drums, reflux accumulators, exchanger shells, and other related vessels that are vulnerable to wet hydrogen sulfide (H_2S) or cyanides are subject to cracks in their welds and in the heat-affected zones of the welds.

In vessels where sludges may settle out, concentration cell corrosion sometimes occurs. The areas contacted by the sludge are most susceptible to corrosion. If the sludge contains acidic components, this corrosion may be rapid.

If steam is injected into a vessel, corrosion and erosion may occur at places directly opposite the steam inlet. Bottom heads and pockets that can collect condensate are also likely to be corroded.

Often a reboiler will be used at the bottom of a tower to maintain a desired temperature. The point where the hot process stream returns to the tower may be noticeably corroded. This is especially true if the process stream contains components that may decompose with heat and form acid compounds, as in alkylation units and soap or detergent plants.

Because of metallurgical changes caused by the heat of welding at welded seams and adjacent areas, corrosion often accelerates there. Most of the cracks that occur in pressure vessels will be found in these areas. Areas opposite inlet streams may be subject to impingement attack or erosion.

Vessels in water service, such as exchangers or coolers, are subjected to maximum corrosion where the water temperatures are highest. Thus, when water is in the tubes of an exchanger, the outlet side of the channel will be most corroded. Figure 29 shows pitting in a channel.

In any type of vessel, corrosion may occur where dissimilar metals are in close contact. The less noble of the two metals will corrode. A carbon steel exchanger channel's gasket surfaces near brass tube sheets will often corrode at a higher rate than it would elsewhere.

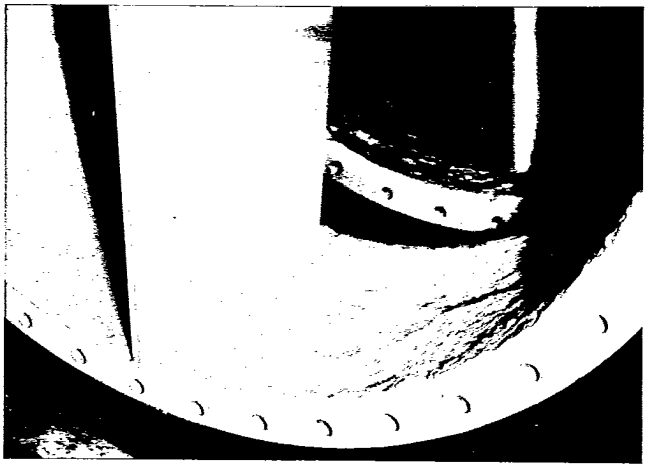


Figure 29—Pitting in Channel

Cracks in vessels are most likely to occur where there are sharp changes in shape or size or near welded seams, especially if a high stress is applied. Nozzles, exchanger channel and shell-cover flanges, baffles in exchanger channels, floating tube-sheet covers, and the like should be checked for cracks.

When materials flow at high velocities in exchanger units, an accelerated attack can be expected if changes are made in the direction of flow. Tube inlets in tubular units, return bends in double-pipe units, and condenser box or air cooler coils are likely to be attacked.

Shells of vessels adjacent to inlet impingement plates are susceptible to erosion. This is especially true when velocities are high.

The preliminary inspection of the vessel interior may indicate that additional cleaning is needed. If large areas are deeply corroded, abrasive blasting may be necessary. Normally, it is not necessary to remove light coatings of rust on more than a spot basis with a wire brush.

The preliminary inspection may reveal unsafe conditions, such as those due to loose internals that may fall or due to badly corroded or broken internal ladders or platforms. These parts must be repaired or removed immediately before a more detailed inspection may proceed.

8.4.4 DETAILED INSPECTION

A detailed inspection should start at one end of the vessel and work toward the other end. A systematic procedure should be followed to avoid overlooking obscure but important items.

All parts of the vessel should be inspected for corrosion, erosion, hydrogen blistering, deformation, cracking, and laminations. A careful record should be made of the types and locations of all deterioration found.

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Thickness measurements should be taken at those locations that show the most deterioration. When deterioration appears to be widespread, enough readings should be taken to assure an accurate determination of the remaining thickness. When deterioration is slight, one thickness measurement on each head and each shell sheet may be sufficient on small vessels, but more measurements should be taken on large vessels. Ultrasonic instruments can be used to obtain the necessary measurements. Other special methods of measuring wall thicknesses are discussed in 8.4.7.

Pitting corrosion can usually be found by scratching suspected areas with a pointed scraper. When extensive and deep pitting or grooving is found, and depth measurements are wanted, the areas may have to be abrasive blasted. The depths of pits or grooves can be measured with a depth gauge, a pit gauge, or (in the case of large pits or wide grooves) a straightedge and a steel rule. A depth can be estimated by extending the lead of a mechanical pencil as a depth gauge. Depressions or pockets that can hold sludge or water should be scraped clean and carefully examined for evidence of corrosion.

A hammer can be used to inspect for thin areas of vessel shells, nozzles, and parts. Naturally, experience is needed before the hammer can be used effectively. When striking the shell, nozzle, or part, an experienced inspector can often find thin spots by listening to the resulting sound and by noting the feel of the hammer as it strikes.

When cracks are suspected or found, their extent can be checked with dye-penetrant, magnetic-particle (wet or dry), or ultrasonic shear-wave inspection methods. To use any of these methods effectively, the suspected areas must be prepared by abrasive blasting, grinding, or other methods acceptable to the inspector. Figure 30 shows a crack in a shell weld.

Welded seams in vessel shells should be closely checked when the service is amine, wet hydrogen sulfide (H_2S), caustic, ammonia, cyclic, high temperature, or another service that may promote cracks. In addition, welds in vessels constructed of high-strength steels (above 70,000 psi tensile) or coarse grain steels should be checked. Welds in vessels constructed of low-chrome materials and in high-temperature service should receive careful inspection. In all cases, cracks may occur in or adjacent to the welded seams. The wet fluorescent magnetic-particle technique should be considered the best means of locating surface indications.

Nozzles connected to the vessel should be visually examined for internal corrosion. The wall thickness of nozzles can best be obtained with ultrasonic instruments. In some cases, a record of inside diameter measurements of nozzles may be desirable. These measurements can be made with a pair of internal, spring-type transfer calipers or with direct-reading, scissor-type, inside diameter calipers. When the piping is disconnected, actual nozzle wall thicknesses can be obtained by calipering around the flange. In this way, any eccentric corrosion of the nozzle will be revealed.

In most instances, inspection of internal equipment should be made when adjacent shell areas are inspected. This may be very difficult in some large vessels.

The supports for trays, baffles, screens, grids, piping, internal stiffeners, and other internal equipment should be inspected carefully. Most of this inspection will be visual. Light tapping with a hammer can be used as a check for soundness. If there appears to be any metal loss, the thickness of the support should be measured and checked against the original thickness. Transfer or direct-reading calipers, micrometers, or ultrasonic thickness instruments can be used for these measurements.

The general condition of trays and related equipment should be noted. Shell and tray surfaces in contact with tray packing should be examined for possible loss of metal by corrosion. The condition of trays and related equipment will not affect the strength of the vessel but will affect the efficiency and continuity of operation. Normally only visual inspection will be required for such equipment. If measurements are required, they can be obtained with calipers or ultrasonic instruments.

The performance of some trays depends on the amount of leakage. If tray leakage is appreciable, then efficiency is lost, and the withdrawal of side streams from the tower or vessel may be almost impossible. Therefore, tray leakage



Figure 30—Crack in Shell Weld

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should be minimized. The process design will usually specify the amount of leakage that can be tolerated. Tests for leakage may be made by filling the tray with water to the height of the overflow weir and observing the time it takes for all of the water to leak through the gasket surfaces of the tray. Excessive leaks can be located by observing the underside of the tray during the test. If difficulty is encountered in determining the location of leaks, plug the weep or drain holes in the low sections of the tray prior to the test. Because of their design, ballast and valve-type trays cannot be checked for leakage.

All internal piping should be thoroughly inspected visually, especially at threaded connections. Hammer testing of the pipe by an experienced inspector is a quick way to determine its condition. The sound, the feel, and any indentation will indicate any thinness or cracking in the pipe. If excessive metal loss is indicated, the remaining wall thickness may be measured.

The internals of vessels such as catalytic reactors are very complicated. Figure 31 is an illustration of this internal equipment. Inspection of this equipment may be mostly visual, although some scraping, picking, and tapping may be necessary. Thickness measurements and corrosion rate calculations may be required in some areas, although operating efficiency rather than strength is the most important consideration.

Erosion usually differs in appearance from corrosion. Figure 21 shows erosion and Figure 22 shows erosion and corrosion. Erosion is characterized by a smooth, bright appearance; marked absence of the erosion product; and metal loss, usually confined to a clearly marked local area. On the other hand, corroded areas (see Figure 20) are not often smooth or bright. See 6.2 for more detailed information on corrosion and 6.3 for more detailed information on erosion.

The shells of exchangers next to bundle baffles and inlet impingement plates should be checked for erosion. Turbulence near the impingement plate and increased velocity around exchanger bundle baffles sometimes cause erosion of the adjacent shell areas. Erosion or corrosion at the baffles of exchangers will often show up as a series of regularly spaced rings when a flashlight beam is directed parallel to the shell surface. Sometimes, a lack of scale will indicate this type of erosion.

Erosion occurs not only in exchangers but in any vessel that has wear plates, baffles, or impingement plates. In catalytic reactors and regenerators, the catalyst and air distribution facilities are especially susceptible to erosion and should be examined closely for this type of attack.

Areas directly above or below the liquid level in vessels containing acidic corrodents are subject to hydrogen blistering. Blisters are most easily found by visual examination. A flashlight beam directed across the metal surface will sometimes reveal blisters; the shadows created by the blisters can be observed. When many small blisters occur, they can often be found by running fingers over the metal surface. The metal thickness of large blisters should be measured so the remaining effective wall thickness can be determined. Usually this can be done with an ultrasonic thickness instrument or by drilling a hole at the highest point of the blister and measuring the thickness with a hook scale. If an ultrasonic thickness instrument is used, the blister size must allow a transducer to be placed on it for an ultrasonic test (UT) reading. Also, if the blister is near a weld, a UT reading may be difficult because of the weld roughness. Figures 32 and 33 illustrate hydrogen blistering.

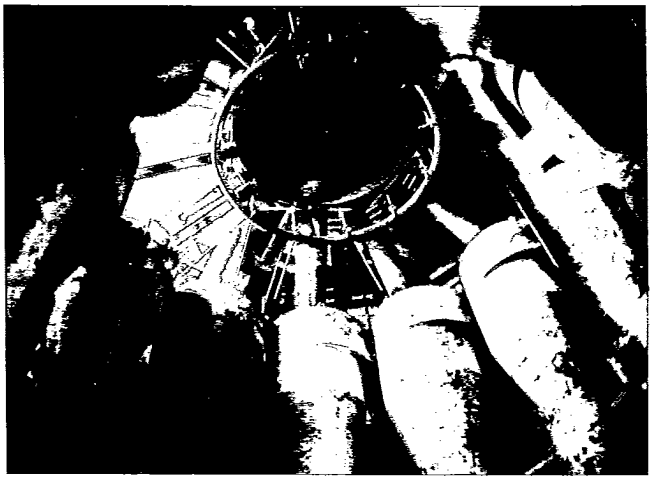


Figure 31—Catalytic-Reactor Internals—Cyclones

Both the shell and the heads of a vessel should be inspected for deformation. Normally the shell is more likely to suffer deformation than the heads. However, some older vessels have heads formed with a small knuckle radius, which may be seriously deformed. Unless dimensions of head parts, such as the crown radius or the knuckle radius, are al-

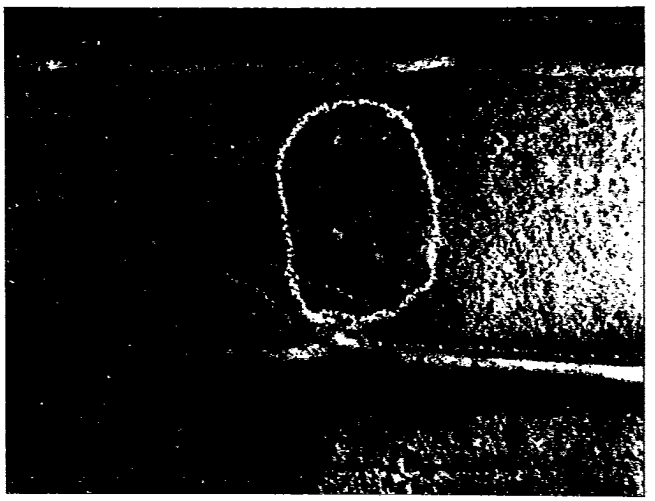


Figure 32—Hydrogen Blistering

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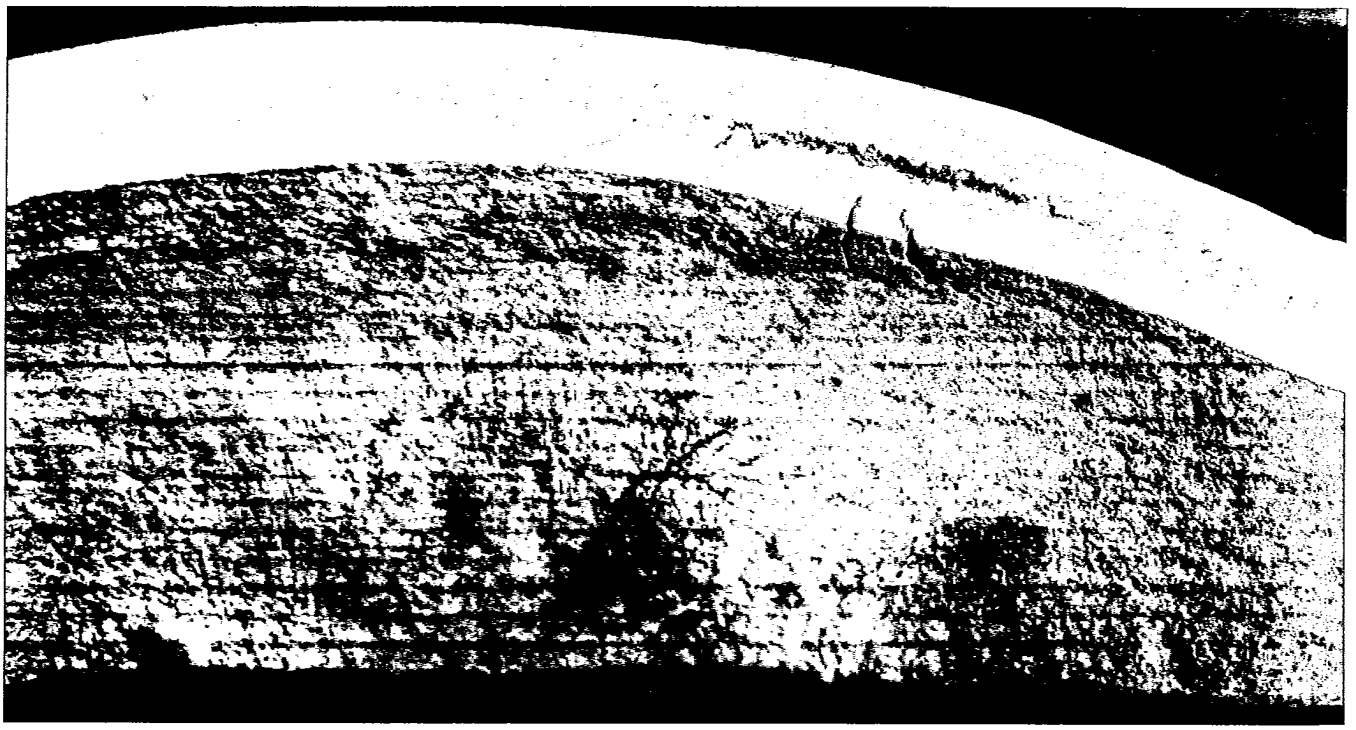


Figure 33—Cross Section Through Hydrogen Blisters Showing Various Types of Crack Propagation

ready on record, these dimensions should be taken and recorded at the time of the first inspection. If deformation is suspected or indicated later, these measurements should be repeated and compared with the original values.

Excessive deformation of the shell by either bulging or collapsing can usually be detected visually from the outside of the vessel, unless it is externally insulated. Out-of-roundness or bulging may be evaluated by measuring the inside diameter of the vessel at the cross section of maximum deformation and comparing it with the inside diameter at the cross section of minimum deformation.

Exchanger shells should be checked closely for any deformation, particularly after repairs or alterations. Out-of-roundness caused by welding can make installation of tube bundles extremely difficult, and extractions after difficult installations can be nearly impossible.

If the out-of-roundness occurs at intervals throughout the length of the vessel, measurements should be taken at each interval to compare with the original shell dimensions or measurements. In this case, the center wire method or the plumb line (or optical plummet) method can be used to measure the deformation. In the center wire method, a steel wire is positioned on the centerline of the vessel and stretched taut. If no manways or nozzles exist in the centers of the heads, a plumb line or an optical plummet may be used. When the deformation is restricted to one side of the vessel, it may be more convenient to measure offsets from a wire stretched parallel and adjacent to the wall rather than along the vessel axis (as in the method shown in Figure 28, but with the brackets and wire inside the shell instead of outside). In horizontal vessels, some special method may be required to hold the wire in position. The wire furnishes a reference line from which to measure the deformation. Sufficient measurements can be taken at intervals along the wire to permit drawing a profile view of the vessel wall. Local deformation can sometimes be measured by placing a straightedge parallel to the vessel axis against the vessel wall and using a steel rule to measure the extent of bulging. The best method of locating suspected deformations is to direct a flashlight beam parallel to the surface. Shadows will appear in depressions and on the unlighted side of internal bulges.

A careful inspection should be made for evidence of cracking. A strong light and a magnifying glass will be helpful when doing this work visually. If cracking is suspected or any evidence of cracking is found using visual means, a more thorough method of investigation must be used. The most sensitive method of locating surface cracking is the wet fluorescent magnetic-particle method. Other valuable methods are the dry magnetic-particle, dye-penetrant, ultrasonic, and radiographic methods.

Vessels containing amines (absorbers, accumulators, coalescers, condensers, coolers, contactors, extractors, filter vessels, flash drums, knockout drums, reactivators, reboilers, reclaimers, regenerators, scrubbers, separators, settlers,

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API RP*572 92 ■ 0732290 0500800 6 ■

skimmers, sour gas drums, stills, strippers, surge tanks, treating towers, and treated fuel gas drums) are subject to cracks in their welds and the heat-affected zones of the welds. Wet fluorescent magnetic-particle testing is a very sensitive inspection method of detecting surface cracks and discontinuities and is the primary recommended inspection method. See API Publication 945 for more detailed information.

Deaerators on boilers should have their welds and heat-affected zones checked for possible deaerator cracking. Wet fluorescent magnetic-particle testing is the primary recommended inspection method.

Supports are almost always welded to the shell. The point of attachment should be examined closely for cracking. A good light and a scraper will usually be sufficient for this examination.

The attachment points of baffles to exchanger channels and heads should also be checked closely for cracks. Usually, visual inspection with the aid of a light, a magnifying glass, a scraper, and a brush is sufficient.

Laminations in vessel plates have an appearance similar to cracks, but they run at a slant to the plate surface, while cracks run at right angles to the surface. If open sufficiently for a thin feeler to be inserted, the angle of the lamination can be observed. If a lamination is suspected but not open enough for a feeler to be inserted, heating to approximately 200°F (75°C) with a torch will usually cause the edge of the lamination to lip upward. In many cases, an ultrasonic flaw detector may be used to trace the lamination.

8.4.5 INSPECTION OF METALLIC LININGS

Many vessels are provided with metallic linings. The primary purpose of these linings is to protect the vessels from the effects of corrosion or erosion. The most important conditions to check for when examining linings are the following:

a. That there is no corrosion.
b. That the linings are properly installed.
c. That no holes or cracks exist.

Special attention should be given to the welds at nozzles or other attachments.

A careful visual examination is usually all that is required when checking a lining for corrosion. Light hammer taps will often disclose loose lining or heavily corroded sections. If corrosion has occurred, it may be necessary to obtain measurements of the remaining thickness. Unless the surface of the lining is relatively rough, these measurements can be made with the ultrasonic thickness-measuring instrument. Another method of checking the thickness of the lining is to remove a small section and check it with calipers. This method provides an opportunity to inspect the surface of the shell behind the lining.

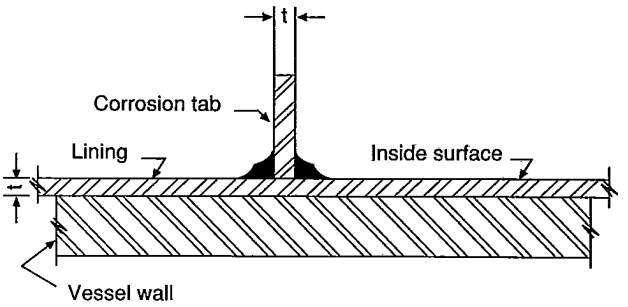
Small 1- by 2-inch (2.5- by 5.0-centimeter) tabs of lining that form a right angle with one leg extending into the vessel may be welded on the lining. The thickness of the protruding leg should be measured at each inspection. Since both sides of the tab are exposed to corrosive action, the loss in thickness would be twice that of the shell lining where only one side is exposed. This permits a fairly accurate check of any general corrosion of the lining. Figure 34 illustrates this lining inspection method.

Cracks in metallic linings can usually be located by visual inspection and light hammering. A cracked section of a liner or a loose liner gives a special tinny sound when tapped with a hammer. If cracking is expected, dye-penetrant methods may be used to supplement visual inspection. With the exception of the straight chrome steels, most of the materials used as linings are primarily nonmagnetic. Magnetic-particle inspection can not be used on austenitic materials.

If cracks are found in a clad liner or an arc-welded liner, the cracked areas should be checked to insure that the cracks do not extend beyond the cladding and into the base metal or parent metal.

Bulges and buckling often occur in metallic linings and usually indicate that cracks or leaks exist in the bulged section of the lining or that pin holes exist in the adjacent welding. The bulges are formed either by the expansion or buildup of a material that seeps behind the lining during operation or by differential expansion. If material seeps behind the lining during operation and cannot escape when the vessel pressure is reduced for a shutdown, the lining may bulge. In vacuum service the lining might bulge in service and depress when the vessel is shut down. This condition may actually wrinkle the lining. When bulges or wrinkling become excessive, it may be necessary to inspect the lining for cracks or pin holes that can be repaired or to replace the lining. Figure 35 illustrates the deterioration of strip-welded linings.

Where a lining leaks, whether corrosion has taken place behind the lining should be determined. In some cases, ultrasonic testing from the outside may be used. Removal of rep-



Note: Corrosion tab of same material and thickness as lining is installed as shown when vessel is lined.

Figure 34—Corrosion Tab Method of Determining Metal Loss on Vessel Linings

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Figure 35—Strip-Liner Deterioration

resentative lining sections to permit visual examination of the vessel wall is always preferred if feasible.

Many reactors in hydrogen service, such as hydrocrackers and hydrotreaters, use complete weld overlay that uses stabilized austenitic stainless steel welding rods or wire as a liner instead of plug- or strip-welded or clad plate. Disbonding from the parent metal can be a problem with this type of lining. Ultrasonic testing, visual checking for bulges, and light tapping with a hammer can reveal this problem.

8.4.6 INSPECTION OF NONMETALLIC LININGS

There are various kinds of nonmetallic linings: glass, plastic, rubber, ceramic, concrete, refractory, and carbon block or brick linings. These materials are used most often for corrosion resistance. Some forms of refractory concrete are used as an internal insulation to keep down the shell temperatures of vessels operating at high temperatures. Refractory tile is also used for insulation.

The effectiveness of these linings in lessening corrosion is appreciably reduced by breaks in the film or coatings. For the most part, inspection should consist of a visual examination for discontinuities in the coatings. These breaks are sometimes called *holidays*. Bulging, blistering, and chipping are all indications that openings exist in the lining. The spark tester method of inspection for leaks in paint, glass, plastic, and rubber linings is quite thorough. A high-voltage, low-current, brush-type electrode is passed over the nonconductive lining. The other end of the circuit is attached to the shell of the vessel. An electric arc will form between the brush electrode and the vessel shell through any holes in the lining. This method can not be used for concrete, brick, tile, or refractory linings.

CAUTION: The voltage used in this inspection method should not exceed the dielectric strength of the coating.

Considerable care should be exercised when working inside vessels lined with glass, rubber, plastic sheets, or paint. These coatings are highly susceptible to mechanical damage. Glass-lined vessels are especially susceptible to damage and are costly and hard to repair.

Concrete and refractory linings may spall and crack in service. Inspection of these linings should be mostly visual. Mechanical damage, such as spalling and large cracks, can be readily seen. Figure 36 illustrates the deterioration of a refractory-tile lining. Minor cracks and areas of porosity are more difficult to find. Light scraping will sometimes reveal such conditions. Bulging can be located visually and is usually accompanied by cracking. In most cases, if corrosion occurs behind a concrete lining, the lining will lose its bond with the steel. The sound and feel of light hammer tapping will usually make such looseness evident. If corrosion behind a lining is suspected, small sections of the lining may be removed. This permits an inspection of the shell and a cross-sectional examination of the lining.

Some refractory-tile linings are hung with a blanket of ceramic fiber or another insulation between the shell and the tile. Broken or missing tiles create lanes for the channeling of any fluid that gets behind the lining. This results in the washing away of some of the insulation. Inspection of tile linings should include a visual inspection of the insulation in the vicinity of broken or missing tile and may be done by removing enough tiles to disclose the damaged areas.

In all cases where bare metal has been exposed because of lining failures, a visual inspection should be made of the exposed metal. If corrosion has taken place, the remaining wall thickness should be measured. Ultrasonic instruments are best suited for this measurement.

Internally insulated vessels are sometimes subject to severe corrosion due to condensation on the shell behind the insulation during operation. If the shell-metal temperatures are near the calculated dew point of the process stream, shell corrosion should be suspected, and the shell should be

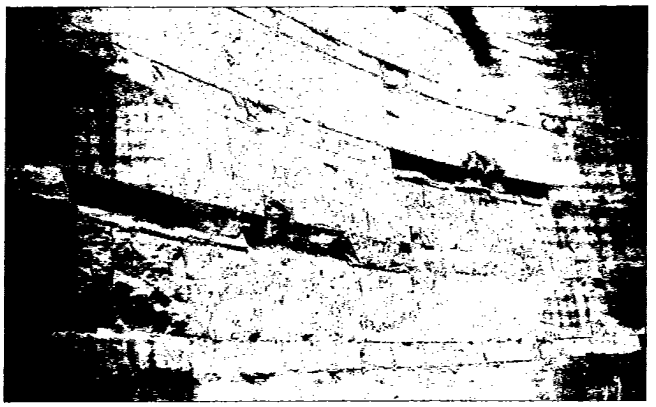


Figure 36—Deteriorated Refractory-Tile Lining

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checked. A frequently used corrective measure is to reduce the internal insulation or to add extra external insulation. Precautions should be taken to assure that design metal temperatures are not exceeded when these measures are used.

8.4.7 THICKNESS–MEASURING METHODS

There are many tools designed for measuring metal thicknesses. The selection of tools used will depend on several factors:

a. The accessibility to both sides of the area to be measured.
b. The desire for nondestructive methods.
c. The time available.
d. The accuracy desired.
e. The economy of the situation.

Ultrasonic instruments are now the primary means of obtaining thickness measurements on equipment. Radiography may also be used in a limited way to determine the thicknesses of vessel parts such as nozzles and connecting piping. Methods such as depth drilling, the use of corrosion buttons, and the use of test holes may be applied at some special locations, but these methods have generally been replaced by more modern methods of thickness gauging such as ultrasonics.

Ultrasonic inspection, both for thickness measurement and for flaw detection, represents an important technique of nondestructive inspection. The technique is fast, providing instantaneous results in most cases. Mechanical vibrations for measurement, analysis, or test purposes are generated by electromechanical transducers. The thickness is gauged by the instrument's ability to accurately measure the time of returning echoes (sound vibrations). The acoustic velocities of most materials used in the petroleum and chemical industries are known. With the knowledge of the time of acoustic travel and the material's acoustic velocity, an accurate thickness can be determined.

Radiographs are taken with a step gauge of known thickness that will show on the developed film of the vessel part in question. By comparing the thickness of the step gauge on film to the thickness of the part on film, the part thickness may be determined.

Corrosion buttons or plugs are fabricated from materials highly resistant to corrosion and are fastened to the vessel wall in sets of two. Losses in thickness are obtained by placing a steel straightedge on the two plugs and measuring the distance from the bottom of the straightedge to the surface of the vessel.

Where the corroded surface is very rough, test holes through the vessel wall may be used to determine thickness.

Depth drilling is used in a similar way to determine corrosion rates. In this method a hole is drilled in the vessel wall where the most corrosion is expected. The depth from the bottom of the hole to the inside surface of the vessel is measured with a depth gauge. Future readings taken at subsequent inspections will permit calculation of the corrosion loss. Between readings, a corrosion-resistant plug is screwed into the hole to protect the bottom of the hole from corrosion.

8.4.8 SPECIAL METHODS OF DETECTING MECHANICAL DEFECTS

Visual examination will reveal most mechanical defects. Magnetic-particle (wet or dry) and dye-penetrant methods may be useful and have been discussed in preceding text. Other methods, such as radiography, shear-wave ultrasonics, etching, and sample removal, are available and may be used when conditions warrant.

Radiography and shear-wave ultrasonics are used to analyze defects, usually in welded seams, that are not visible on the surface of the metal.

Etching of small areas may sometimes be used to find small surface cracks. First, the surface must be abrasive-grit blasted clean. Then, etching solution, usually an acid, is used to wash the suspected area. Because of the nature of the resulting reaction, any cracks will stand out in contrast to the surrounding area.

Sample removal can be used to spot check welds and to investigate cracks, laminations, and other flaws. Small metal samples from the affected area are removed with a trepan or weld probe tools. The sample is then analyzed under a microscope or with an ordinary magnifying glass. If they can be adequately cleaned, the filings obtained during the cutting operation may be used in making a chemical analysis of the metal. The hole left in the vessel wall by sample removal must be carefully repaired, and the repair must be thoroughly inspected. The decision to remove samples should be made by someone who knows how to analyze the problems related to the repair of the sample holes.

8.4.9 METALLURGICAL CHANGES AND IN-SITU ANALYSIS OF METALS

The methods used to detect mechanical changes can also be used to detect metallurgical changes. In-place metallography can be used to detect these changes with portable polishing equipment and using replica transfer techniques. Hardness, chemical spot, and magnetic tests are three other methods of detecting metallurgical changes.

Portable hardness testers can be used to detect the effects of faulty heat-treating, carburization, nitriding, decarburization, and other processes that cause changes in hardness.

Local chemical tests are most often used to detect the installation of materials other than those specified. Chemicals such as nitric acid in various concentrations are used. A spot is cleaned on the metal surface and a drop of a chemical is place on the surface. An experienced observer can observe the reactions to the acid of the metal being tested and identify the metal. Eddy-current, x-ray fluorescence, radiation,

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API RP*572 92 ■ 0732290 0500803 1 ■

and portable light-emission spectroscopic instruments are also used for metal identification.

Because normally nonmagnetic steel usually becomes magnetic when carburized, carburization of austenitic stainless steel can sometimes be detected by a magnet.

8.5 Testing

8.5.1 HAMMER TESTING

In hammer testing, an inspector's hammer is used to supplement visual inspection. The hammer is used to do the following jobs:

a. To locate thin sections in vessel walls, heads, and attachments.
b. To check tightness of rivets, bolts, brackets, and the like.
c. To check for cracks in metallic linings.
d. To check for lack of bond in concrete or refractory linings.
e. To remove scale accumulations for spot inspection.

The hammer is used for these jobs by lightly striking or tapping the object being inspected and observing the sound, feel, and indentation resulting from the blow. The proper striking force to be used for the various jobs can be learned only through experience. Hammer testing is used much less today than previously. It is not recommended to hammer test objects under pressure. Also, piping upstream of a catalyst bed should not be hammered, as hammering could dislodge scale or debris and cause plugging.

8.5.2 PRESSURE AND VACUUM TESTING

When a pressure vessel is fabricated, it is tested for integrity and tightness in accordance with the standard or construction code to which it was built. (In addition to determining the vessel's integrity and tightness, the pressure test can result in a beneficial stress redistribution at defects.) These methods of testing may also be used subsequently to inspect for leaks and to check repair work. When major repair work—such as replacing a head, a large nozzle, or a section of the shell plate—is performed, the vessel should be tested as if it were just installed. In certain circumstances, the applicable construction code requirements for inspection of vessels in service also require periodic pressure testing, even though no repair work has been necessary. For code rules concerning tests of vessels in service, see API 510 and NBBPVI NB-23. The ASME Code, although a new vessel fabrication code, may also be followed in principle in many cases.

A large vessel and its structural supports may not necessarily be designed to support the weight of the vessel when it is filled with water. Whether is can support this weight should be determined before a hydrostatic test is made. If the vessel or its supports are inadequate for a hydrostatic test, then a pneumatic test may be considered.

Pressure testing consists of filling a vessel with liquid or gas and building up an internal pressure to a desired level. The pressure and procedures used should be in accordance with the applicable construction code requirements consistent with the existing thickness of the vessel and the appropriate joint efficiencies. (As noted in preceding text, sometimes the rules for inspection in service require periodic pressure testing even if no repair work has been necessary.) When pressure vessels form a component part of an operating unit, the entire unit is sometimes pressure tested. Water or oil is used as a testing medium and the charge pumps of the unit are used to provide the test pressure. While the vessel or vessels are under pressure, the external surfaces are given a thorough visual examination for leaks and signs of deformation.

In recent years, acoustic emission analysis has been developed for use with pressure testing. When acoustic emission equipment is used on a vessel under pressure, it is possible to determine the overall structural integrity of the vessel. This method can be especially useful for vessels of complex design or where the vessel contents cannot be easily removed to permit an internal inspection.

When testing pneumatically, an ultrasonic sound detector, a soap solution, or both should be used to aid visual inspection. The soap solution is brushed over the seams and joints of the vessel. The vessel is then examined for bubbles as an indication of leakage.

An ultrasonic sound detector may be used to pick up leaks in joints and the like that can not be reached with a soap solution without scaffolds or similar equipment. Very small leaks may be detected and located with the sound detector.

Often a vessel that operates at a vacuum may be pressure tested. When feasible, pressure testing is the preferred testing method as leaks from an internal pressure source are more easily located. When pressure testing is not feasible, a vacuum vessel can be tested for leaks with evacuators or vacuum pumps that are installed in the unit and used to create a vacuum. If the vacuum can be held for a specified time after closing off the evacuators or vacuum pumps, it is likely that the vessel is free of leaks. If the vacuum cannot be held, leaks are present. However, since this method gives no indication of the locations of leaks, a search, which may be difficult, must then be made to locate the leaks.

Consideration should be given to the temperature at which testing is done. Many of the common steels used in fabrication exhibit severe reduction of impact resistance at low temperatures. API 510 recommends that vessels constructed with these steels be tested either at temperatures not less than those recommended by the applicable section of the ASME Code or at 70°F (20°C) and, in any case, at not more than 120°F (50°C), unless there is information on the brittle characteristics of the vessel material to indicate the acceptability of a lower test temperature or the need for a higher test temperature (see API Publication 920).

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When conducting hydrostatic or pneumatic pressure tests, it is a good safety practice for all personnel not connected with the test to remain away from the area until the test is completed and the pressure is released. The number of inspection personnel in the area should be limited to the number necessary to run the test. When making pneumatic pressure tests, the recommendations set forth in the ASME Code should be followed.

8.5.3 TESTING EXCHANGERS

When an exchanger is removed from service, it is often desirable to apply a test to either the shell side or the tube side before dismantling. A leak may be detected by observation of a drain point, such as a disconnected lower nozzle or an open bleeder. Usually, the test must be run for some time before a small leak will show up. If the exchanger leaks, it is then partially dismantled and the test reapplied. For example, when testing a floating-head exchanger with the pressure in the tubes, removal of the shell cover will reveal the source if the leak is in the gasket, stay bolts, or tube rolls at the floating head. This test will not normally distinguish between tube roll leaks at the stationary tube sheet and those at penetrated tube walls, as these parts are not visible while the tube bundle is in the shell. A shell test applied to a floating-head exchanger with the channel cover off will reveal leaking tube rolls at the stationary tube sheet but will not identify the source of other leakage. In most cases, exchangers that do not use floating heads are so constructed that a shell side test applied to the partially dismantled exchanger will enable individual detection of leaking tubes and their plugging. The plugging is done by driving a tapered plug made of a solid metal similar to the metal of the tube into each end of the tube. Also, leaking tube rolls at either end can be detected and rerolled. Exchangers with floating heads do not permit individual detection of leaking tubes or access to both ends of tubes during a shell side test. A test ring is sometimes used for these exchangers. This is a device which temporarily converts the arrangement of the partially dismantled exchanger to a dual fixed tube-sheet arrangement.

In some cases, leak testing is performed at each downtime. If leaking tubes are found, the leaks are located and plugged, and the bundle is put back in service. This procedure should be repeated until no new leaks are discovered; several repetitions may be required. If, at some point, a sufficient number of tubes have been plugged to interfere with the use of the exchanger, the bundle should be retubed.

When leakage is encountered for the first time in a given service, inspection may be performed to determine the nature of the deterioration. After historical records have been built up, inspection is performed only when the number of plugged tubes indicates that the replacement point may be approaching. When a decision is made to retube, inspection is employed to determine which parts can be salvaged and reused and which require replacement.

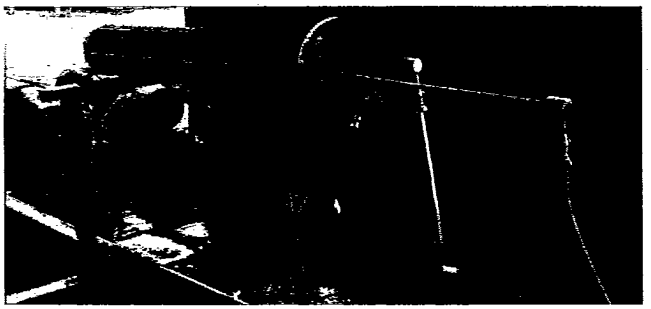
It is customary to test an exchanger at assembly. Where retubing has been performed, a test may be applied to the partially assembled exchanger to detect roll leaks individually. In any case, a final test on both the shell and tube sides is normally applied to the assembled exchanger.

Frequently, a bundle will be tested while it is out of the shell. In this case, the channel and floating tube-sheet covers are left in place. This method makes observation for leaks easier but necessitates a separate shell test.

When any of the parts are under test pressure, the external surfaces, rolled joints, and gasketed joints are given a thorough visual examination. Leaks and distortion of parts may be found by pressure testing.

Special equipment is available to test exchanger tubes individually. An example of this equipment is shown in Figure 37.

The pressures to be used when testing will depend on the operating and design pressures of the unit. These pressures should be determined locally in accordance with individual practice or jurisdictional requirements. Before applying pressure to the shell side only of an exchanger, the inspector should be sure that the tubes of the bundle are of sufficient wall thickness to withstand the external pressure. In addition, the channel side test pressure and the shell side test pressure



Step 1



Step 2

Figure 37—Steps in Using Special Equipment to Test Individual Tubes

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should be checked against one another, and care should be taken that one side or the other of the exchanger is not excessively pressured when testing. Particular care should be taken with any high-pressure exchangers where the tube sheets were designed on the basis of differential pressure.

8.6 Limits of Thickness

The limits of corrosion and any other kind of deterioration that may be tolerated must be known, or an inspection will lose much of its value. The two most important factors of this problem are the following:

a. The retiring thickness of the part considered.
b. The rate of deterioration.

Before determining the limiting or retiring thicknesses of parts of any pressure vessel, which code and edition of that code it is to be rated under and whether there are any regulations regarding limits and allowable repairs must be determined.

There are a great many variables, such as size, shape, material, and method of construction, that affect the minimum allowable thickness. For this reason it is not possible in this document to present a set of precalculated minimum or retiring thicknesses. API 510 contains additional guidance on the rating of pressure vessels.

When corrosion or erosion is causing deterioration, the rate of metal loss can usually be obtained by comparing consecutive inspection records. Sometimes a graph to show this information is drawn and is kept with the vessel records. In many cases, computerization of inspection data has been helpful in quickly determining corrosion rates and in estimating retirement dates. The ability to predict when a vessel will reach a retiring thickness is important. Scheduling of repairs and replacements will be greatly influenced by such predictions.

When the limit of safe thickness is approached or reached, decisive action is necessary. In some cases decisions will have to be made quickly without much time for study or for consideration and review by others. The minimum thicknesses or the methods of calculating the thicknesses should be known in advance for each vessel. Remember that different parts of a vessel may have different retiring thicknesses. As mentioned before, computer treatment of the data could be valuable.

Most vessels are built with some thickness in vessel walls and heads over that required to withstand the internal operating pressures. This excess thickness may result from any of the following contributions:

a. Excess thickness deliberately added in the design as a corrosion allowance.
b. Excess thickness as a result of using a nominal plate thickness rather than the exact, smaller value calculated.
c. Excess thickness as a result of setting minimum plate thicknesses for construction purposes.
d. Excess thickness resulting from a change in vessel service: a reduction of the safety valve setting, the maximum metal temperature, or both.

When the total excess thickness and the corrosion rate are known, the date when repairs or replacement will be needed for any vessel can be predicted with reasonable accuracy.

CAUTION: In some cases, the excess thickness of the shell or head plates is used by the designer as nozzle reinforcement.

Since the ASME Code is a design and construction standard for vessels, the methods of calculating the retirement thicknesses of many accessories of pressure vessels are not covered. Some of these parts are trays, internal tray supports, valves, grids, baffles, ladders, and platforms. For some of this equipment, there are generally accepted methods of setting the retiring thicknesses. Minimum thicknesses should be developed for all this equipment. The results of possible failure of equipment should be considered when setting these limits. Safety is the prime factor affecting retiring thickness. After safety, continuous and efficient operations become a factor. (API Recommended Practice 574 should be consulted for inspection of some of the parts mentioned in the preceding text.)

Normally, no minimum thickness is set for applied metallic linings. As long as the lining remains free of leaks and does not require excessive repairs, it should be satisfactory for further service.

In the case of exchangers, minimum thicknesses should be developed for tubes, tube sheets, channels, covers, and other pressure-containing exchanger parts. The results of possible failure of such parts should be considered when setting these limits. Safety is the primary factor affecting retiring thickness for this equipment. Normally, failure of internal parts, such as the various components of the tube bundle, does not involve a hazard; hence, continuous, efficient operation is the governing factor in establishing retirement limits of internal parts. Some parts, such as baffles, may be continued in service until failure, and tubes need not be plugged or replaced until actual perforation occurs.

SECTION 9—METHODS OF REPAIR

Although repair and maintenance are not parts of inspection, repairs that affect the pressure rating of a vessel and that require reinspection for safety reasons are of concern.

Before any repairs are made to a vessel, the applicable codes and standards under which it is to be rated should be studied to assure that the method of repair will not violate appropriate requirements. API 510 sets forth minimum petroleum and chemical process industry repair requirements and is

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API RP*572 92 ■ 0732290 0500806 7 ■

recognized by several jurisdictions as the proper code for repair or alteration of petroleum or chemical pressure vessels.

Note: Some jurisdictions require that welded repairs and alterations be done by an organization with an appropriate NBBPVI "R" stamp, usually in accordance with NBBPVI NB-23 and accompanied by the completion and filing of NBBPVI Form R-1 with the jurisdiction.

The defects requiring repair and the repair procedures employed should be recorded in the field notebook and later in the permanent records of the vessel (see API 510 for sample repair and alteration record sheets or refer to jurisdictional requirements). Most repairs on the shell and heads of a vessel are made to maintain the strength and safety of the vessel, will therefore require reinspection, and may require radiography, stress relieving, or both. A quick visual check of all other repairs is also desirable to make sure that they have been completed.

It is important that the source of the problem requiring the repair is determined. Treating the source of the condition causing deterioration will, in many cases, prevent future problems.

Repairs made by welding to the shell and heads of a vessel should be inspected. The inspection should include a check for completion and quality. Normally, a visual examination will be sufficient for minor repairs; however, magnetic-particle and dye-penetrant methods should be used on major repairs, and if required by the applicable construction code, a radiographic or ultrasonic shear-wave examination should also be made.

A pressure test should be applied after all major repairs unless in the judgement of the inspector a sound technical case can be made to justify waiving the test. In cases where a pressure test is waived, other nondestructive testing or inspection is required.

The repair of sample holes left by a trepan or weld probe tool must be closely inspected. The weld quality in such repairs is likely to be poor unless carefully controlled. Therefore, the removal of samples for weld inspection should be avoided if possible.

Sections of shell plates may be replaced to remove locally deteriorated areas. The joint efficiency of the patch should be equal to or greater than the efficiency of the original joints in the shell.

Cracks in vessel walls or heads may be repaired by chipping, by flame, arc, or mechanical gouging, or by grinding the crack from end to end and then welding. Care should be used in flame and arc gouging, as heat may cause the crack to enlarge or lengthen. If a crack extends completely through the plate, it may be expedient to cut a groove from both sides of the plate. In any event, complete removal of the crack is absolutely essential before welding is begun. Magnetic-particle or dye-penetrant techniques should be employed to assure removal of the crack. If several cracks occur in any one plate, it may be wise to replace the entire plate. Repairs of cracks by welding should be checked carefully. If the cavity caused by grinding out a defect is not deep and the remaining metal provides adequate strength and corrosion protection, the repair may be completed without welding by beveling the edges of the cavity.

Scattered pits in pressure vessels are best repaired by welding. For temporary repair, some proprietary epoxy base materials can be packed into pits to prevent further corrosion. This material must be capable of resisting the service conditions. In all cases, pits should be well cleaned, preferably by abrasive-grit blasting, before repairs are made.

Note: When considering the use of this method, the inspector must be satisfied that the pits are not large enough or close enough together to represent a general thinning of the vessel component—see the subsection on corrosion and minimum thickness evaluation in API 510.

Repair of appurtenances such as platforms, ladders, and stairways will usually consist of replacing excessively worn parts. Stairway treads that have been worn smooth can be roughened by placing weld beads on the worn surfaces. Also, proprietary coatings containing gritty material for application on the treads are available.

Linings are repaired by replacing parts that are corroded through or cracked. Repairs of metallic linings require welding. Visual inspection of welding after thorough slag removal will normally be sufficient to check weld quality, unless code requirements specify a radiographic, dye-penetrant, magnetic-particle, or other examination of the weld.

SECTION 10—RECORDS AND REPORTS

10.1 Records

Inspection records are required by API 510, NBBPVI NB-23, and jurisdictions. As they form the basis of a scheduled maintenance program, they are very important. A complete record file should contain three types of information:

a. Basic data.
b. Field notes.
c. The data that accumulates in the "continuous file."

Basic data includes the manufacturer's drawings, data reports and specifications, design information, and the results of any material tests and analyses.

Field notes consist of notes and measurements recorded on site either on prepared forms (see Appendix B) or in a

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API RP*572 92 ■ 0732290 0500807 9 ■

written or electronic field notebook. These notes should include in rough form a record of the condition of all parts inspected and the repairs required.

The continuous file includes all information on the vessel's operating history, descriptions and measurements from previous inspections, corrosion rate tables (if any), and records of repairs and replacements.

As indicated in preceding text, some companies have developed software for the computerized storage, calculation, and retrieval of inspection data. When their data is kept up-to-date, these programs are very effective in establishing corrosion rates, retirement dates, and schedules. The programs permit quick and comprehensive examination of all accumulated inspection data. Some of these programs can produce an inspection report that may be issued to all interested parties.

10.2 Reports

Copies of reports recommending repairs should be sent to all management groups, which would normally include engineering, operating, and maintenance departments. These reports should include the location, extent, and reasons for recommended repairs.

General inspection reports (see Appendix B) may be sent to interested parties, such as the operating, maintenance, and engineering departments. Who the interested parties are will depend on the organization of the plant or company. These reports should include metal thickness measurements, corrosion rates, descriptions of the conditions found, repairs required, and allowable operating conditions, estimations of remaining life, and any recommendations. Occasionally special reports covering unusual conditions may be circulated.

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APPENDIX A—EXCHANGERS

A.1 General

Exchangers are used to reduce the temperature of one fluid by transferring heat to another fluid without mixing the fluids. Exchangers are called condensers when the temperature of a vapor is reduced to the point where some or all of the vapor becomes liquid by the transfer of heat to another fluid, usually water. When a hot fluid is cooled to a lower desired temperature by the transfer of heat to another fluid, usually water, the exchanger is usually referred to as a cooler. When air is used to reduce the temperature of a hot liquid to a lower desired temperature, the exchanger is referred to as an air cooler. Figures 6 and 7 illustrate heat exchanger parts and types.

A.2 Shell and Tube–Bundle Exchangers

A.2.1 GENERAL

There are several types of shell and tube-bundle exchangers. Usually, the tubes are attached to the tube sheet by rolling. A properly rolled tube is shown in Figure A-1. The tubes may be rolled and welded or attached by packing glands. The physical characteristics of fluids such as the temperature determine the type of fluid used for a particular service. A description of some of the types of exchangers commonly used and the factors influencing their selection follow.

A.2.2 ONE FIXED TUBE SHEET WITH A FLOATING HEAD

One type of exchanger consists of a cylindrical shell flanged on both ends, a tube bundle with a tube sheet on both ends, a channel, a channel cover, a floating-head cover for one end of the tube bundle, and a shell cover. The diameter of one tube sheet of the tube bundle is small enough to pass through the cylindrical shell. The diameter of the other tube sheet is large enough to bear on a gasketed surface of one shell flange or may be an integral part of the channel. The bundle is inserted in the shell with the large tube sheet against one shell flange. The channel is bolted to the shell flange that holds the tube sheet in place. The channel and floating heads may be divided so that incoming liquid flows through some of the tubes and returns through other tubes to the channel. The number of divisions and the number of tube flow passes will vary with the design. The flow through the shell is directed by baffles as desired. Since the floating tube end is free to move in the shell, this type of construction permits free expansion and contraction with changes in temperature. This is the type of heat exchanger most commonly used.

A.2.3 TWO FIXED TUBE SHEETS

The construction details of an exchanger with two fixed tube sheets are similar to those of the floating tube-sheet type; however, both tube sheets are fixed and the tubes are installed and rolled after the tube sheets are in place. The shell side can not be exposed for cleaning. Therefore, it is limited to either clean service or service susceptible to chemical cleaning. Because both tube sheets are fixed, the exchanger is limited to little expansion and contraction unless an expansion joint is provided in the shell.

A.2.4 ONE FIXED TUBE SHEET WITH U TUBES

A U-tube exchanger has one fixed tube sheet with the tubes bent in the form of a long U in place of the floating head. These exchangers have the same freedom of expansion and contraction as those of the floating-head type. Clean service is usually limited to the tube side because of the difficulty of mechanically cleaning the inside of the U tubes. Chemical cleaning, abrasive-grit blasting, or hydroblasting

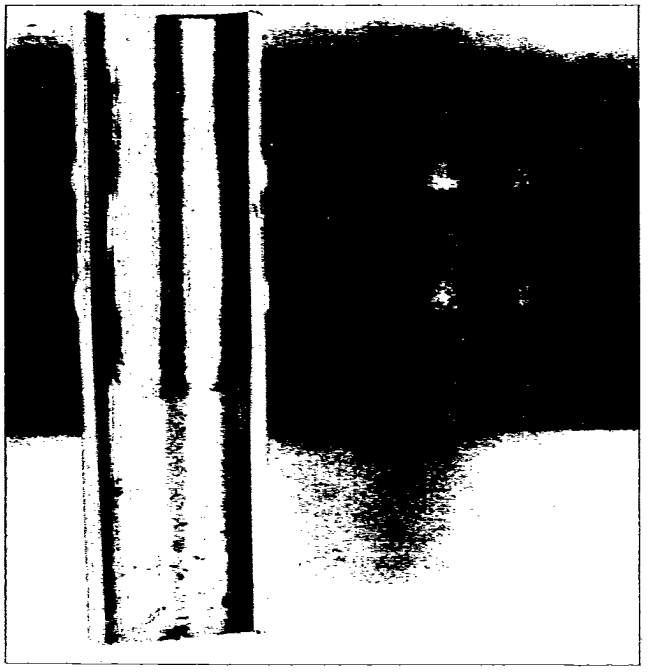


Figure A-1—Properly Rolled Tube

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can be used successfully if care is taken not to allow the tubes to become completely plugged.

A.2.5 DOUBLE–TUBE–SHEET EXCHANGERS

In certain services where even minute leakage of one fluid into another can not be tolerated, double-tube-sheet construction of the exchanger is sometimes employed. As the name implies, two tube sheets are used together with only a small distance, usually 1 inch or less, between them. The tubes are rolled into both tube sheets. The outer tube sheet is attached to the channel, and the inner tube sheet is fixed to the shell. The purpose of this arrangement is to cause any leakage from the tube roll to bleed off into the space between the two sheets, thus preventing contamination of one fluid by the other. This construction is applicable only where there is no floating tube sheet. Hence, it can be used only with a U-tube exchanger.

A.2.6 REBOILERS AND EVAPORATORS

The construction details of reboilers and evaporators are the same as those of any other exchanger with one fixed tube sheet, with the exception that horizontal reboilers have a large vapor space above the tube bundle. They are used to produce vapor from liquids by passing a hot fluid through the tubes.

A.2.7 WATER HEATERS

Water heaters may be either the floating-head type, the U-tube type, or the fixed tube-sheet type. They are used to heat water for boiler feed or for other purposes by exchanging heat from a hot fluid.

A.2.8 CONSTRUCTION

Exchangers are equipped with baffles or support plates, the type and design of which vary with the service and heat load the exchanger is meant to handle. Pass partitions are usually installed in the channels and sometimes in the floating tube-sheet covers to provide multiple flow through the tubes. The flow through the shell may be single pass, or longitudinal baffles may be installed to provide multiple passes. The baffling used in the shell will determine the location and number of shell nozzles required. Figures 6 and 7 show various channel and shell baffle arrangements. Frequently, an impingement baffle plate or rod baffle is located below the shell inlet nozzle to prevent impingement of the incoming fluid on the adjacent tubes.

The tubes may be arranged in the tube sheet on either a square pitch or a triangular pitch. When the fluid circulating around the outside of the tubes may coke or form other dirty deposits on the tubes, the square pitch is generally used. The square pitch arrangement permits better access for cleaning between the tubes.

A.3 Exposed Tube Bundles

A.3.1 GENERAL

Exposed tube bundles are used for condensing or cooling and may be located under spraying water or may be completely submerged. They also may be used as heaters, particularly in tanks where they are submerged in the liquid.

A.3.2 AN EXPOSED TUBE BUNDLE UNDER A COOLING TOWER

Exposed tubes arranged in compact bundles can be placed under a cooling tower; in this arrangement, the water from the tower flows over the tubes, and heated water is returned to the top of the tower for cooling and reuse. This placement of the tube bundles is most effective in a climate with a low relative humidity resulting in maximum evaporative effect.

A.3.3 AN EXPOSED TUBE BUNDLE UNDER SPRAY HEADS

Spray heads may be installed above an exposed tube bundle to provide an even distribution of water over the tubes (this represents a modification of the method described in A.3.2). A receiving tank is located below the tube bundle for use mainly when the water is naturally cool enough to permit recirculation without additional cooling. Where water is plentiful, this type of cooler may be used without a receiving tank, permitting the used water to drain into a treatment system.

A.3.4 SUBMERGED EXPOSED TUBE SECTIONS

When exposed tube sections are submerged, the sections are mounted either vertically or horizontally within a box. The hot fluid enters the top of the headers in vertical installations and the top section in horizontal installations. In either installation, the cooled fluid leaves at the bottom. Cool water enters near the bottom of the box, and the warmed water overflows a weir near the top of the box. This arrangement produces counter current flow resulting in maximum cooling with a minimum use of water.

Submerged sections are used primarily when a hot fluid leaving the cooler might result in a dangerous condition if the water supply should fail. The large volume of water in the cooler box would give partial cooling for an extended period and allow time for an orderly shutdown of the operation if necessary.

A.4 Storage Tank Heaters

The tube-bundle version of the tank heater is built in three general types for the following installations:

a. Installation outside the tank.
b. Installation partially within the tank.
c. Installation entirely inside the tank.

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The first two are installations of suction line heaters, and the third (see Figure A-2) heats the entire contents of the tank.

A.5 Air–Cooled Exchangers

An air-cooled unit is similar to an exposed tube bundle unit; however, air is used as the cooling medium. A bank of tubes is located in a steel framework through which air is circulated by a fan placed either above or below the tube bank (a fan above the tube bank is usually referred to as an *induced draft air cooler* and a fan below the tube bank is usually referred to as a *forced draft air cooler*). These coolers may be used for the condensing or cooling of vapors and liquids and are installed where water is scarce or for other reasons. Figure A-3 illustrates air-cooled exchangers. (API Standard 661 covers the minimum requirements for design, materials, fabrication, inspection, testing, and preparation for initial delivery.)

A.6 Pipe Coils

A.6.1 GENERAL

Pipe coils are of two types:

a. Double-pipe coils.
b. Single-pipe coils.

A.6.2 DOUBLE–PIPE COILS

A.6.2.1 General

Double-pipe coils are used when the surface required is small, because they are more economical than the shell or tube type of exchanger in such service. They are also used where extremely high pressures are encountered, since their small diameter and cylindrical shape require a minimum wall thickness.

A.6.2.2 Clean–Service Double–Pipe Coils

Clean-service double-pipe coils consist of tubes within tubes (see Figure A-4). The internal tubes are connected at one end by return bends that are enclosed by return bends connecting the external tubes of the same coil unit. At the opposite end, the internal tubes project beyond the external tubes and through a tight closure that prevents leakage. Internal tubes terminate in piping or are connected to adjacent units with exposed return bends. The external tubes are connected to piping or adjoining external tubes by branch-flanged nozzles.

A.6.2.3 Dirty–Service Double–Pipe Coils

Dirty-service double-pipe coils (scraper-type coils) are identical to clean-service double-pipe coils with the exception that a scraper is added to the inside of the internal tube. Each internal tube is equipped with scrapers mounted on a rod or shaft extending the full length of the tube. The rod projects through the return bends at each end. To prevent leakage, a bearing for the rod is capped at one end, and a bearing and a stuffing box are used at the other end. The rod extends through the stuffing box, and a sprocket is mounted on the end of the rod. The rods and scrapers are rotated by a sprocket chain driven by some form of prime mover, usually an electric motor.

A.6.3 SINGLE–PIPE COILS

A.6.3.1 General

Single-pipe coils are used in several different ways, but all are continuous runs of pipe through which flows a medium to be cooled or heated.

A.6.3.2 Condenser or Cooler Coils

A condenser or cooler coil consists of a continuous pipe coil or a series of pipe coils installed in a box through which cold water flows. The pipe coil or coils rest on supports in the box and are free to move with any expansion or contraction. Water enters near the bottom of the box and overflows a weir near the top.

A.6.3.3 Chilling Coils

Chilling coils are pipe coils installed in cylindrical vessels to cool a product to below atmospheric temperature. Usually a refrigerant is circulated through the coils to accomplish the cooling. The pipe may be coiled near the internal periphery of the vessel and extend from the bottom to the top or may be arranged as a flat, spiral coil near the bottom of the vessel.

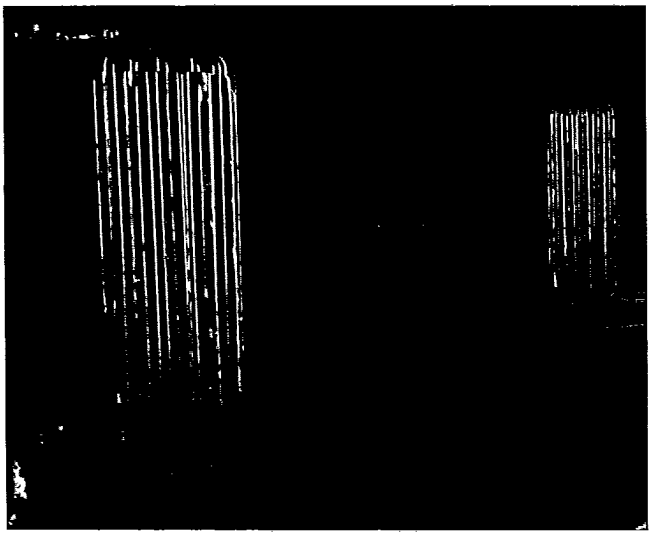


Figure A-2—Tube-Bundle Type of Tank Heater Entirely Inside Tank

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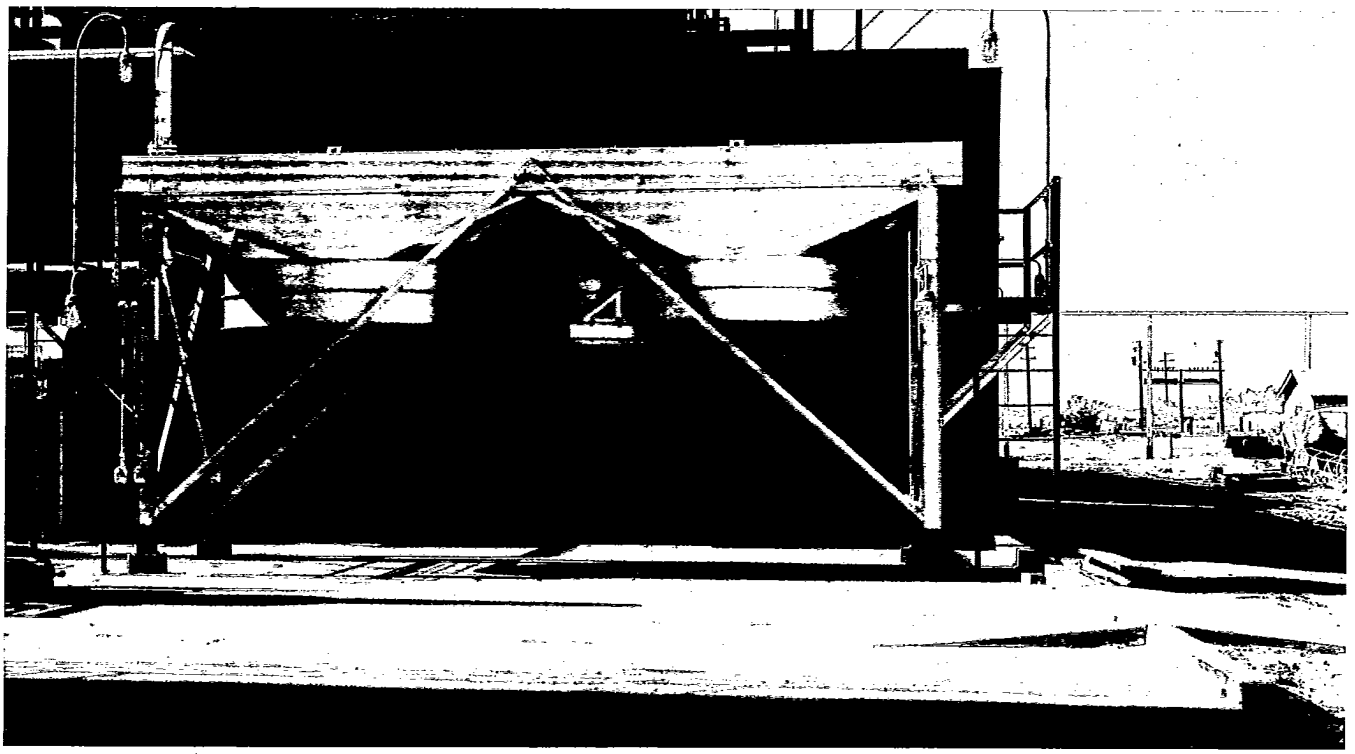


Figure A-3—Air-Cooled Exchangers

A.6.3.4 Flat–Type Tank Heater Coils

The flat-type coil extends over most of the bottom of a storage tank and is a continuous coil with return bends connecting the straight runs of pipe. Steam enters one end of the coil, and condensate is drained at the other end through a steam trap. The coil rests on low supports at the bottom of the tank and slopes gently from the inlet to the outlet to facilitate drainage of the condensate. The pipe is usually made of steel, and generally all joints are welded to minimize the probability of leakage.

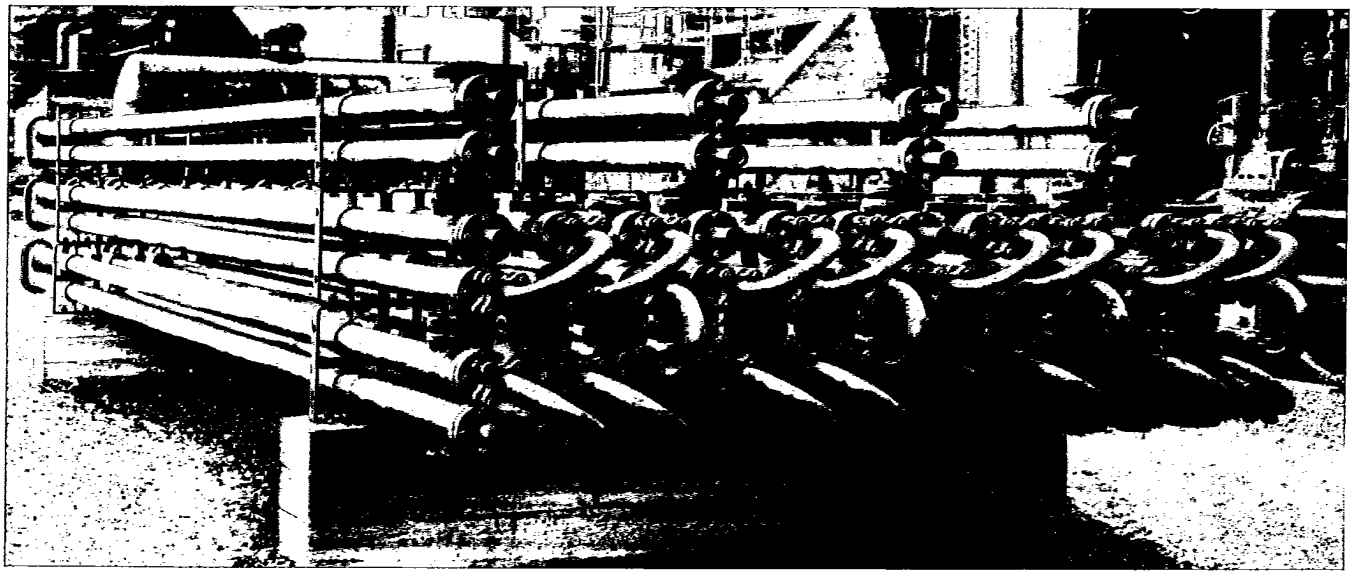


Figure A-4—Clean-Service Double-Pipe Coils

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A.6.3.5 Box–Type Tank Heater Coils

The box-type coil is constructed in a rectangular shape, as shown in Figure A-5, and extends diametrically from the tank outlet to within a few feet of the opposite side of the tank. The coil is enclosed in a box made of steel or wood. The end of the box opposite the tank outlet remains open to permit the entrance of oil. The oil flows through the box, around the coil, and to the tank outlet. Steam enters the top of the coil and flows downward to the outlet where condensate is drained through a steam trap. The entire coil is sloped gently from the inlet to the outlet to facilitate drainage of the condensate.

A.7 Extended Surface or Fin-Type Tubes

Extended surface or fin-type tubes are used quite extensively for more efficient heat exchange, especially when the exchange is between two fluids having widely different heat conductivities. The addition of the extended surface requires less internal tube surface. Consequently, an exchanger smaller than would be required if plain tubes were used is necessary. The use of fin-type tubes in a double-pipe coil is shown in Figure A-6.

A.8 Plate–Type Exchangers

The plate-type exchanger is also constructed with an extended surface, making use of alternating layers of thin plates and corrugated sections. Integral channel and manifold sections enclose the open ends. The process material flows into the corrugated openings. Because the flow openings are small, they are easily clogged by dirt and products of corrosion. This is one of the reasons these units are constructed of materials that are highly resistant to corrosion. A plate-type exchanger installation for storage tank heating service is illustrated in Figure A-7.

A.9 Inspection of Exchanger Bundles

A.9.1 GENERAL

The first step in bundle inspection is a general visual inspection that may establish general corrosion patterns. If possible, bundles should be checked when they are first pulled from the shells, because the color, type, amount, and location of scales and deposits often help to pinpoint corrosion problems. An overall, heavy scale buildup on steel tubes may indicate general tube corrosion. The lack of any scale or deposit on tubes near the shell inlet may indicate an erosion problem. A green scale or deposit on copper base tubes indicates that these tubes are corroding. As an inspector gains experience, these scales and deposits will become a useful inspection guide.

While visually inspecting a bundle, the inspector should make use of a pointed scraper to pick at suspected areas next to tube sheets and baffles. These areas may not have been cleaned completely. Picking in these areas will sometimes disclose grooving of tubes and enlargement of baffle holes. Figure A-8 shows tubes thinned at baffles.

Tapping the tubes with a light (4–8-ounce) ball peen hammer or inspection hammer during the visual check will often help to locate thinned tubes. This method is especially useful when inspecting light-wall tubes of small outside diameters. The amount of rebound and the sound of the blow give an indication of the tube wall thickness. This method will become more helpful as experience is gained with the hammer.

The inside of the tubes can be partially checked at the ends with flashlight extensions, fiber optic scopes, borescopes, and special probes. The special probes are slender ⅛-inch rods with pointed tips bent at 90 degrees to the

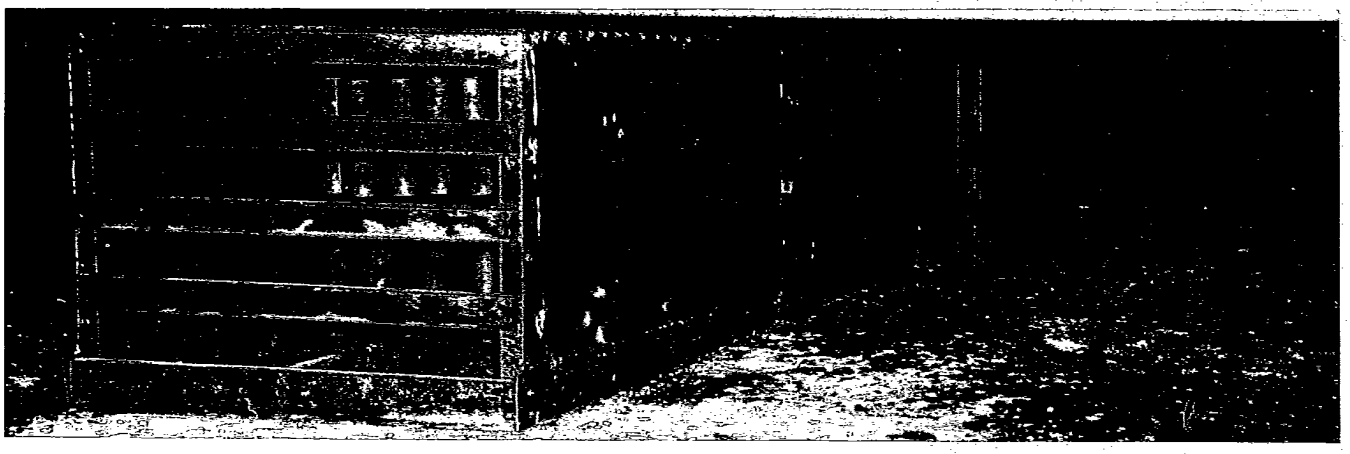


Figure A-5—Tank Suction Heater With Everything But Forward End Enclosed; Shell Suction Nozzle Enclosed in Far End

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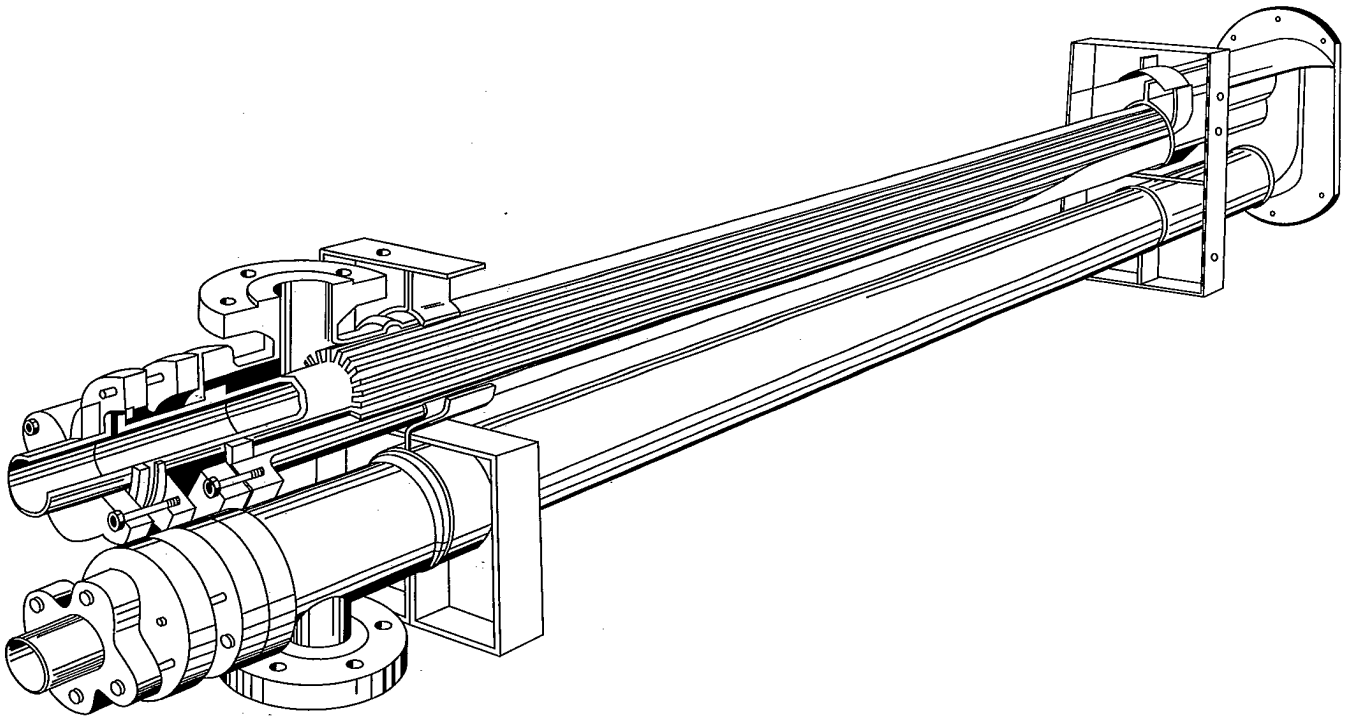


Figure A-6—Fin-Type Tubes in Double-Pipe Coil

axis of the rod. With these tools, it is possible to locate pitting and corrosion near the tube ends.

Obviously, only the outer tubes of a bundle can be thoroughly inspected externally, and without a borescope or a fiber optic scope, only the ends of the tubes can be inspected internally. If a complete inspection of the tubes for defects is required, it can be made using eddy-current methods or ultrasonic methods (for internal-rotary, ultrasonic thickness measurements).

Tubes may also be removed from the bundle and split for visual inspection. There are devices available for pulling a single tube from a bundle.

Removal of one or more tubes at random will permit sectioning and a more thorough inspection for determining the probable service life of the remainder of the bundle. When bundles are retubed, a close inspection of tubes removed will help to identify the causes of failure and improve future service.

The baffles, tie rods, tube sheets, and floating-head cover should be visually inspected for corrosion and distortion. Gasket surfaces should be checked for gouge marks and corrosion. A scraper will be useful when making this inspection. Sufficient gasketed surface should remain to make a tight seal possible when the joints are completed.

Tube sheets and covers can be checked for distortion by placing a straightedge against them. Distortion of tube sheets can result from the overrolling or improper rolling of tubes, thermal expansion, explosions, rough handling, or overpressuring during a hydrotest.

Tube-sheet and floating-head thicknesses can be measured with mechanical calipers. Except in critical locations, continuous records of such readings are not usually kept. However, the original thickness readings of these parts should be recorded. Thickness readings of tie rods and baffles are not generally taken. The condition of these parts is determined by a visual inspection.

Tube wall thicknesses should be measured and recorded at each inspection. It is sufficient to measure the inside and outside diameters and to thus determine the wall thickness. Eccentric corrosion or wear noted during the visual inspection should be taken into account in determining the remaining life of the tubes.

It is difficult to accurately measure the outside diameter of fin-type tubes, except at tube ends, and the outside diameter of tubes in fixed tube sheets can not be measured at all. Some situations may require the tube wall thickness of these tubes. The internal-rotary, ultrasonic thickness-measuring devices can be used to measure the thickness. In some cases, it may be necessary to remove sample tubes. Tube removal is also employed when special examinations, such as metallurgical and chemical ones, are needed to check for the dezincification of brass tubes, the depth of etching or fine cracks, or any high-temperature metallurgical changes.

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A.9.2 LIKELY LOCATIONS OF CORROSION

The locations where corrosion should be expected depend on the service of the equipment. However, there are certain locations that should be watched under most conditions of service.

The outside surface of tubes opposite shell inlet nozzles may be subject to erosion or impingement corrosion. When a mildly corrosive substance flows on the shell side of the tube bundles, the maximum corrosion often occurs at these inlet areas. The next most likely point of attack under the same conditions is adjacent to the baffles and tube sheets. Any deterioration here is probably erosion-corrosion (see Figure A-9).

When a high-temperature material flows into the tube inlet pass, the back of the stationary tube sheets or tubes immediately adjacent to it may suffer extensive corrosion.

When process conditions allow a sludge or similar deposit to form, it will generally settle along the bottom of the shell. If the deposit contains a corrosive material, the maximum corrosion will occur along the bottom of the shell and the bottom tubes.

In water service, the greatest corrosion will occur where the water temperature is highest. Thus, when the water is in the tubes, the outlet side of the channel will be the location of greatest corrosion. Figure 29 shows pitting in a channel.

Also in water service, when exchanger parts are made of gray cast iron, they should be checked for graphitic corrosion. This type of attack is most often in water-service channels or along the bottoms of shells where sour water might collect. It can be found by scraping at suspected areas with a stiff scraper. Whether the attack is serious depends on its location and depth. Quite often, pass partitions can be almost completely corroded and still function efficiently, unless the carbon shell is broken or chipped.

In any type of exchanger, corrosion may occur where dissimilar metals are in close contact. The less noble of the two metals will corrode. Thus, carbon steel channel gasket surfaces near brass tube sheets will often corrode at a higher rate than they would otherwise.

Cracks are most likely to occur where there are sharp changes in shape or size or near welded seams, especially if a high stress is applied to the piece. Parts such as nozzles and shell flanges should be checked for cracks if excessive stresses have been applied to a unit.

When materials flow at high velocities in exchanger units, greater attacks can be expected if changes are made in the di-

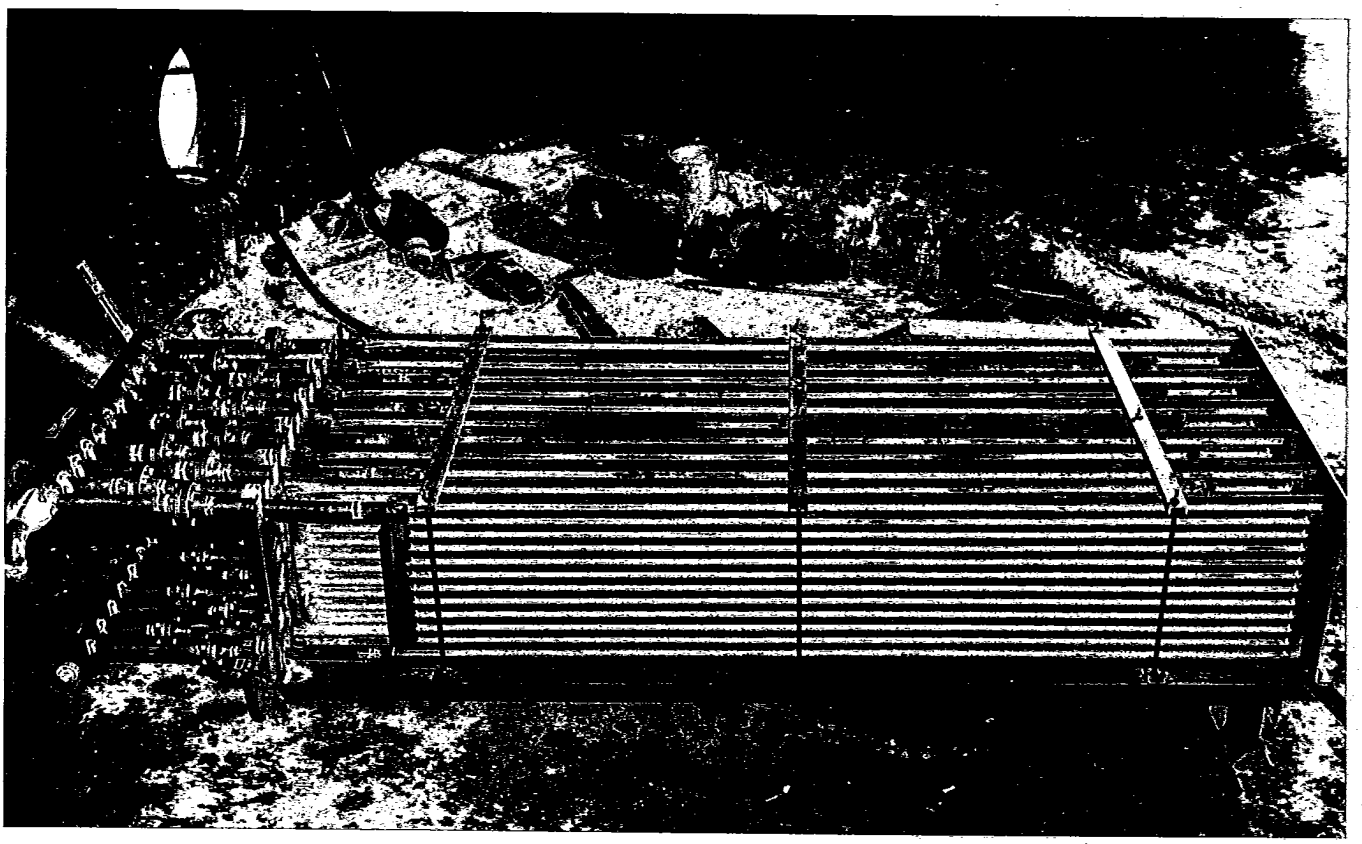


Figure A-7—Plate-Type Exchanger

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Figure A-8—Tubes Thinned at Baffles

rection of flow. These greater attacks would occur on or near such parts as tube inlets in tubular units and return bends in double-pipe units and condenser box coils.

Shells adjacent to inlet impingement plates and bundle baffles are susceptible to erosion, especially when velocities are high.

A distinctive prussion blue on bundle tubes indicates the presence of ferriferrocyanide. Hydrogen blistering is likely to be found on the exchanger shell near this color. A long straightedge may prove useful in determining the existence of blistering. Irregularities of the surface show up when the straightedge is placed on it. (The straightedge is also useful when investigating pitting.)

A.10 Inspection of Coils and Double–Pipe Exchanger Shells

Basically, coils in open condenser boxes and double-pipe exchanger shells are composed of pipe. They should be inspected according to the procedures detailed in API Recommended Practice 574. (See Appendix B for a sample form for making an inspection report on a double-pipe coil exchanger.)

First, a thorough visual inspection should be made, including a complete hammering of the pipe. A scraper may be used to detect external pitting, a common defect found on the outside of coils in condenser boxes.

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Following the visual inspection, thickness measurements should be taken. It is generally sufficient to use calipers to measure the open ends of double-pipe exchanger shells. To measure the wall thicknesses of coils and the middle section of double-pipe shells, ultrasonic and eddy-current devices can be used.

The enclosures of condensers and cooler boxes are made of concrete or light-gauge carbon steel. These enclosures should be visually inspected when the enclosed coil is inspected. When the container is made of carbon steel, the hammer is the most useful inspection tool available to aid the visual check. Thin spots in the container wall can be found by hitting the wall with the hammer. Calipers can be used to measure the wall thickness at the open top. If measurements below the top are required, the nondestructive instruments can be used or the method of test-hole drilling can be applied. Concrete walls are best inspected by picking at selected points with a scraper to check for spalling, cracks, or soft spots.

A.11 Inspection of Extended Plate Exchangers

Extended plate exchangers are designed so that the flow openings between the plates are quite small. For this reason and because of the inaccessibility of the unit interior, these exchangers are usually built of alloys highly resistant to corrosion in their expected service. In most cases, the alloys used will also be highly resistant to corrosion in refinery atmospheres. Visual examination, except as discussed in 8.5.3, will not reveal much. The outer surfaces can be checked for nicks, cuts, gouges, or other forms of mechanical damage and for bulging from internal failures. Good lighting is essential for this inspection and will also prove valuable when performing the soap tests discussed in 8.5.2.

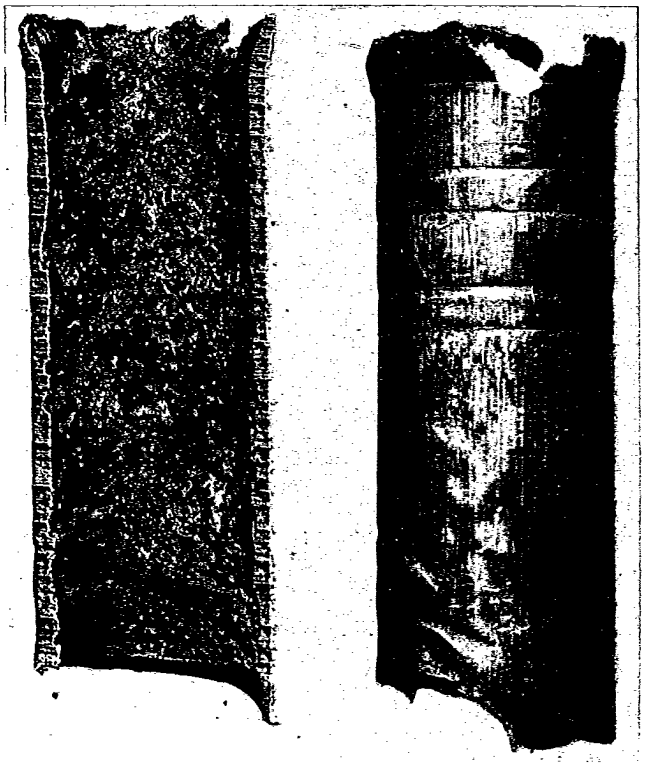


Figure A-9—Erosion-Corrosion Attack at Tube Ends

These units are usually built with integral channels and distribution manifolds, the thickness of which can be accurately measured with the ultrasonic instruments and then recorded. It is not advisable to use drilling equipment on the exchangers because the equipment could easily be damaged at these points. Welding of the alloys used in the units, such as aluminum and austenitic stainless steel alloys, requires welder skills not always readily available.

Light tapping with a small (8-ounce) hammer is useful in looking for cracked or broken parts on the exposed portions of the extended plate exchangers. The sound of the blow gives a clue to the condition. Cracked plates or manifold sections give off a tinny sound, which can be recognized more easily as more experience with the use of a hammer is acquired.

A.12 Inspection of Air–Cooled Exchangers

Refer to API Standard 661 for descriptions, minimum design criteria, and general information regarding air-cooled exchangers. API Standard 510 and the principles of API Standard 661 are to be followed in any ratings, repairs, and alterations of this type of exchanger. (See Appendix B for a sample form for making an inspection report on an air-cooled exchanger.)

Tubes are enclosed in fins and can not be inspected from the exterior. The best methods of inspecting the tubes are the internal-rotary, ultrasonic thickness-testing method and the eddy-current-testing method. Both of these methods work from the interior of the tubes. With competent operators and clean tubes, thicknesses and defects can be found using these methods. The tubes must be thoroughly cleaned before either method is effective.

The external fins of the tubes should be checked for cleanliness. If the fins need cleaning, washing with clean water alone or clean water with soap may be sufficient. If not, care should be taken in selecting a cleaning solution. Usually the fins are aluminum and could be harmed if the wrong cleaning medium is used.

The exterior of the tubes should be inspected between the tube sheet and the start of the fins. Exchangers in intermittent service or in service cool enough to allow moisture collection in this area are subject to external corrosion severe enough to cause leaks in this area. Coatings applied to this area will alleviate the problem of corrosion.

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API RP*572 92 ■ 0732290 0500817 1 ■

The insides of the tubes may be visually inspected near the tube-sheet ends of the air cooler. Fiber optic devices and borescopes are excellent devices for this type of inspection. A probe rod ⅛ inch (3.2 millimeters) or less in diameter and approximately 36 inches (91.4 centimeters) in length with a pointed tip bent at 90 degrees to the axis of the rod may also help to locate pits or corrosion at the tube ends.

Erosion-corrosion at the tube inlets is a common problem with air-cooled heat exchangers. This damage can be found by visual inspection through the header-box plug holes, or directly if the header box has a removable cover plate. If suitable conditions exist, reflecting sunlight into the tubes with a mirror is useful in inspecting for erosion-corrosion.

The box-type header ends of the air cooler should be inspected using the same techniques as those recommended for a pressure vessel. In addition, the sharp change in direction caused by its rectangular construction should be carefully checked for cracking. The header boxes with removable cover plates are obviously the easiest to inspect. A fiber optic scope may be the only way to check a header that has plug-type closures as opposed to a cover plate.

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APPENDIX B—SAMPLE RECORD FORMS

The *inspection record for pressure vessel in service* illustrates a form used to store data gathered during the inspection of a pressure vessel in service. Most plants develop a more detailed set of forms that also includes other pertinent data.

The *permanent pressure vessel record* illustrates a form that is used to record all the basic data of an individual pressure vessel and that becomes the permanent record for that vessel.

The *vessel inspection sheet* illustrates a form used as a progressive record of thicknesses, from which a corrosion rate can be calculated. Three versions of this form are included. One is blank. The other two show sketches that might be made for different types of pressure vessels. Information on only one pressure vessel should be recorded on any individual copy of this form.

Normally, an inspector would use one copy of this form to record field data, and another copy would become an office record. An inspector might use this form without a sketch when inspecting a vessel for which no basic data is available. In this case, he would make a sketch of the vessel on the form, including all pertinent dimensions and data he can secure in the field.

The *record of all pressure vessels on an operating unit* illustrates a form used to record and report the actual physical conditions and the allowable operating conditions of all pressure vessels on an operating unit.

The *exchanger inspection field data sheet,* the *exchanger data record,* the *exchanger inspection report form,* the *air cooler exchanger inspection report form,* and the *double-pipe exchanger inspection report form* illustrate other forms.

Note: Computer storage and retrieval of data in a format similar to that of the sample forms is acceptable and may be advantageous in many cases.

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INSPECTION RECORD FOR PRESSURE VESSEL IN SERVICE

OWNER OR USER NO. _____ OWNER _____ DATE _____

JURISDICTION OR

NATIONAL BOARD NO. _____ OWNER _____ DATE _____

MANUFACTURER _____

MANUFACTURER'S SERIAL NO. _____

DESIGN PRESSURE _____ TEMPERATURE _____

MAXIMUM ALLOWABLE ORIGINAL HYDROSTATIC

WORKING PRESSURE _____ TEST PRESSURE _____

ORIGINAL THICKNESS A _____ B _____ C _____ D _____

CORROSION ALLOWANCE A _____ B _____ C _____ D _____

DATE OF INSPECTION	THICKNESS AT CRITICAL POINTS				MAXIMUM METAL TEMPERATURE AT CRITICAL POINTS				MINIMUM ALLOWABLE METAL THICKNESS AT CRITICAL POINTS				DATE OF NEXT INSPECTION	SIGNATURE OF INSPECTOR
	A	B	C	D	A	B	C	D	A	B	C	D		

DESCRIPTION OF LOCATION _____ DATE _____

DESCRIPTION OF LOCATION _____ DATE _____

DESCRIPTION OF LOCATION _____ DATE _____

DESCRIPTION OF LOCATION _____ DATE _____

DESCRIPTION OF LOCATION _____ DATE _____

NOTE: MANUFACTURER'S DRAWING CAN BE USED TO SHOW THE LOCATION OF A, B, C, AND D.

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API RP*572 92 ■ 0732290 0500820 1 ■

PERMANENT PRESSURE VESSEL RECORD

NAME OF UNIT _____
VESSEL NAME _____
LOCATION _____
ORIGINAL ITEM NO. _____
DATE _____

HISTORY

ESTIMATE NO. _____
ORDER NO. _____
MANUFACTURED BY _____
MANUFACTURER'S SERIAL NO. _____
MANUFACTURER'S INSPECTOR _____

MANUFACTURER'S TEST PRESSURE _____
DATE RECEIVED _____
DATE INSTALLED _____
COMPANY NO. _____
COMPANY INSPECTOR _____

DESCRIPTION

GENERAL
DRAWING NO. _____
FABRICATOR'S _____
CONTRACTOR'S _____
COMPANY _____
POSITION (VERTICAL OR HORIZONTAL) _____
CODE CONSTRUCTED _____
CODE _____ YEAR _____
CODE STAMP _____
MATERIAL SPECIFIED AND GRADE OR TYPE _____
BASE _____
LINING _____
THICKNESS _____
STRESS RELIEVED (ORIGINAL) _____
RADIOGRAPHED (ORIGINAL) _____
COMPLETE _____
WELD INTERSECTIONS _____
SIZE
NOMINAL INSIDE DIAMETER _____
LENGTH BASE LINE TO BASE LINE _____
DESIGN _____
PRESSURE, PSI _____
TEMPERATURE, °F _____
STRESS, PSI _____
MAXIMUM ALLOWABLE OPERATING PRESSURE, PSI _____
MAXIMUM ALLOWABLE TEMPERATURE, °F _____
LIMITED BY _____

SHELL
TYPE OF CONSTRUCTION _____
JOINT EFFICIENCY _____
TYPE OF SUPPORT _____
INTERIOR OR EXTERIOR STIFFENERS _____
ORIGINAL THICKNESS _____
CORROSION ALLOWANCE _____
MANWAYS
NO. _____
SIZE _____
FLANGE RATING _____
REINFORCEMENT
FACTORY OR FIELD _____

TOP HEAD TYPE
ELLIPTICAL _____ HEMISPHERICAL _____
DISHED
CROWN REGION _____ KNUCKLE REGION _____
CONICAL (ANGLE) _____
FLAT _____
JOINT EFFICIENCY _____
ORIGINAL THICKNESS _____
CORROSION ALLOWANCE _____
MANWAYS
NO. _____
SIZE _____
FLANGE RATING _____
REINFORCEMENT
FACTORY OR FIELD _____

BOTTOM HEAD TYPE
ELLIPTICAL _____ HEMISPHERICAL _____
DISHED
CROWN REGION _____ KNUCKLE REGION _____
CONICAL (ANGLE) _____
FLAT _____
JOINT EFFICIENCY _____
ORIGINAL THICKNESS _____
CORROSION ALLOWANCE _____
MANWAYS
NO. _____
SIZE _____
FLANGE RATING _____
REINFORCEMENT
FACTORY OR FIELD _____
NOZZLES
MINIMUM FLANGE RATING _____
TYPE FACING _____
OPENINGS REINFORCED _____

REMARKS _____

NOTE: A COPY OF THIS SHEET SHALL BE PREPARED FOR EACH INDIVIDUAL VESSEL IN A UNIT. IF NEW VESSELS ARE INSTALLED OR ANY CHANGES ARE MADE TO PRESENT VESSELS AFFECTING THE DESCRIPTION ITEMS, A NEW OR REVISED COPY OF THIS SHEET SHALL BE SUBMITTED WITH THE CURRENT INSPECTION REPORT.

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VESSEL INSPECTION SHEET

UNIT _____

NAME OF VESSEL _____

DIAMETER _____

LENGTH _____

VESSEL NO. _____

DRAWINGS			
	SHELL	LINING	INTERNAL
FABRICATOR			
CONTRACTOR			
COMPANY			
SKETCH			

WALL THICKNESS MEASUREMENTS												
POINT NO.		ORIGINAL THICKNESS	MINIMUM ALLOWABLE THICKNESS	DATE								
INSPECTION INTERVALS	NO.											
	1											
	2											
	3											
	4											
	5											
	6											
	7											
	8											
	9											

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VESSEL INSPECTION SHEET

UNIT _____

NAME OF VESSEL _____

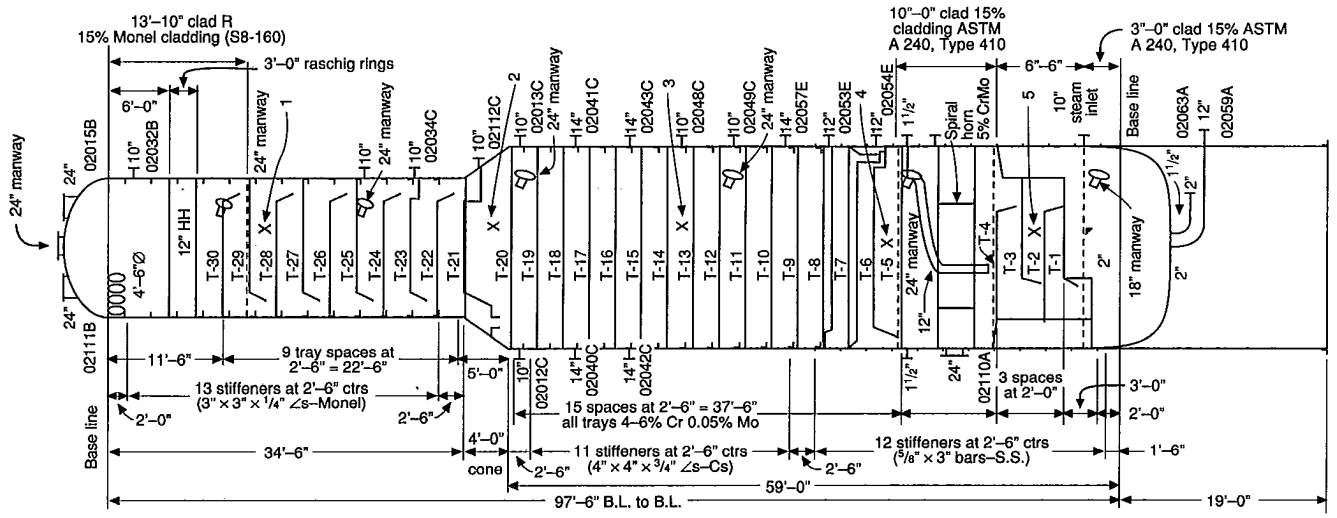
DIAMETER _____

LENGTH _____

VESSEL NO. _____

DRAWINGS			
	SHELL	LINING	INTERNAL
FABRICATOR			
CONTRACTOR			
COMPANY			

SKETCH



WALL THICKNESS MEASUREMENTS

POINT NO.		ORIGINAL THICKNESS	MINIMUM ALLOWABLE THICKNESS	DATE							
INSPECTION INTERVALS	NO.										
	1										
	2										
	3										
	4										
	5										
	6										
	7										
	8										
	9										

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API RP*572 92 ■ 0732290 0500823 7 ■

VESSEL INSPECTION SHEET

UNIT _____

NAME OF VESSEL _____

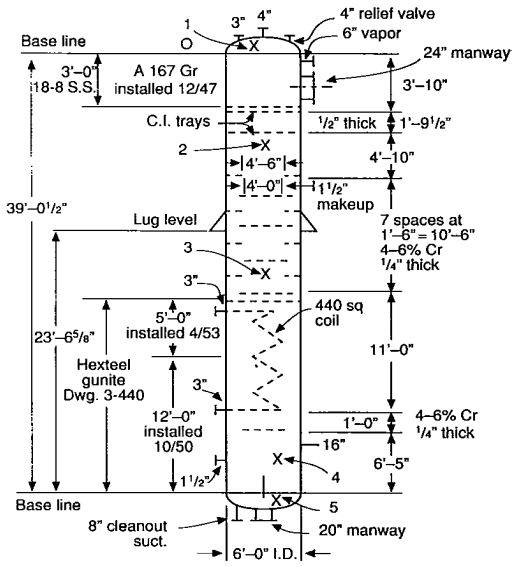
DIAMETER _____

LENGTH _____

VESSEL NO. _____

DRAWINGS			
	SHELL	LINING	INTERNAL
FABRICATOR			
CONTRACTOR			
COMPANY			

SKETCH



WALL THICKNESS MEASUREMENTS											
POINT NO.		ORIGINAL THICKNESS	MINIMUM ALLOWABLE THICKNESS	DATE							
INSPECTION INTERVALS	NO.										
	1										
	2										
	3										
	4										
	5										
	6										
	7										
	8										
	9										

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RECORD OF ALL PRESSURE VESSELS ON AN OPERATING UNIT (SHEET 1)

NAME OF UNIT _____

LOCATION _____

INSPECTION AND TEST NO. _____

DATE _____

NAME OF VESSEL				
COMPANY VESSEL AND SKETCH NO.				
OPERATING DATA				
HOURS UNDER PRESSURE TO DATE				
SERVICE DATA				
AVERAGE MAXIMUM OPERATING PRESSURE, PSI				
AVERAGE MAXIMUM OPERATING TEMPERATURE, °F				
TOP				
BOTTOM				
NO. TRAYS				
NO. BAFFLES				
NO. COILS				
INSPECTION AND TEST DATA				
INSPECTOR				
NOMINAL INSIDE DIAMETER				
MINIMUM THICKNESS				
SHELL				
LOCATION				
TOP HEAD				
BOTTOM HEAD				
JOINT EFFICIENCY				
SHELL				
HEADS				
HEAD FACTOR				
TOP				
BOTTOM				
LAST INSPECTION				
INSIDE WELDS				
OUTSIDE WELDS				
UNDER INSULATION				
EXTENT INSPECTION				

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API RP*572 92 ■ 0732290 0500825 0 ■

RECORD OF ALL PRESSURE VESSELS ON AN OPERATING UNIT (SHEET 2)

NAME OF UNIT _____

LOCATION _____

INSPECTION AND TEST NO. _____

DATE _____

NAME OF VESSEL				
COMPANY VESSEL AND SKETCH NO.				
TEST PRESSURE				
VESSEL, PSI				
COILS, PSI				
TEST MEDIUM				
TIME PRESSURE HELD –(1) + (2)				
MAXIMUM ALLOWABLE OPERATING PRESSURE, PSI				
MAXIMUM ALLOWABLE OPERATING				
TEMPERATURE, °F				
LIMITED BY				
WORKING STRESS AT OPERATING				
TEMPERATURE, PSI				
APPROVED OPERATING PRESSURE, PSI				
APPROVED OPERATING TEMPERATURE, °F				
SAFETY VALVE SETTING, PSI				
PROTECTIVE LINING DATA				
DRAWING NO.				
DATE INSTALLED				
MATERIAL AND TYPE				
SECTION LINED				
DATE REPAIRED				
EXTENT OF REPAIRS				
DATE PREVIOUS LINING REMOVED				
CAUSE OF REMOVAL				
REMARKS				

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EXCHANGER INSPECTION FIELD DATA SHEET

ORIGINAL THICKNESS																				
	DRILL POINT LOCATIONS WITH THICKNESS MEASURED IN 0.01 INCHES																			
DATE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

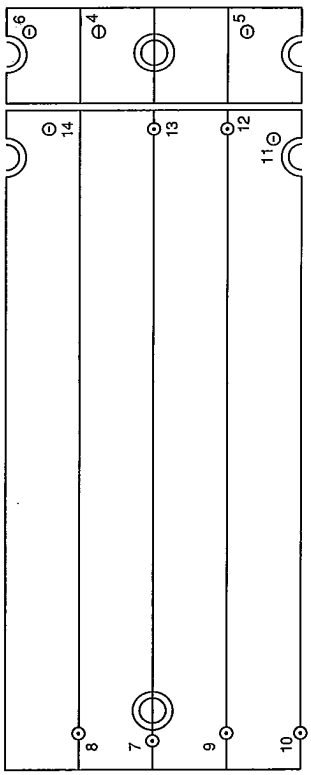
ORIGINAL THICKNESS																				
	NOZZLE LOCATIONS WITH NECK THICKNESS MEASURED IN 0.01 INCHES																			
DATE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

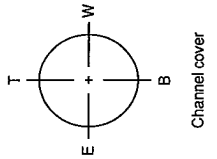
LEGEND
⊙ REMOVABLE TEST PLUG
● WELDED TEST PLUG
⊖ CALIPERED
⊙ TELL TALE HOLE

REVISIONS

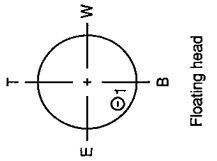
Mfr. _____ Type _____
Design Data:

	Tubes	Shell
Fluid thru		
Temperature		
Pressure		
Size		
Material		

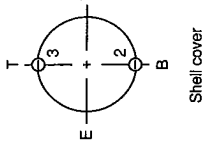




Channel cover



Floating head



Shell cover

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EXCHANGER DATA RECORD

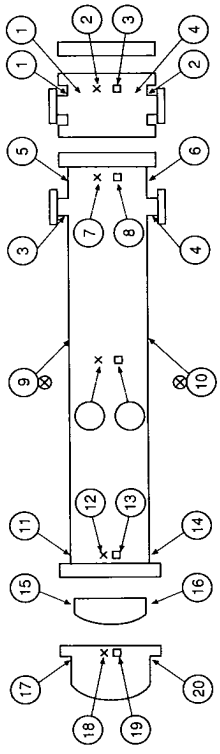
ORIGINAL THICKNESS		DRILL POINT LOCATIONS WITH THICKNESS MEASURED IN 0.01 INCHES																			
	DATE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

ORIGINAL THICKNESS		NOZZLE LOCATIONS WITH NECK THICKNESS MEASURED IN 0.01 INCHES																			
	DATE	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

LEGEND
⊙ REMOVABLE TEST PLUG
● WELDED TEST PLUG
⊖ CALIPERED
⊙ TELL TALE HOLE
□ SIDE SHOWN
X REVERSE SIDE

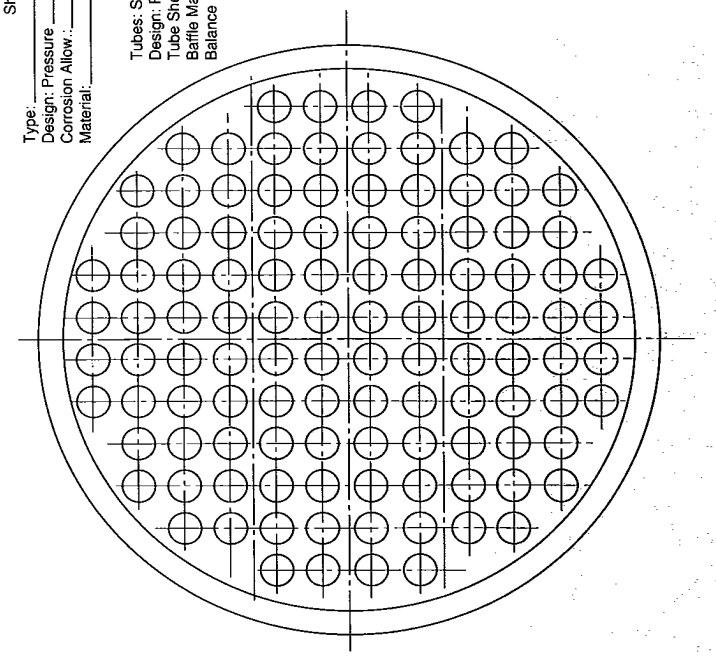
REVISIONS

SCALE SHOWN
DRAWN BY
CHECKED BY
DATE
APPROVED BY
DRAWING NO.



Shell
Type:
Design: Pressure Temp.
Corrosion Allow:
Material:

Bundle
Tubes: Size Mat.
Design: Press. Temp.
Tube Sheet Mat.
Baffle Mat.
Balance Mat.



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EXCHANGER INSPECTION REPORT FORM

TO: _____ DATE _____

_____

UNIT _____ EQUIPMENT _____ SERVICE _____

DOWN TIME _____ RETURNED TO SERVICE _____ DATE OF LAST INSPECTION _____

REASON _____

_____

INSPECTION COMMENTS _____

_____

	MINIMUM THICKNESS			
	ORIGINAL	LAST INSPECTION	PRESENT	RETIRING THICKNESS
SHELL				
SHELL COVER				
CHANNEL				
CHANNEL COVER				
TOP NOZZLE (SHELL)				
BOTTOM NOZZLE (SHELL)				
TOP NOZZLE (CHANNEL)				
BOTTOM NOZZLE (CHANNEL)				
TUBES				

RECOMMENDATIONS _____

_____

WORK REQUESTED BY _____ INSPECTED BY _____

CC: _____ SIGNED _____

CHIEF INSPECTOR

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AIR COOLER EXCHANGER INSPECTION REPORT FORM

NAME OF UNIT _____

_____ PLANT

EXCHANGER NAME _____

_____

ORIG. ITEM NO. _____

PLANT NO. _____

PLUG TYPE—REMOVABLE HEADER

DATE ORIG. EXCH. INST. _____

DATE PREV. BDLE. INST. _____

CAUSE REMOVAL _____

DATE PRES. BDLE. INST. _____

MAT'L HANDLED _____

DESIGN—PRESS. PSI _____

TEMP. °F _____

OPER.—PRESS. PSI _____

TEMP. °F _____

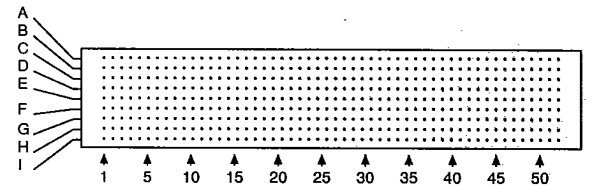
TUBE—MATERIAL _____

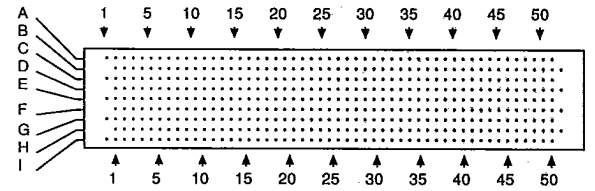
—DIAMETER _____

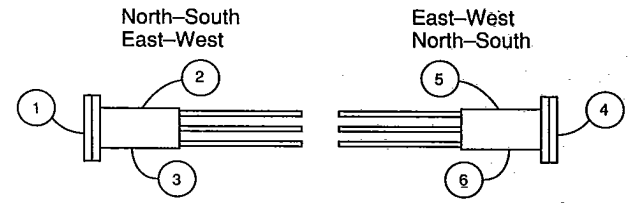
—THICK. _____

—LENGTH _____

FIN—MATERIAL _____







THICKNESS DATA

POINT NO.		ORIG. THK.	PREV. MIN.	DATE					
INSP. INT.	NO.								
	1								
	2								
	3								
	4								
	5								
	6								

INSP. & TEST DATA						
DATE _____						
GENERAL CONDITION _____						
CHANNEL SECTION _____						
TUBE SECTION _____						
FINS _____						
FAN _____						
FAN HOUSING _____						
NO. TUBES PLUGGED _____						
CHEMICALLY CLEANED _____						
MAX. ALLOW. PRESS. PSI _____						
TEMP. °F _____						
TEST PRESS. PSI _____						
APPROVED - PRESS. PSI _____						
TEMP. °F _____						

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DOUBLE–PIPE EXCHANGER INSPECTION REPORT FORM

NAME OF UNIT _____

_____ PLANT

EXCHANGER NAME _____

_____

ORIG. ITEM NO. _____

PLANT NO. _____

SERVICE DATA

DATE INT. TUBE INST. _____

DATE PREV. INT. TUBE _____

INSTALLED _____

REMOVED _____

CAUSE REMOVAL _____

DATE EXTER. TUBE

INSTALLED _____

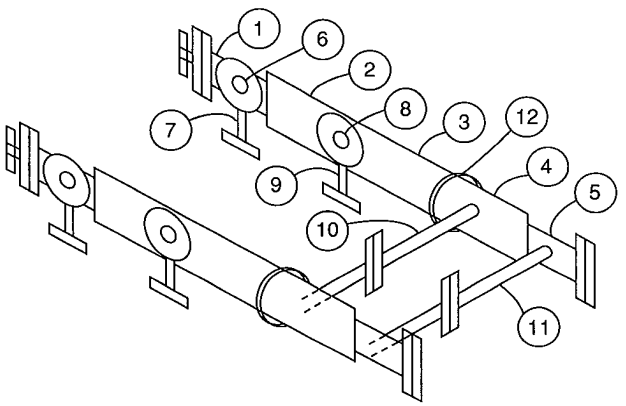
REMOVED _____

CAUSE REMOVAL _____

OPERATING DATA

TYPE OF SERVICE _____

	INT. PIPE	EXT. PIPE
STATE OF MEDIUM		
MED. OR MAT'L HAND.		
DESIGN—PRESS. PSI		
TEMP. °F		
OPER.—PRESS. PSI		
TEMP. °F		



THICKNESS DATA

POINT NO.		ORIG. THK.	PREV. MIN.	DATE					
INSP. INT.	NO.								
	1								
	2								
	3								
	4								
	5								
	6								
	7								
	8								
	9								
	10								
	11								
	12								
	13								

INSP. & TEST DATA

DATE												
INSPECTION INTERVAL												
LIMITING—INT. TUBE THK.												
LIMITING—EXT. TUBE THK.												
GENERAL CONDITION												
INT. TUBE SECTION												
EXT. TUBE SECTION												
INT. TUBE MATERIAL												
EXT. TUBE MATERIAL												
DATE LAST INSPECTED												
CHEMICALLY CLEANED												
INT. MAX. ALL.—PRESS. PSI												
TEMP. °F												
EXT. MAX. ALL.—PRESS PSI												
TEMP °F												
INT. TEST PRESS. PSI												
EXT. TEST PRESS. PSI												
	INT. TUBE	EXT. TUBE	INT. TUBE	EXT. TUBE	INT. TUBE	EXT TUBE	INT. TUBE	EXT TUBE	INT. TUBE	EXT TUBE	INT. TUBE	EXT TUBE

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Order No. 822-57200

1-1300—2/92—7.5C (1E)

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